



Avalara AvaTax Mapper

Guide for Extending Avalara AvaTax for Salesforce

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1.0 Acronyms and Abbreviations

FIPS	Federal Information Processing Standards
ECMS	Exemption Certificate Management System
UPC	Universal Product Code
CPQ	Configure Price Quote
SOQL	Salesforce Object Query Language
NA	Not Applicable
SST	Streamlined Sales Tax
VAT	Value Added Tax
GST	Goods and Service Tax
RSFs	Retail Sales Fees
PIFs	Property Improvement Fund
PDF	Portable Document Format
JPG	Joint Photographic Group
SKU	Stock Keeping Unit
XML	Extensible Markup Language
REST	Representational State Transfer

2.0 AvaTax Mapper

AvaTax Mapper is a framework that helps you to create the Address Validation and Tax Calculation request based upon the AvaTax XML configuration file.

It includes a large library for analyzing the AvaTax XML configuration file and then formulates the service request models.

It supports a variety of Salesforce-based products like Salesforce CPQ+, Salesforce Billing, Salesforce Lightning Order Management, and Salesforce Sales Cloud.

Note: In the examples provided in this document, AVA_SFCLLOUD is the namespace given only to the Salesforce Sales Cloud package. For other AvaTax Salesforce packages, refer to the respective package namespace instead of AVA_SFCLLOUD.

3.0 Architecture

3.1. XML Analyzer

The XML analyzer offers a platform for reading the AvaTax XML configuration file.

It extracts relevant data and transforms it into the understandable data type.

The map/dictionary plays an important role in creating the service request objects required by the AvaTax services.

3.2. Tax Calculator

The Tax Calculator class handles all the complexities of creating the tax calculation request from an AvaTax XML configuration file.

The package offers Hooks, which can be extended to override the default flow of the AvaTax Mapper package, so that it can fit into the user's business process as appropriate.

The following extension methods are provided for the tax calculation:

1. Before Tax Calculation
2. Tax Validation
3. Override Request Line
4. Before Update
5. After Update

3.3. Address Validator

The Address Validator class handles all the complexities of creating address validation request and displays the validated address on the separate page.

It provides an option to the user to update the original address.

3.4. Hook Manager

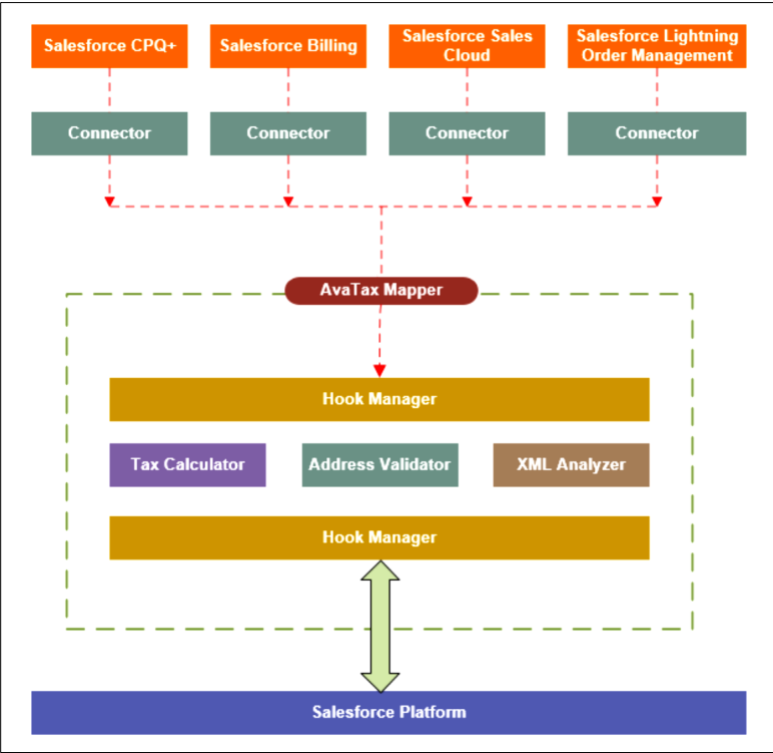
The Hook Manager is the core of providing an extension, so that the customer(s)/SI partner(s) can modify the web request that goes to the AvaTax engine.

The administrator or developer needs to create a class that implements a specific interface to make certain changes to the system to handle custom business requirements.

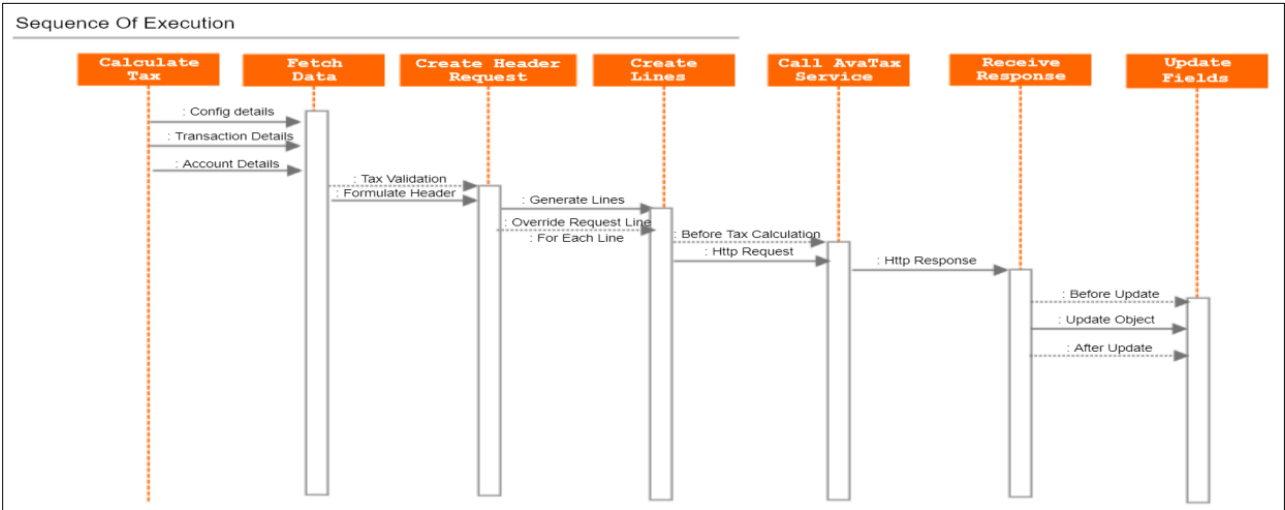
The user needs to enter the class name in the AvaTax XML configuration file.

The Hook Manager considers the extensions provided in the AvaTax XML configuration file.

The following sequence diagram gives you more information on the Hook Manager and the extensions:



The following sequence diagram allows to elaborate more about the Hook Manager and the available extensions:



4.0 AvaTax XML Configuration Guideline

The AvaTax XML configuration file helps to specify "Request | Response" parameter required to consume the AvaTax service.

The AvaTax XML configuration file is a part of the AvaTax extension packages.

The AvaTax XML configuration file formulates the request and receives a response for the Tax Calculation, Address Validation, and the AvaTax Exemption Document.

The AvaTax XML configuration file describes about the main configuration settings.

4.1. Configuration File Section

The AvaTax XML configuration file is divided into the following nodes:

4.1.1. Beginning Node

The Beginning Node specifies the beginning of the AvaTax XML configuration file.

For Example, `<avatax> </avatax>`.

4.1.2. Feature Node

The Feature Node specifies the beginning of the function name that needs to be covered under the AvaTax XML configuration file.

Based on the value and purpose, the AvaTax Salesforce core package decides, which the AvaTax API needs to call. For Example, `<addressvalidation></addressvalidation>`.

4.1.3. Object Node

The Object Node specifies Salesforce object for the Feature Node. For Example,

`<Account></Account>`.

XML Scaffold

```
<avatax>
<FeatureNode1>
<ObjectNode1>
<ObjectComponent1>
<Hook> </Hook>
<QueryNode> </QueryNode>
<ParameterNode> </ParameterNode>
<MappingNode> </MappingNode>
<UpdateMappingNode> </UpdateMappingNode>
</ObjectComponent1>
</ObjectNode1>
. . .
<ObjectNode n> </ObjectNode n>
</FeatureNode1>
. . .
<FeatureNode n> </FeatureNode n>
</avatax>
```

4.1.4. Object Component

The node specifies the Object Component that needs to be considered for the Feature Node. For Example, Billing Address from Account to be considered for Address Validation.

Sr. No.	Feature Node	Object Node	Object Component
1.	Address Validation	Account	Billing

The Account object contains two addresses as Billing and Shipping.

The Object Component contains below nodes:

1. Query Node
2. Parameter Node
3. Mapping Node
4. Update Mapping Node

Note: The Billing address is taken as an example to explain the Object Component Nodes.

4.1.4.1 Query Node

The Query Node specifies the SOQL query for fetching the attributes of the Object Component which are specified in the Object Component Node. This SOQL query can be use further for the Mapping Node.

```
<query>
select BillingStreet, BillingCity,BillingState,BillingCountry,Billingpostalcode from Account where
id = '{AccountId}'
</query>
```

4.1.4.2 Parameter Node

The Parameters Node specifies the attributes that must be used for the WHERE clause in the SOQL query that is specified in the Query Node.

```
<query>
select BillingStreet, BillingCity,BillingState,BillingCountry,Billingpostalcode from Account where id =
'{AccountId}'
</query>
<parameters>
<AccountId>Id</AccountId>
</parameters>
```

Note: The ID - <AccountId>Id</AccountId> here is the record id of the object.

4.1.4.3 Mapping Node

The Mapping Node specifies the parameters of the request required to create a Feature Node request.

Each parameter in the node is a request feature parameter.

```
<Mapping>
<line1>BillingStreet</line1>
<city>BillingCity</city>
<region> BillingState </region>
<country>BillingCountry</country>
<postalcode> Billingpostalcode </postalcode>
</Mapping>
```

Note: The fields marked in Red are **AvaTax** Request Model fields, and the fields marked in Teal are the **Salesforce** standard fields.

4.1.4.4 UpdateMapping Node

The UpdateMapping Node specifies the response field, where the service response needs to be shown.

```
<UpdateMapping>
<line1>BillingStreet</line1>
<city>BillingCity</city>
<region> BillingState </region>
<country>BillingCountry</country>
<postalcode> Billingpostalcode </postalcode>
</UpdateMapping>
```

Note: The fields marked in **Red** are **AvaTax** Request Model fields, and the fields marked in **Teal** are the **Salesforce** standard fields.

4.2. Sample Tax Calculation XML

The following nodes and the AvaTax XML configuration file formulate the request and receives the response for the Tax Calculation.

4.2.1. Header Node

The Header Node is used to specify the header of the tax calculation request. The Header Node consists of the following nodes:

4.2.1.1 Query Node

The Query Node specifies the SOQL query for fetching the attributes of the Object Component which are specified in the Object Component Node. This SOQL query can be use further for the Mapping Node.

```
<query>
select Id, AccountId,SalesTax__c,CloseDate,LeadSource from Opportunity where id =
'{'OpportunityId}'
</query>
```

4.2.1.2 Parameter Node

The Parameters Node specifies the attributes that must be used for the WHERE clause in the SOQL query that is specified in the Query Node.

```
<query>
select Id, AccountId,SalesTax__c,CloseDate,LeadSource from Opportunity where id =
'{{OpportunityId}}'
</query>
<parameters>
<OpportunityId>Id</OpportunityId>
</parameters>
```

Note: The Id - < OpportunityId >ID</ OpportunityId > here is the record ID of the object.

4.2.1.3 Mapping Node

- The Mapping Node specifies the parameters of the request required to create a Feature Node request.
- Each parameter in the node is a request feature parameter.

```
<Mapping>
<code>Id</code>
<customerCode>AccountId</customerCode>
<description>LeadSource</description>
</Mapping>
```

- **Note:** The fields marked in Red are **AvaTax** Request Model fields, and the fields marked in Teal are the **Salesforce** standard fields.

4.2.1.4 UpdateMapping Node

The UpdateMapping Node specifies the response field, where the service response needs to be shown.

```
<UpdateMapping>
< SalesTax__c >totalTax</SalesTax__c>
</UpdateMapping>
```

Note: The fields marked in Red are **Salesforce** fields, and the fields marked in Teal are the **AvaTax** response model fields.

4.2.2. Line Node

The Line Node is used to specify the tax calculation request line. The node of the line consists of the following nodes:

4.2.2.1 Query Node

The Query Node specifies the SOQL query for fetching the attributes of the Object Component which are specified in the Object Component Node. This SOQL query can be use further for the Mapping Node.

```
<query>
select Id, Pricebookentry.product2.ProductCode, Description, SalesTax__c,
Quantity, TotalPrice, UnitPrice from OpportunityLineItem where OpportunityId = '{OpportunityId}'
</query>
```

4.2.2.2 Parameter Node

The Parameters Node specifies the attributes that must be used for the WHERE clause in the SOQL query that is specified in the Query Node.

```
<query>
select Id, Pricebookentry.product2.ProductCode, Description, SalesTax__c,
Quantity, TotalPrice, UnitPrice from OpportunityLineItem where OpportunityId =
'{{OpportunityId}}'
</query>
<parameters>
<OpportunityLineId>Id</ OpportunityLineId >
</parameters>
```

Note: The ID - < OpportunityLineId >Id</ OpportunityLineId > here is the record ID of the object.

4.2.2.3 Mapping Node

The Mapping Node specifies the parameters of the request required to create a Feature Node request. Each parameter in the node is a request feature parameter.

```
<Mapping>
<quantity> Quantity </quantity>
<itemCode> Pricebookentry.product2.ProductCode </itemCode>
<amount> TotalPrice </amount>
</Mapping>
```

Note: The fields marked in Red are **AvaTax** Request Model fields, and the fields marked in Teal are the **Salesforce** standard fields.

4.2.2.4 Update Mapping Node

The UpdateMapping Node specifies the response field, where service response needs to be shown.

```
<UpdateMapping>  
<SalesTax__c>totalTax</SalesTax__c>  
</UpdateMapping>
```

Note: The fields marked in Red are **Salesforce** fields, and the fields marked in Teal are the **AvaTax** response model fields.

5.0 How to Use AvaTax Mapper

5.1. Address Validation

- The Address Validation follows an easy and effective approach with the AvaTax Mapper.
- It allows you to validate single and multiple addresses. For Example, if you need to have Address Validation on Account object, then follow the below mentioned steps:

5.1.1. Address Validation for Multiple Addresses In A Single Click Event

The user needs to create a button on the object where the Address Validation is required.

1. On the Select Field Type, click the drop-down arrow and then select the object.
2. On the Content Source, click the drop-down arrow and then select the URL.

The screenshot shows the 'Custom Button or Link Edit' window. The 'Label' is 'Validate Billing Address', 'Name' is 'AvaTax_Validate_Billing_Ac', and 'Namespace Prefix' is 'AVA_SFCLLOUD'. The 'Display Type' is set to 'Detail Page Button'. The 'Behavior' is 'Display in existing window without sidebar'. The 'Content Source' is 'URL'. Below these fields, the 'Select Field Type' is 'Account' and the 'Insert Field' is '-- Insert Merge Field --'. The 'Insert Operator' is set to '&'. The URL in the content source field is `/apex/AVA_MAPPER_addressValidation?id={!Account.Id}&sType=SFConfig`.

The screenshot shows the 'Custom Button or Link Edit' window. The 'Label' is 'Validate Addresses', 'Name' is 'Validate_Addresses', and 'Namespace Prefix' is 'AVA_SFCLLOUD'. The 'Display Type' is set to 'Detail Page Button'. The 'Behavior' is 'Display in existing window without sidebar'. The 'Content Source' is 'URL'. Below these fields, the 'Select Field Type' is 'Account' and the 'Insert Field' is '-- Insert Merge Field --'. The 'Insert Operator' is set to '&'. The URL in the content source field is `/apex/addressValidation?id={!Account.Id}&sType=CPQ`. A red arrow points from the 'Content Source' dropdown to the URL field.

Custom Button or Link Edit [Save] [Quick Save] [Preview] [Cancel]

Label:

Name:

Description:

Display Type:
☐ Detail Page Link [View example](#)
☒ Detail Page Button [View example](#)
☐ List Button [View example](#)

Behavior: [View Behavior Options](#)

Content Source:

Select Field Type: Insert Field: Insert Operator:

3. In the URL box enter the data `"/apex/AVA_MAPPER__addressValidation?id={!Account.Id}&objectAPIName=AVA_SFCLCLOUD__AvaTax_Mapping__c&fieldAPIName=AVA_SFCLCLOUD__Field_Mapping__c"`.
4. Click Save.

Note:

1. The `AVA_SFCLCLOUD__AvaTax_Mapping__c` specifies the AvaTax Connector's AvaTax Mappings Custom Object API Name and `AVA_SFCLCLOUD__Field_Mapping__c` specifies the AvaTax Connector's AvaTax Mappings Custom Field API Name which is available in the AvaTax Connector package.
2. `{!Account.id}` specifies the Object ID of the object where the Address Validation is required. For address validation on the custom object, please specify the Custom Object API Name instead of Account.
3. For the Salesforce field with respect to AvaTax field mappings, please refer to the addressvalidation node from the AvaTax XML.
4. If available, then add the environment namespace before the object name if you need to integrate on the Custom Object. For Example: If you need Address Validation on Credit Note (Custom Object), then the URL is `"/apex/AVA_MAPPER__addressValidation?id={!AVA_SFCLCLOUD__CreditNote__c.Id}&objectAPIName=AVA_SFCLCLOUD__AvaTax_Mapping__c&fieldAPIName=AVA_SFCLCLOUD__Field_Mapping__c"` here, the `CreditNote__c` is the Custom Object API name and `AVA_SFCLCLOUD` is the Salesforce environment namespace.

5.1.2. Address Validation For Single Address In A Single Click Event

The user needs to create a button on the object where the Address Validation is required.

1. On the Select Field Type, click the drop-down arrow and then select the object.
2. On the Content Source, click the drop-down arrow and then select the URL.

Custom Button or Link Edit [Save] [Quick Save] [Preview] [Cancel]

Label: Validate Billing Address

Name: AvaTax_Validate_Billing_Ac [i]

Namespace Prefix: AVA_SFCLCLOUD

Description: [Text Area]

Display Type:

☐ Detail Page Link View example

☒ Detail Page Button View example

☐ List Button View example

Behavior: Display in existing window without sidebar [View Behavior Options]

Content Source: URL

Select Field Type: Account [v] Insert Field: -- Insert Merge Field -- [v] Insert Operator: [v]

/apex/AVA_MAPPER__addressValidation?id={!Account.Id}&sType=SFCConfig&adrType=Billing

3. In the URL box enter the data /apex/addressValidation?id={!Account.Id}&objectAPIName= AVA_SFCLCLOUD__AvaTax_Mapping__c&fieldAPIName= AVA_SFCLCLOUD__Field_Mapping__c&adrType=Billing
4. Click Save.

Note:

1. The difference between validating multiple and single address validation is the addition of "adrType" in the URL.
2. The "adrType" refers the address which will be validated. You must keep the same name as it is mentioned in the AvaTax package configuration file.

5.1.3. Address Validation For Multiple Records of a Single Object In A Single Click Event

The user needs to create a list button on the object where the Address Validation is required.

1. On the Select Field Type, click the drop-down arrow and then select the object.
2. On the Content Source, click the drop-down arrow and then select URL.

Account Custom Button or Link
New Button or Link

Custom Button or Link Edit Save Quick Save Preview Cancel

Label

Name ⓘ

Namespace Prefix AVA_MAPPER

Description

Display Type
☐ Detail Page Link [View example](#)
☐ Detail Page Button [View example](#)
☒ List Button [View example](#)
☒ Display Checkboxes (for Multi-Record Selection)

Behavior [View Behavior Options](#)

Content Source

Select Field Type Insert Field

Custom Button or Link Edit Save Quick Save Preview Cancel

Label

Name ⓘ

Description

Display Type
☐ Detail Page Link [View example](#)
☒ Detail Page Button [View example](#)
☐ List Button [View example](#)

Behavior [View Behavior Options](#)

Content Source

Select Field Type Insert Field

3. In the URL box enter the data,
/apex/AVA_MAPPER__bulkAddressValidation?sType=ConfigFile.
4. The ConfigFile specifies the configuration file of the AvaTax package which is available in the Salesforce static resources section.
5. Click Save.

6. In the Account List View, select the button(s) and then click the arrow button to add the List Buttons.

Edit Search Layout

Accounts List View

Customize the buttons on the Accounts list view:

To remove any standard buttons, remove the check next to the button.
To add custom buttons, select them and click Add.

Standard Buttons

☒ New [New]

☒ Open in Quip [OpenListInQuip]

☒ Get More Accounts [DataDotComGetAccountsMultiAddAction]

☒ Discover Companies [DiscoveryGetAccountsAction]

☒ Add to Call List [MassCreateCallList]

☒ Printable View [PrintableListView]

☒ Send Message [SendBulkMessageAction]

Custom Buttons

Available Buttons

--None--

Add

Remove

Selected Buttons

Validate Address

Up

Down

7. Navigate to the selected object to view the added List Button.



All Accounts

Edit | Delete | Create New View

New Account

Validate Address



☐

Action

☐

Edit | Del

Burlington Textiles Corp ...

☐

Edit | Del

Dickenson plc

Account Name

Account Site

Billing State/Province

8. Select the checkbox of the record(s), and then click the Validate Address button to validate the address of the selected records. You can select multiple records and can perform the address validation for multiple records.

Example: Address Validation on the Custom Object (Credit Note).

1. In Salesforce create the custom object as Credit Note and then browse down to the Custom Fields & Relationships.

Custom Object
Credit Note

[Standard Fields \[4\]](#) | [Custom Fields & Relationships \[8\]](#) | [Validation Rules \[0\]](#) | [Page Layouts \[1\]](#)
[Record Types \[0\]](#) | [Apex Sharing Reasons \[0\]](#) | [...](#)

Custom Object Definition Detail Edit Delete

Singular Label	Credit Note
Plural Label	Credit Notes
Object Name	Credit_Note
Namespace Prefix	AVA_SFCLCLOUD
API Name	AVA_SFCLCLOUD__Credit_Note__c

2. Click New and create the custom field for Credit Note.

Custom Fields & Relationships New Field Dependencies

Action	Field Label	API Name	Data Type	Indexed	Controlling Field
Edit Del	Account	AVA_SFCLCLOUD__Account__c	Lookup(Account)	✓	
Edit Del	Sales Tax	AVA_SFCLCLOUD__Tax__c	Currency(16, 2)		
Edit Del	Shipping City	AVA_SFCLCLOUD__City__c	Text(50)		
Edit Del	Shipping Country	AVA_SFCLCLOUD__Country__c	Text(50)		
Edit Del	Shipping Postal Code	AVA_SFCLCLOUD__Zip__c	Text(50)		
Edit Del	Shipping State	AVA_SFCLCLOUD__State__c	Text(50)		
Edit Del	Shipping Street	AVA_SFCLCLOUD__Street__c	Text(50)		
Edit Del	Status	AVA_SFCLCLOUD__Status__c	Text(50)		

3. Create a button on the Credit Note object. For Example: Validate Address button.

Credit Note Custom Button or Link
Validate Address

[« Back to Custom Object: Credit Note](#)

Custom Button or Link Detail Edit Window Open Properties Delete Where is this used?

Label	Validate Address	Object Name	Credit Note
Name	Validate_Address	Link Encoding	Unicode (UTF-8)
Namespace Prefix	AVA_SFCLCLOUD	Display Type	Detail Page Button
Behavior	Display in existing window without sidebar		
Button or Link URL	/apex/AVA_MAPPER__addressValidation?id={!AVA_SFCLCLOUD__Credit_Note__c.Id}&sType=SFConfig		
Height (in pixels)	600		
Description			
Created By	Sales Cloud Production, 12/11/2020 2:21 AM	Modified By	Sales Cloud Production

Edit Window Open Properties Delete Where is this used?

4. Keep the button or URL link as
/apex/AVA_MAPPER__addressValidation?id={!AVA_SFCLLOUD__Credit_Note__c.Id}&s
Type=SFConfig.
5. Navigate to Installed Packages and then click the Configure link.

Installed Packages

On AppExchange you can browse, test drive, download, and install pre-built apps and components right into your salesforce.com environment.

Apps and components are installed in packages. Any custom apps, tabs, and custom objects are initially marked as "In Development" and the other features in setup or as a group by clicking Deploy.

Depending on the links next to an installed package, you can take different actions from this page.

To remove a package, click **Uninstall**. To manage your package licenses, click **Manage Licenses**.

Action	Package Name	Publisher	Version Number	Namespace Prefix
Uninstall Configure	 AvaTax for SF Sales Cloud	Avalara, Inc.	5.0 (Beta 3)	AVA_SFCLLOUD
Uninstall	 Avalara AvaTax Salesforce Mapper	Avalara, Inc.	1.66 (Beta 1)	AVA_MAPPER

6. Select the **Additional settings** tab and then select **Open mapper studio**. The Avalara mapper studio opens in a new tab.

 Avalara for Salesforce Sales Cloud

Account credentials **Additional settings** Unlink integration company

[Open mapper studio](#)

Multi-company settings

Subsidiary object name

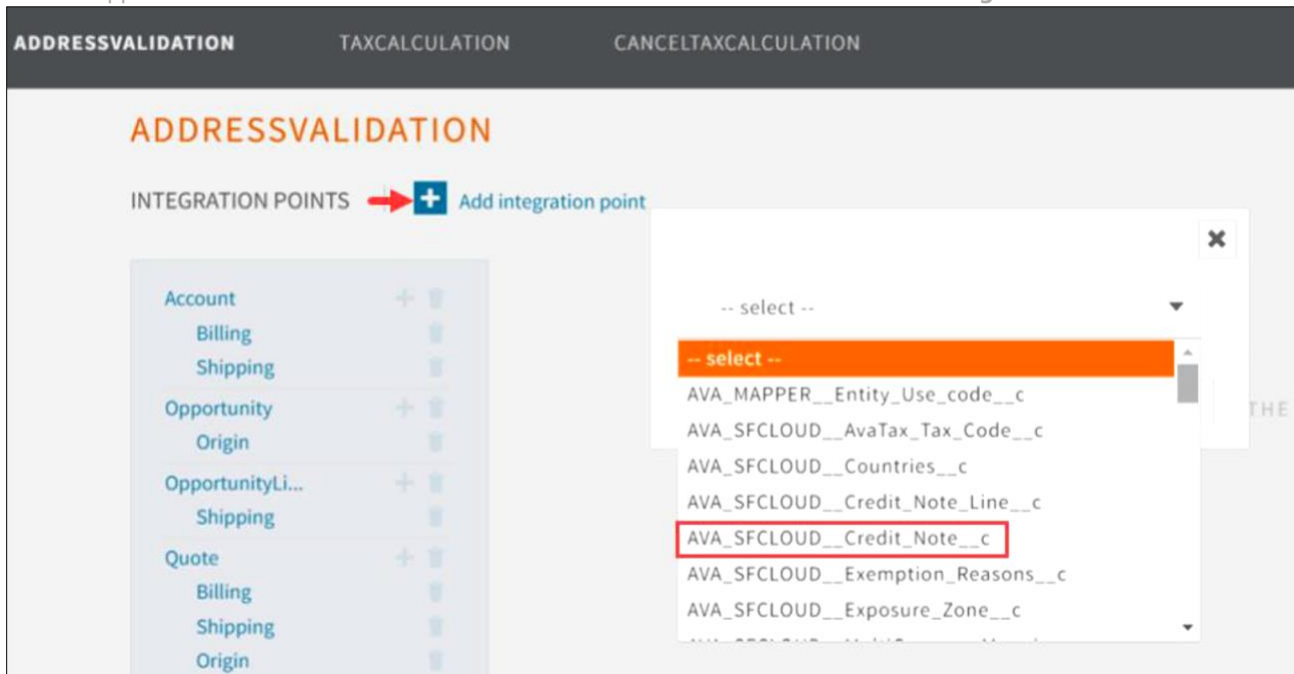
Subsidiary field name on account

[Display mapping](#) [Save](#)

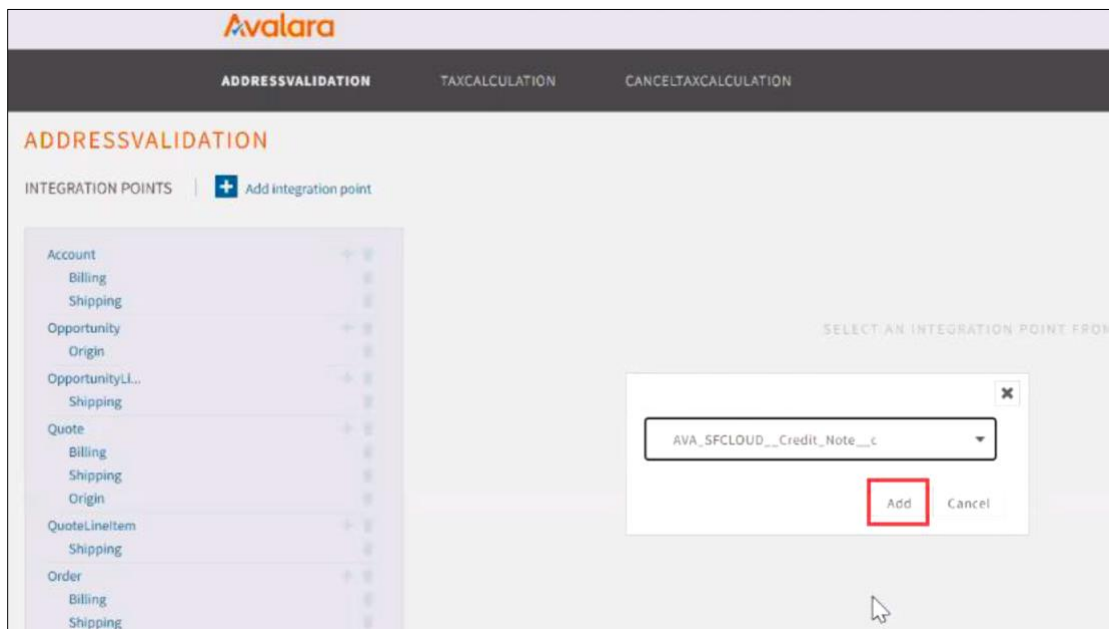
[Close](#)

7. Select ADDRESSVALIDATION and then click the Plus icon of the Add integration point to open the ADD NEW INTEGRATION POINT dialog box.

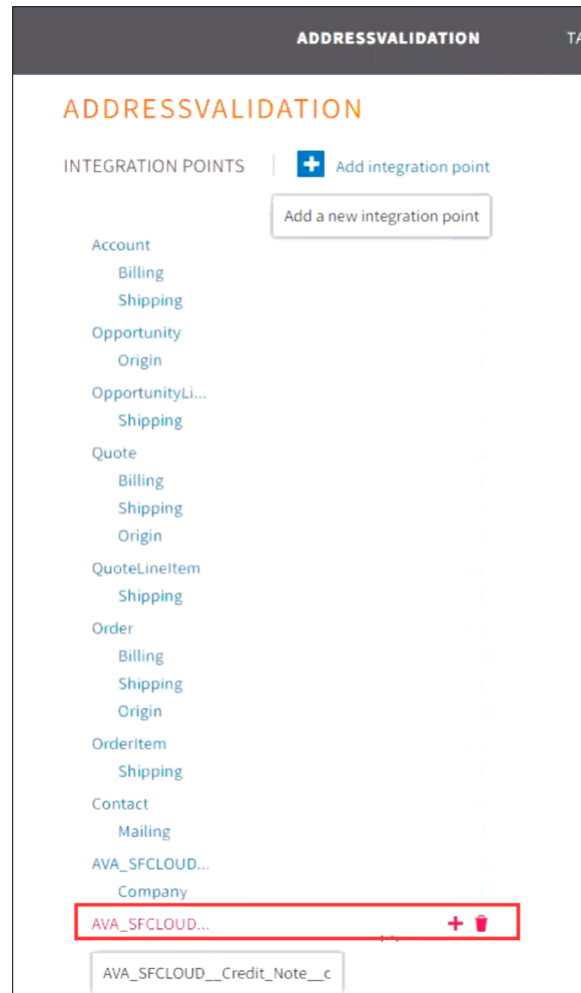


8. Browse the list of the API names and select the recently created custom object API name as "AVA_SFCLCLOUD__Credit_Note__c".

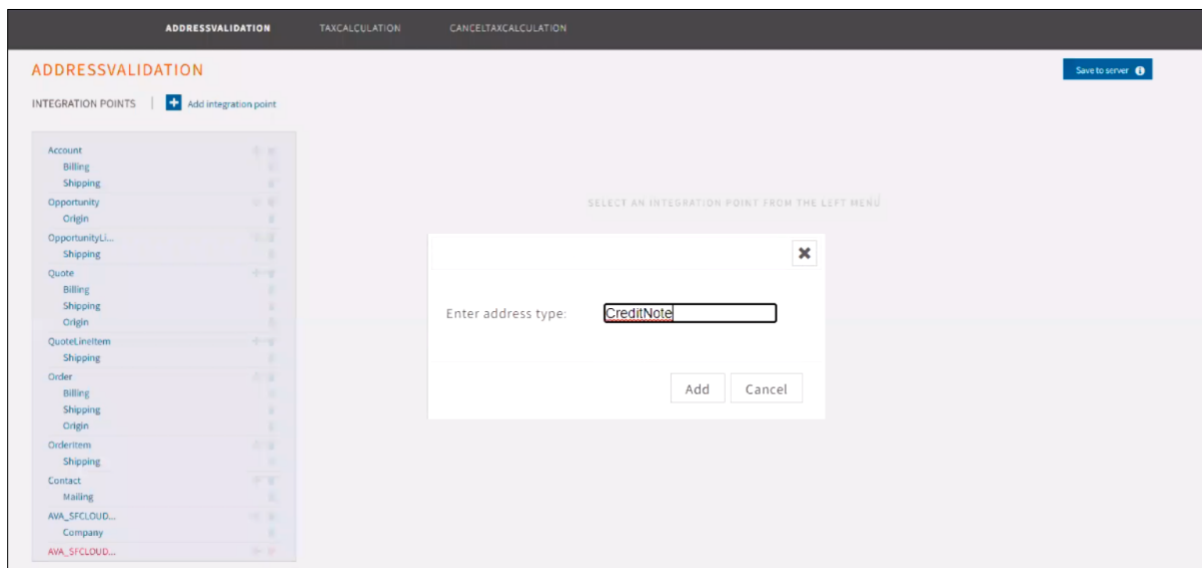
9. Click Add.



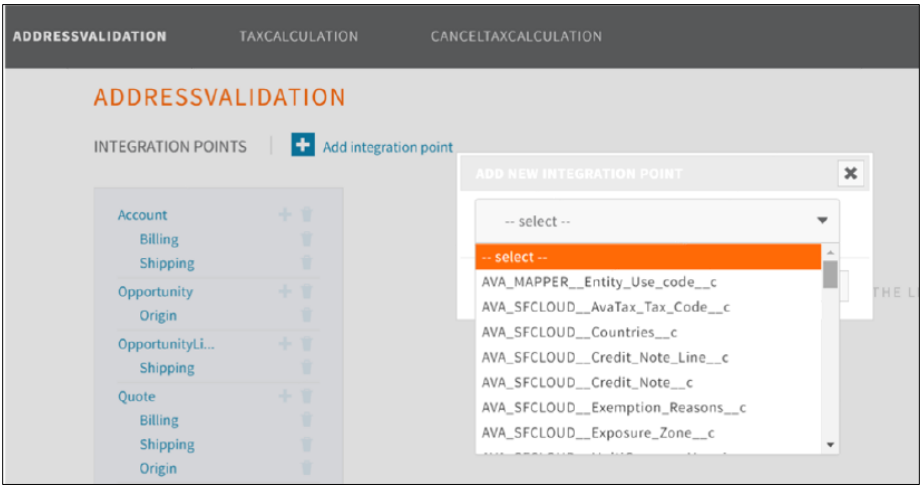
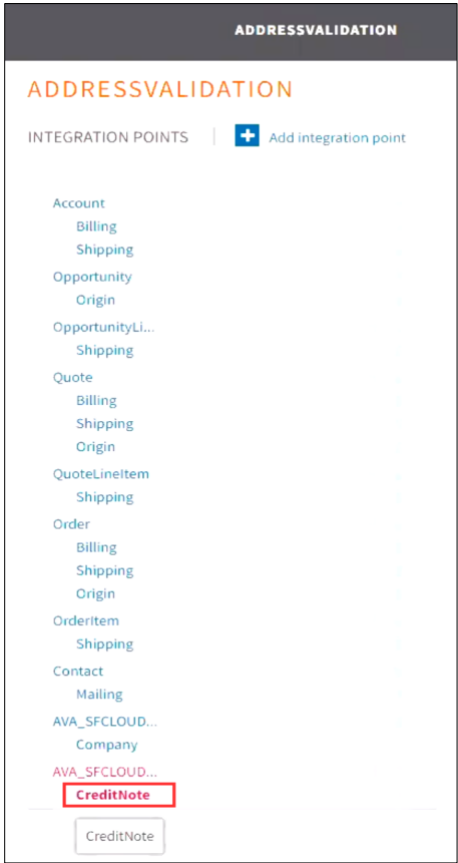
10. Browse to the recently added integration point and click the Plus icon.



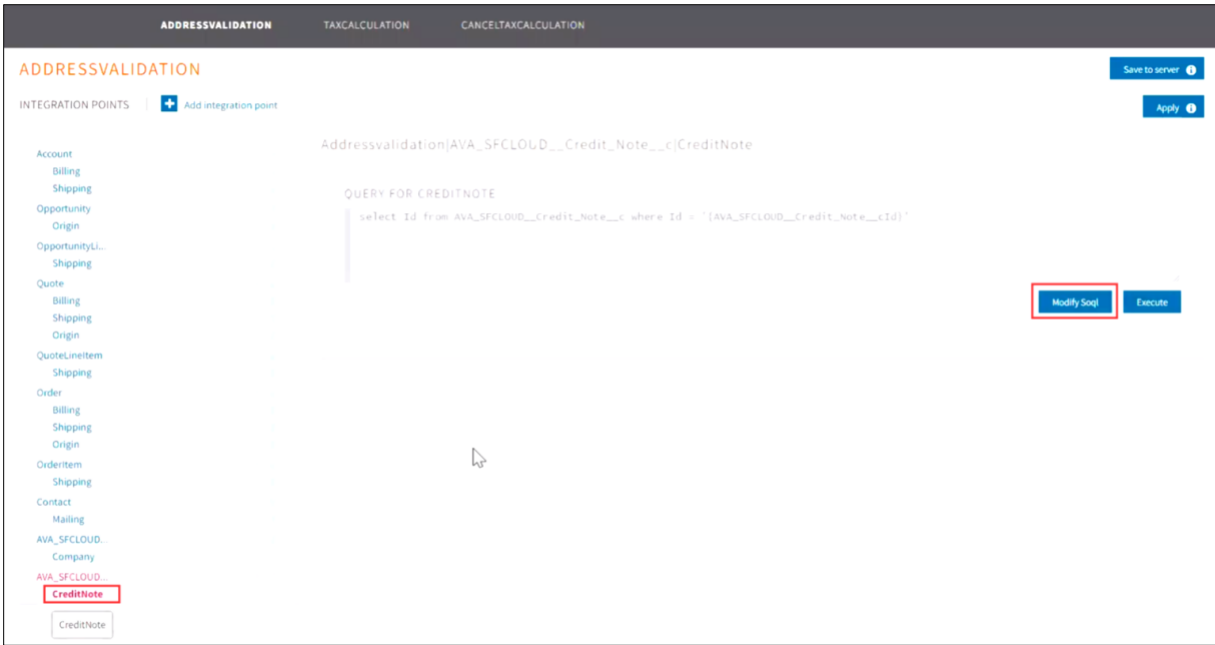
11. Enter the preferred address type and then click on the Add button. For Example: CreditNote



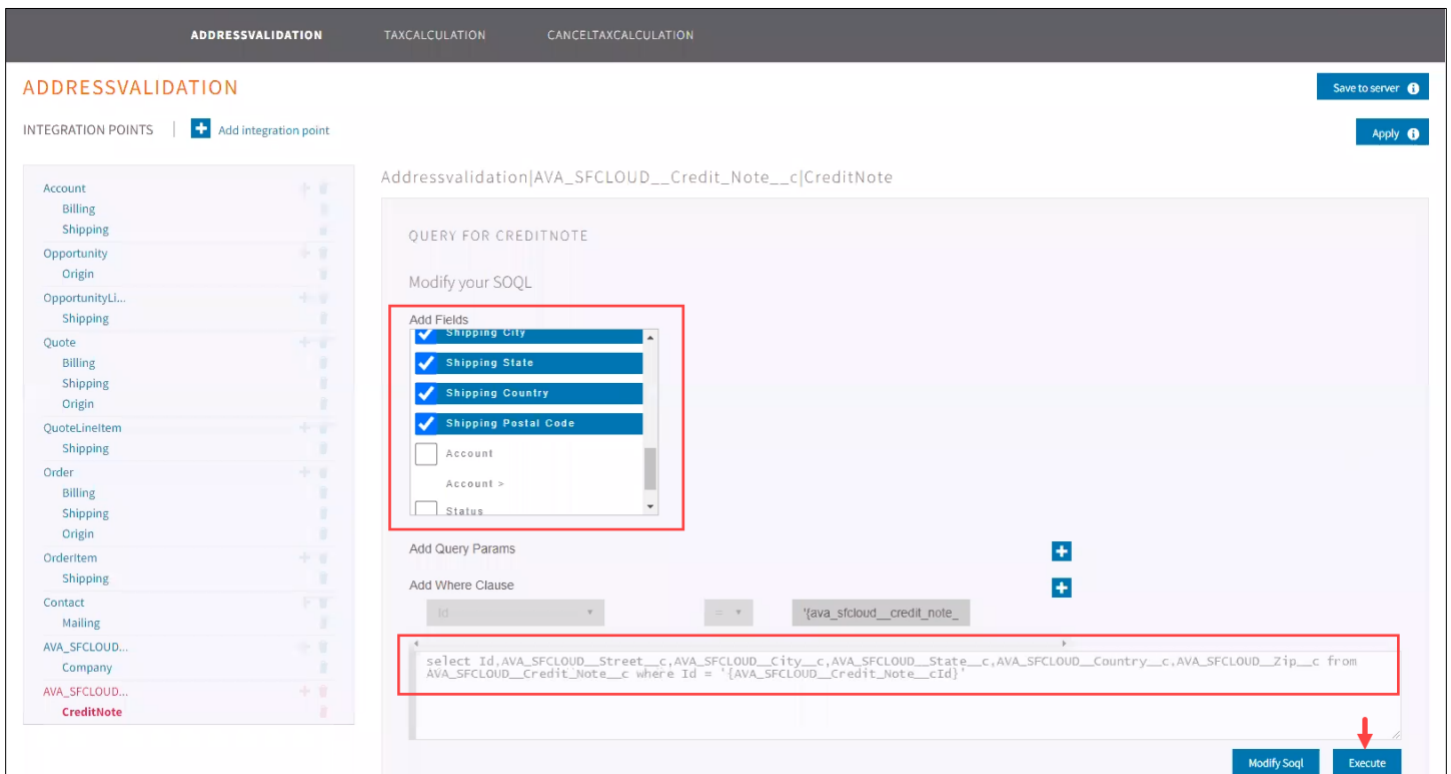
12. Click the recently created address type to open the QUERY FOR CREDITNOTE dialog box.
For Example: CreditNote



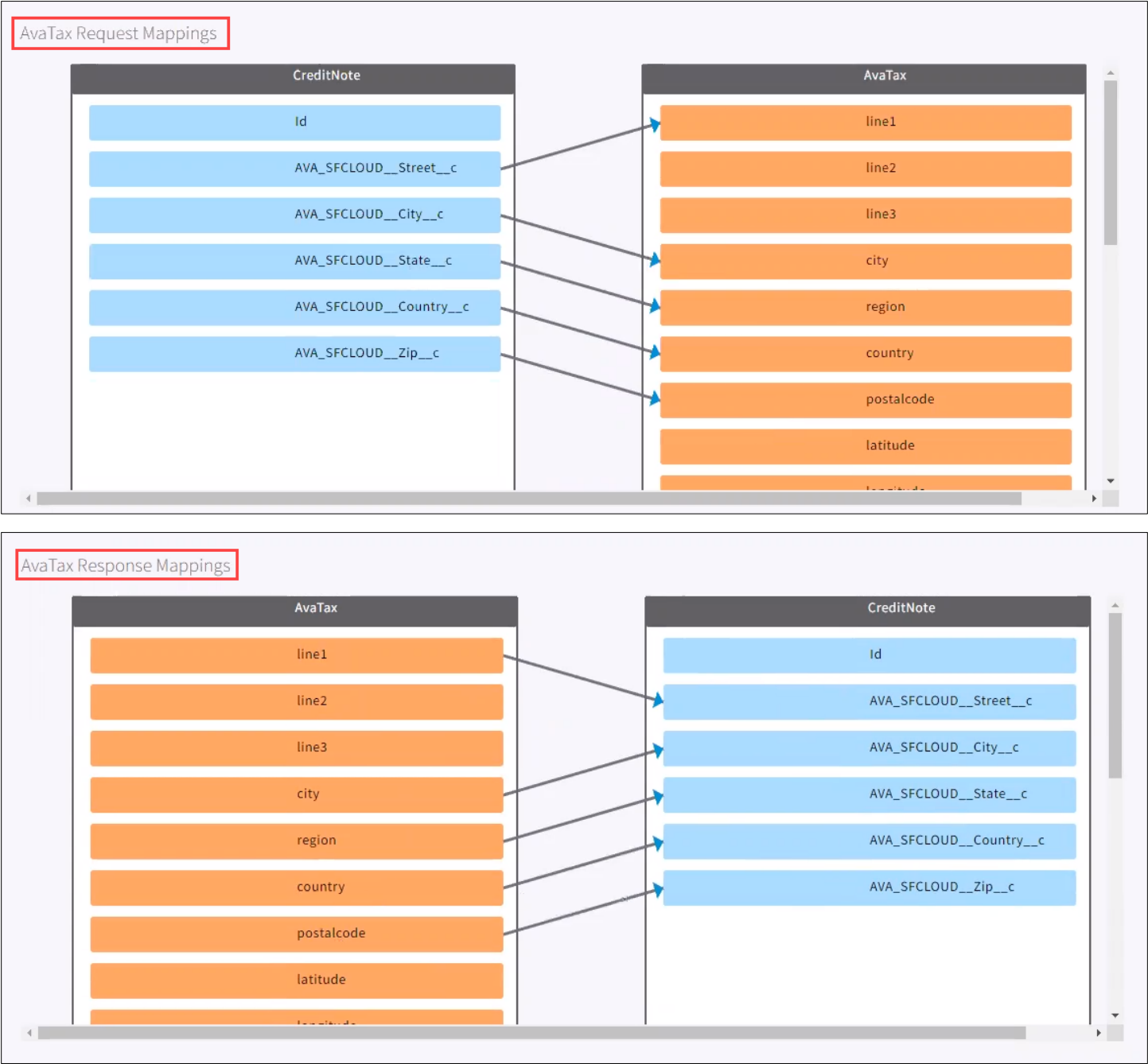
13. Click Modify Soql button.



14. From the Add Fields list, select the address fields for address validation and then click on Execute.



15. Browse down to the “AvaTax Request Mappings” and “AvaTax Response Mappings” to map the Salesforce custom fields (CreditNote) with the AvaTax fields.



16. Click Save to server button to save the changes in the Salesforce AvaTax Custom Object AvaTax_Mapping__c.
17. A record would be created in this custom object which will have the Mapping contents in XML format in the Field Mapping field.

Note: The AVA_SFCLCLOUD__Credit_Note__c node has been added in the addressvalidation node in the SFConfig file.

ADDRESSVALIDATION

INTEGRATION POINTS | + Add integration point

Addressvalidation|AVA_SFCLCLOUD__Credit_Note__c|CreditNote

QUERY FOR CREDITNOTE

Modify your SOQL

Add Fields

- ☒ Shipping City
- ☒ Shipping State
- ☒ Shipping Country
- ☒ Shipping Postal Code
- ☐ Account
- ☐ Account >
- ☐ Status

Add Query Params

Add Where Clause

Id = '{ava_sfcloud_credit_note_

select Id,AVA_SFCLCLOUD__Street__c,AVA_SFCLCLOUD__City__c,AVA_SFCLCLOUD__State__c,AVA_SFCLCLOUD__Country__c,AVA_SFCLCLOUD__Zip__c from AVA_SFCLCLOUD__Credit_Note__c where Id = '{AVA_SFCLCLOUD__Credit_Note__cId}'

Modify Soql | Execute

Save to server | Apply

18. Click Yes, Persist button to confirm the changes.

ADDRESSVALIDATION

INTEGRATION POINTS | + Add integration point

Addressvalidation|AVA_SFCLCLOUD__Credit_Note__c|CreditNote

QUERY FOR CREDITNOTE

Modify your SOQL

Add Fields

- ☒ Shipping City
- ☒ Shipping State
- ☒ Shipping Country
- ☒ Shipping Postal Code
- ☐ Account
- ☐ Account >
- ☐ Status

Add Query Params

Add Where Clause

Id = '{ava_sfcloud_credit_note_

select Id,AVA_SFCLCLOUD__Street__c,AVA_SFCLCLOUD__City__c,AVA_SFCLCLOUD__State__c,AVA_SFCLCLOUD__Country__c,AVA_SFCLCLOUD__Zip__c from AVA_SFCLCLOUD__Credit_Note__c where Id = '{AVA_SFCLCLOUD__Credit_Note__cId}'

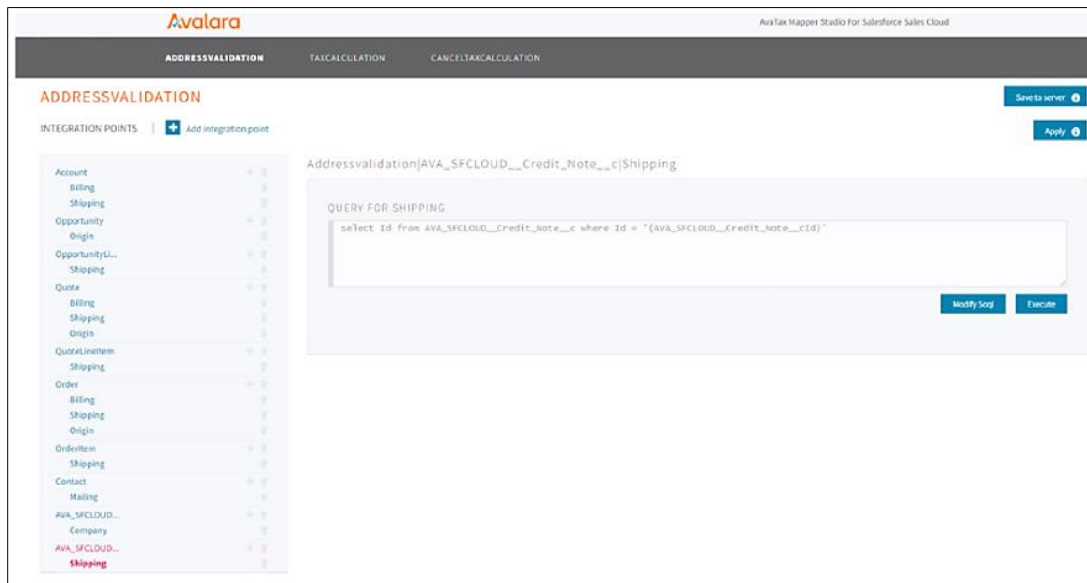
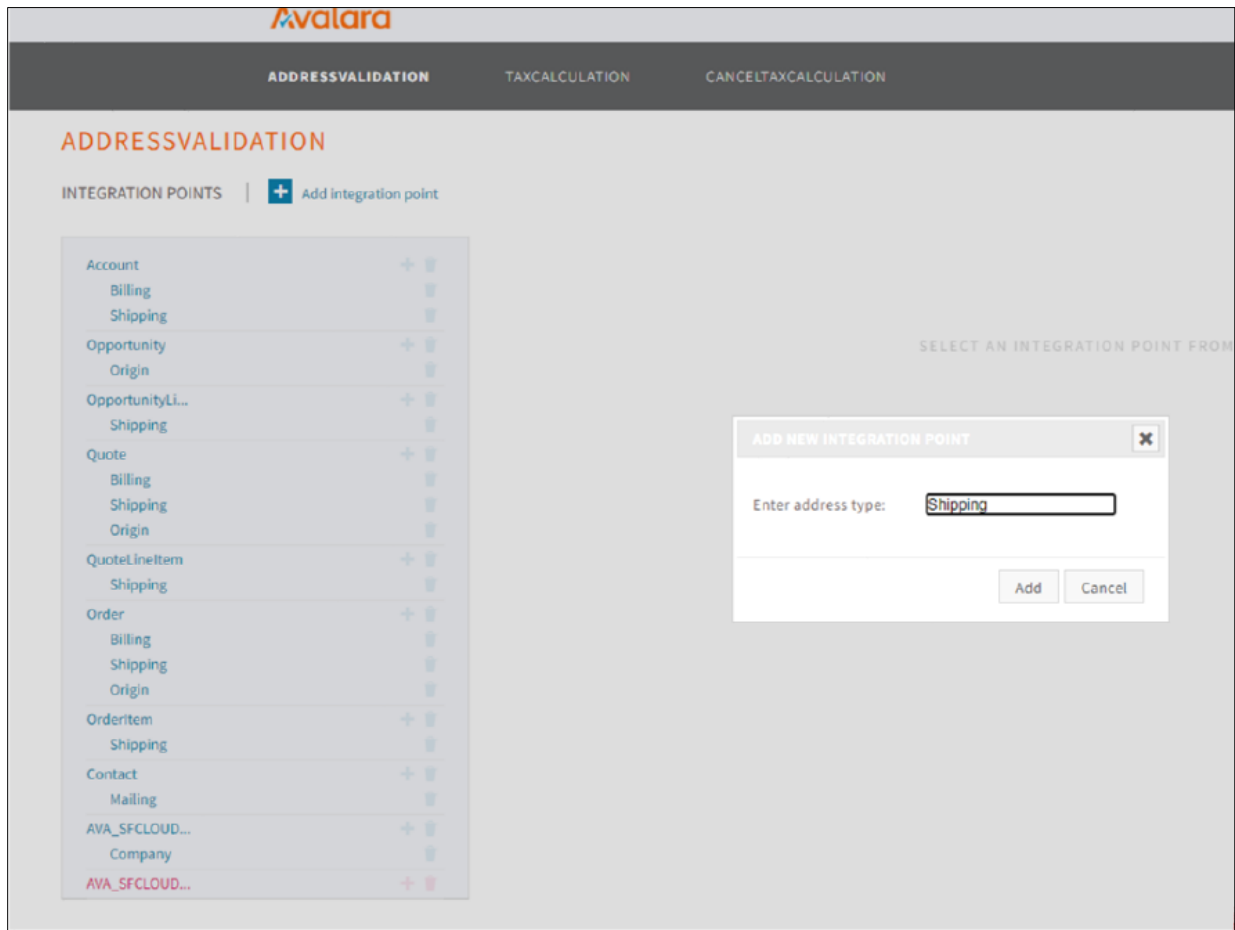
Modify Soql | Execute

Save to server | Apply

Persist changes to the server? This action can not be undone.

Yes, Persist | Cancel

19. Click on the plus icon of newly added node for Credit Note object and enter the address type as "Shipping" and click on the Add button.



AvaTax XML

```

<avatax>
<addressvalidation>
<AVA_SFCLLOUD__Credit_Note__c>
<hooks>
<avaconfig>AVA_SFCLLOUD.ConfigProviderForAddressValidation</avaconfig>
</hooks>
<Shipping>
<query>select
Id,AVA_SFCLLOUD__Street__c,AVA_SFCLLOUD__City__c,AVA_SFCLLOUD__State__c,AVA_SFCLLOUD__Cou
ntry__c,AVA_SFCLLOUD__Zip__c from AVA_SFCLLOUD__Credit_Note__c where Id =
'{AVA_SFCLLOUD__Credit_Note__cId}'</query>
<parameters>
<AVA_SFCLLOUD__Credit_Note__cId>Id</AVA_SFCLLOUD__Credit_Note__cId>
</parameters>
<Mapping>
<line1>AVA_SFCLLOUD__Street__c</line1>
<city>AVA_SFCLLOUD__City__c</city>
<region>AVA_SFCLLOUD__State__c</region>
<country>AVA_SFCLLOUD__Country__c</country>
<postalcode>AVA_SFCLLOUD__Zip__c</postalcode>
</Mapping>
<UpdateMapping>
<line1>AVA_SFCLLOUD__Street__c</line1>
<city>AVA_SFCLLOUD__City__c</city>
<region>AVA_SFCLLOUD__State__c</region>
<country>AVA_SFCLLOUD__Country__c</country>
<postalcode>AVA_SFCLLOUD__Zip__c</postalcode>
</UpdateMapping>
</Shipping>
</AVA_SFCLLOUD__Credit_Note__c>
</addressvalidation>
</avatax>

```

20. Go to Credit Note object and click on the Validate Address button, if the Validate Address button is not available then drag and drop the button from the edit layout section.
21. After clicking the Validate Address button, the Validate Address/es screen appears which consists of original address and the validated address.
22. Click Update Validated Address to update the original address with the validated address on the Credit Note record.

5.1.4. Custom Tax Calculation based on Input Fields

This sample code can be used to implement custom address validation as per the requirements like Web Component etc.

Sample code:

```
public class HGCustomAddressController {

    // Modify the below method to receive attributes and pass those to the model's receptive address
    fields
    public static void validateAddress() {
        AVA_MAPPER.AvaTaxClient client = AVA_MAPPER.AvaTaxUtilities.getClient(new
AvaTaxAddressConfig());
        AVA_MAPPER.AddressValidationInfo model = new AVA_MAPPER.AddressValidationInfo();
        model.line1 = '900 Winslow way E';
        model.city = 'BI';
        model.region = 'WA';
        model.postalCode = '98110';
        model.country = 'United States';
        AVA_MAPPER.AddressResolutionModel result = client.resolveAddressPost(model);
        System.debug('HG: result ' + result);
        if (result.statusCode == 200 || result.statusCode == 201) {
            System.debug('HG: validatedAddress : ' + result.validatedAddresses);
        }
    }

    public class AvaTaxAddressConfig extends AVA_MAPPER.ConfigurationBase
    {
        public AvaTaxAddressConfig()
        {
            AVA_SFCLLOUD__AvaTax__c avaTaxInstance =
AVA_SFCLLOUD__AvaTax__c.getInstance('AvaTax Config');
            appName = 'SF Sales Cloud || 9.0'; // Updated based on the connector
            appVersion = 'a0oUz00000'; // Copy from Custom Labels
            connectorId = 'a0n40000000'; // Copy from Custom Labels
            username = '55555555'; // Enter valid Avatax Account Id
            password = '55555555'; // Enter valid Avatas License Key
            companyCode = avaTaxInstance.AVA_SFCLLOUD__Company_Code__c;
            isAddressValidationEnabled = avaTaxInstance.AVA_SFCLLOUD__Validate_Addresses__c;
            isTaxCalculationEnabled =
avaTaxInstance.AVA_SFCLLOUD__Enable_AvaTax_Tax_Calculation__c;
            environment = avaTaxInstance.AVA_SFCLLOUD__Sandbox__c ?
AVA_MAPPER.AvaTaxEnvironment.Sandbox : AVA_MAPPER.AvaTaxEnvironment.Production;
        }
    }
}
```

The Tax Calculation follows an easy and effective approach using the AvaTax Mapper.

Example: Tax Calculation on the Custom Object (Credit Note and Credit Note Line). The AVA_SFCLCLOUD is used as the Salesforce environment namespace. You need to change it as per your environment namespace or keep it blank if it is not available.

1. In Salesforce create the custom object as Credit Note and Credit Note Line and then browse done to the Custom Fields & Relationships.

Custom Object
Credit Note

Standard Fields (4) | Custom Fields & Relationships (8) | Validation Rules (0) | Page Layouts (1) | Record Types (0) | Apex Sharing Reasons (0) | A

Custom Object Definition Detail

EditDelete

Singular Label	Credit Note
Plural Label	Credit Notes
Object Name	Credit_Note
Namespace Prefix	AVA_SFCLCLOUD
API Name	AVA_SFCLCLOUD__Credit_Note__c

Custom Object
Credit Note Line

Standard Fields (4) | Custom Fields & Relationships (5) | Validation Rules (0) | Page Layouts (1) | Apex Sharing Reasons (0) | Apex Sharin

Custom Object Definition Detail

EditDelete

Singular Label	Credit Note Line
Plural Label	Credit Note Lines
Object Name	Credit_Note_Line
Namespace Prefix	AVA_SFCLCLOUD
API Name	AVA_SFCLCLOUD__Credit_Note_Line__c

2. Click New and create the custom field for Credit Note and Credit Note Line.

Custom Fields & Relationships

NewField Dependencies

Action	Field Label	API Name	Data Type	Indexed	Controlling Field
Edit Del	Account	AVA_SFCLCLOUD__Account__c	Lookup(Account)	✓	
Edit Del	Sales Tax	AVA_SFCLCLOUD__Tax__c	Currency(16, 2)		
Edit Del	Shipping City	AVA_SFCLCLOUD__City__c	Text(50)		
Edit Del	Shipping Country	AVA_SFCLCLOUD__Country__c	Text(50)		
Edit Del	Shipping Postal Code	AVA_SFCLCLOUD__Zip__c	Text(50)		
Edit Del	Shipping State	AVA_SFCLCLOUD__State__c	Text(50)		
Edit Del	Shipping Street	AVA_SFCLCLOUD__Street__c	Text(50)		
Edit Del	Status	AVA_SFCLCLOUD__Status__c	Text(50)		

Custom Fields & Relationships

NewField Dependencies

Action	Field Label	API Name	Data Type	Indexed	Controlling Field
Edit Del	Amount	AVA_SFCLCLOUD__Amount__c	Currency(16, 2)		
Edit Del	Credit Note	AVA_SFCLCLOUD__Credit_Note__c	Lookup(Credit Note)	✓	
Edit Del	Product	AVA_SFCLCLOUD__Product__c	Lookup(Product)	✓	
Edit Del	Quantity	AVA_SFCLCLOUD__Quantity__c	Text(10)		
Edit Del	Tax	AVA_SFCLCLOUD__Tax__c	Currency(16, 2)		

Note: The Data Type for tax related fields must be Currency type.

3. Create an Apex Class to calculate the tax.

Apex Class

```
//Class should be Global so that AvaTax for Mapper package would be able to access the class
global with sharing class HG_OrderTaxCalculator {

    public static Id recordId;

    public HG_OrderTaxCalculator() {
    }

    public HG_OrderTaxCalculator(ApexPages.StandardController controller) {
        recordId = controller.getRecord().id;
    }

    public PageReference calculateTaxButton() {
        calculateTax();
        return redirectPage();
    }

    static public void calculateTax() {
        try {
            Map<String, String> optionalParams = new Map<String, String>();
            AVA_SFCLLOUD.ConfigurationProvider config = new AVA_SFCLLOUD.ConfigurationProvider();
            //AVA_SFPCPQ.ConfigProviderCPQ config = new AVA_SFPCPQ.ConfigProviderCPQ(); //For AvaTax
CPQ+ Package
            // Please modify the below namespace as per the connector
            AVA_MAPPER.TaxCalculator taxCalculatorObj = new AVA_MAPPER.TaxCalculator(
                config.hookExtension(),
                'AVA_SFCLLOUD__AvaTax_Mapping__c',
                'AVA_SFCLLOUD__Field_Mapping__c'
            );
            AVA_MAPPER.TaxCalculationInput taxCalcInput = new AVA_MAPPER.TaxCalculationInput();
            taxCalcInput.recordId = recordId;
            taxCalcInput.controller = 'order'; //If the environment has namespace, append it to the object
API name. Value should be in small letters. This value should match in Config.xml mapping
            taxCalcInput.optionalParams = optionalParams;
            taxCalcInput.isMultiCommit = false;
            // Uncomment below code if you want the transactions to be saved on AvaTax
            // taxCalcInput.docType = AVA_MAPPER.DocumentType.SalesInvoice;
            // Uncomment below code if you want the transactions to be committed on AvaTax
            // taxCalcInput.commitFlag = true;
            System.debug('taxCalcInput: ' + taxCalcInput);
            Ava_Mapper.TransactionModel transResult = taxCalculatorObj.calculateTax(taxCalcInput);
            System.debug('transResult: ' + transResult);
            if (transResult.statusCode == 200 || transResult.statusCode == 201) {
                System.debug('success');
                //Tax calculation is successful. Add business logic as required to display the details.
            } else {
                //There is some error in Tax calculation. Error details will be available in below debug logs.
                System.debug('transResult.statusCode: ' + transResult.statusCode);
                System.debug('AVA_MAPPER.ErrorInfo error: ' + transResult.error);
            }
        } catch (Exception e) {
            system.debug('In exception');
        }
    }

    public PageReference redirectPage() {
```



```

    PageReference pageRef = new PageReference('/') + recordId);
    pageRef.setRedirect(true);
    return pageRef;
}
}

//Test class for HG_OrderTaxCalculator class
@isTest
public class HG_OrderTaxCalculatorTest {
    @testSetup
    static void setupData() {
        AVA_SFCLCLOUD__AvaTax__c settings = new AVA_SFCLCLOUD__AvaTax__c();
        settings.Name = 'AvaTax Config';
        settings.AVA_SFCLCLOUD__Street__c = '900 winslow way e';
        settings.AVA_SFCLCLOUD__Line2__c = 'street 2';
        settings.AVA_SFCLCLOUD__Line3__c = 'street 3';
        settings.AVA_SFCLCLOUD__City__c = 'bi';
        settings.AVA_SFCLCLOUD__State__c = 'WA';
        settings.AVA_SFCLCLOUD__Postal_Code__c = '98110';
        settings.AVA_SFCLCLOUD__Country__c = 'US';
        insert settings;
    }
    @isTest
    static void orderTaxCalculatorTest() {
        PageReference pageRef = Page.Order_Calculate_Tax;
        Test.setCurrentPage(pageRef);
        HG_OrderTaxCalculator creditNoteTaxCalc = new HG_OrderTaxCalculator();
        creditNoteTaxCalc.calculateTaxButton();
    }
}

```

4. Create a Visualforce Page and add the apex class as controller extension.

Visualforce Page

creditNoteTaxCalculator

Page Detail [Edit](#) [Delete](#) [Clone](#) [Where is this used?](#) [Show Dependencies](#) [Preview](#)

Label	Credit Note Tax Calculator	Name	creditNoteTaxCalculator
Namespace Prefix	AVA_SFCLCLOUD	Available for Lightning Experience, Lightning Communities, and the mobile app	☒
Require CSRF protection on GET requests	☐	Description	
Created By	Sales Cloud Production, 12/23/20		
Last Modified By	Sales Cloud Production, 12/23/2020 12:02 AM		

Visualforce Markup **Version Settings**

```

<apex:page standardController="Credit_Note__c" extensions="CreditNoteTaxCalculator" action="{!calculateTaxButton}">
</apex:page>

```

```
<apex:page standardController="Credit_Note__c"
extensions="CreditNoteTaxCalculator" action="{!calculateTaxButton}">
</apex:page>
```

- 5. Create a button on an Object, and then select the Visualforce Page on the "Content Source" field.
- 6. On the Content, click on the drop-down arrow and select the Visualforce Page.

Credit Note Custom Button or Link

Calculate Tax

[« Back to Custom Object: Credit Note](#)

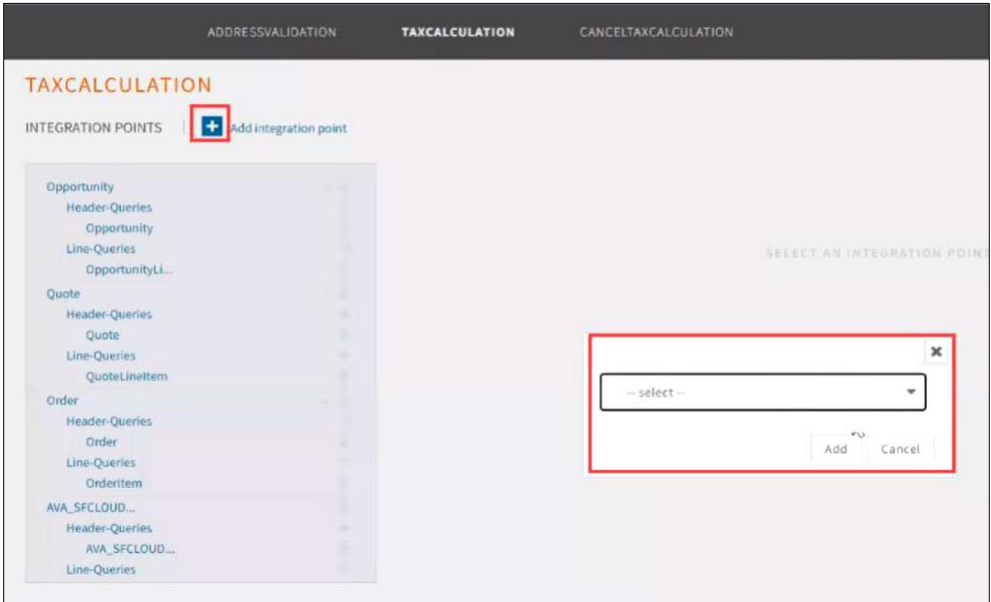
Custom Button or Link Detail

EditWindow Open PropertiesDeleteWhere is this used?

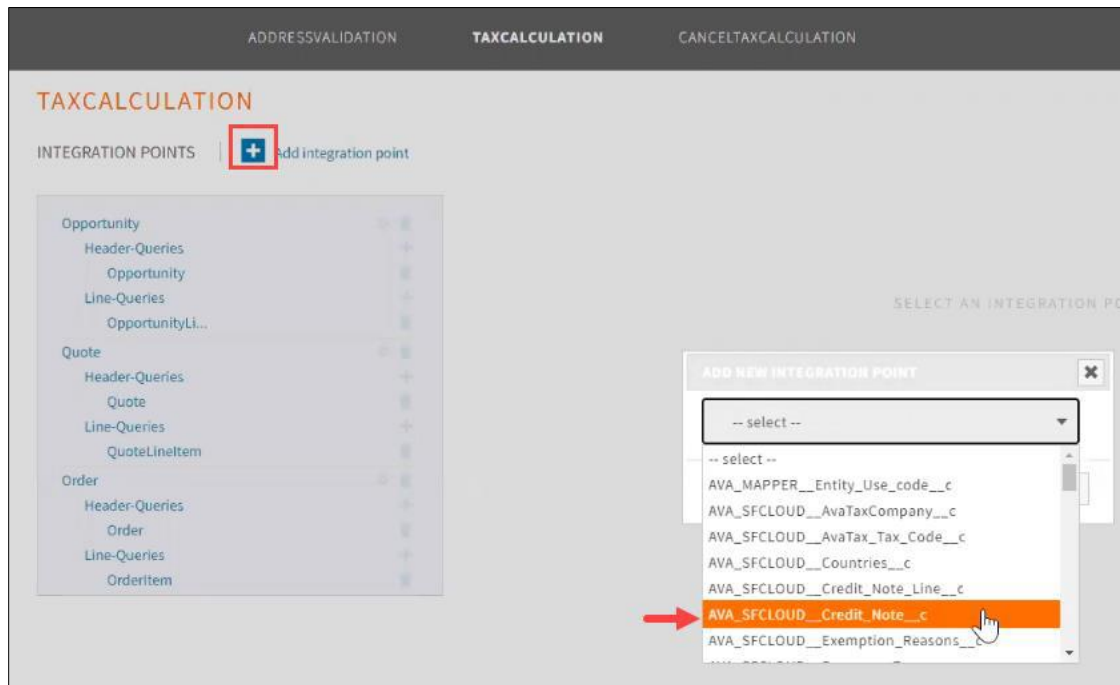
Label	Calculate Tax	Object Name	Credit Note
Name	Calculate_Tax		
Namespace Prefix	AVA_SFCLLOUD	Display Type	Detail Page Button
Behavior	Display in existing window without sidebar		
Content Source	Visualforce Page	Visualforce Page	Credit Note Tax Calculator
Height (in pixels)	600		
Description			
Created By	Sales Cloud Production, 12/23/2020 1:00 AM	Modified By	Sales Cloud Production, 12/23/202

EditWindow Open PropertiesDeleteWhere is this used?

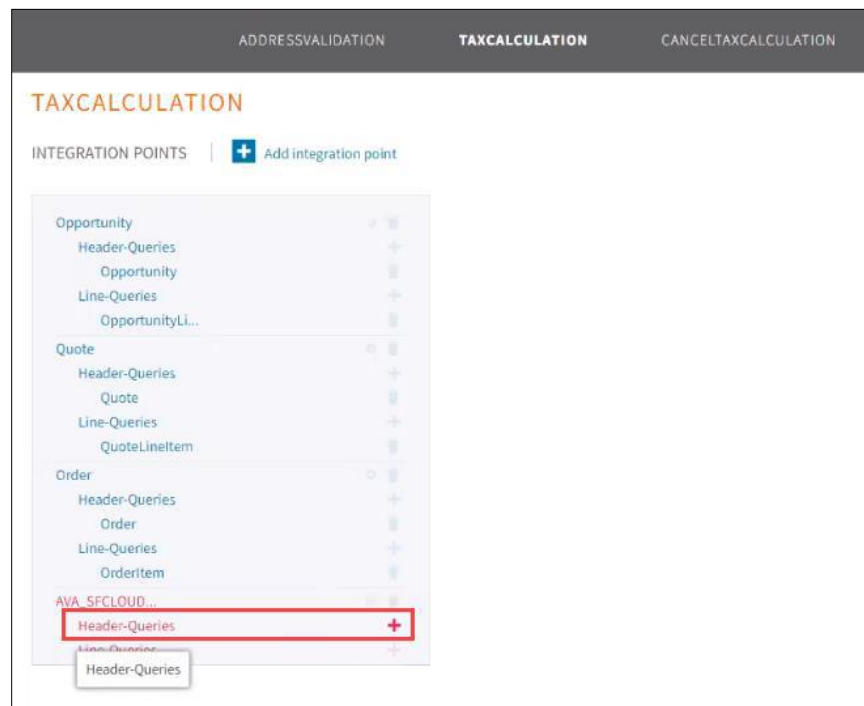
- 7. Follow the step 5 and 6.
- 8. Select TAXCALCULATION and then click on the Plus icon of the Add integration point to open the ADD NEW INTEGRATION POINT dialog box.



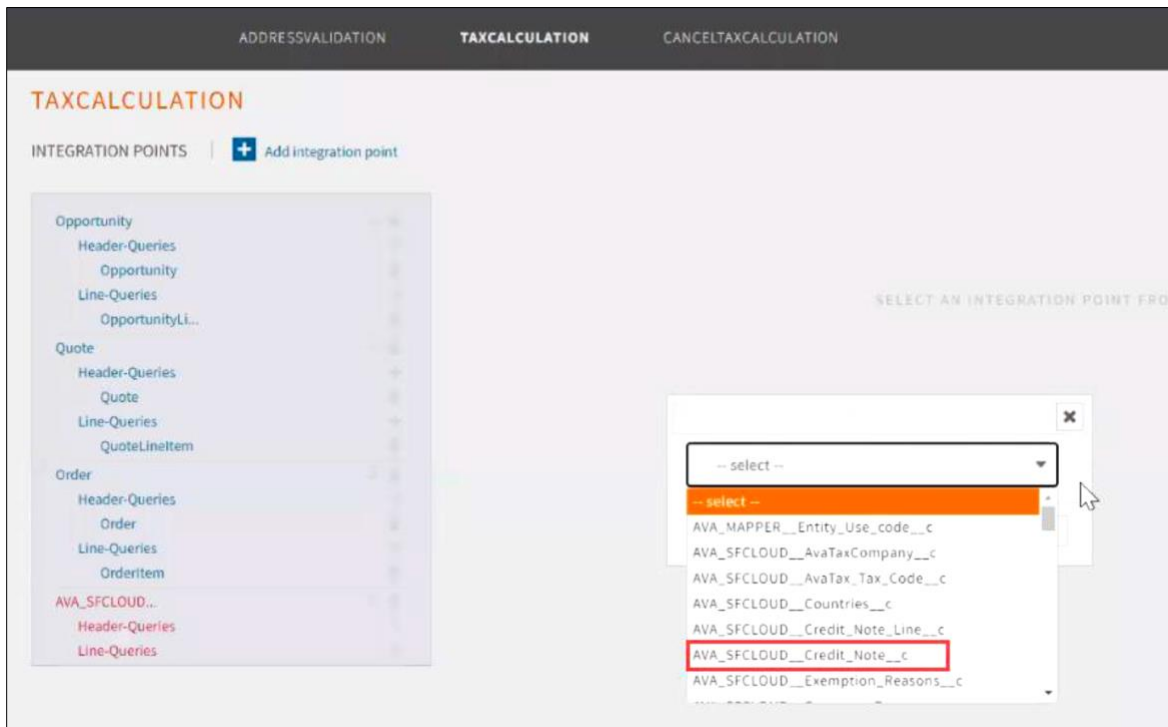
9. Browse the list of the API names and select the recently created custom object API name as "AVA_SFCLCLOUD__Credit_Note__c and then click on the Add button".



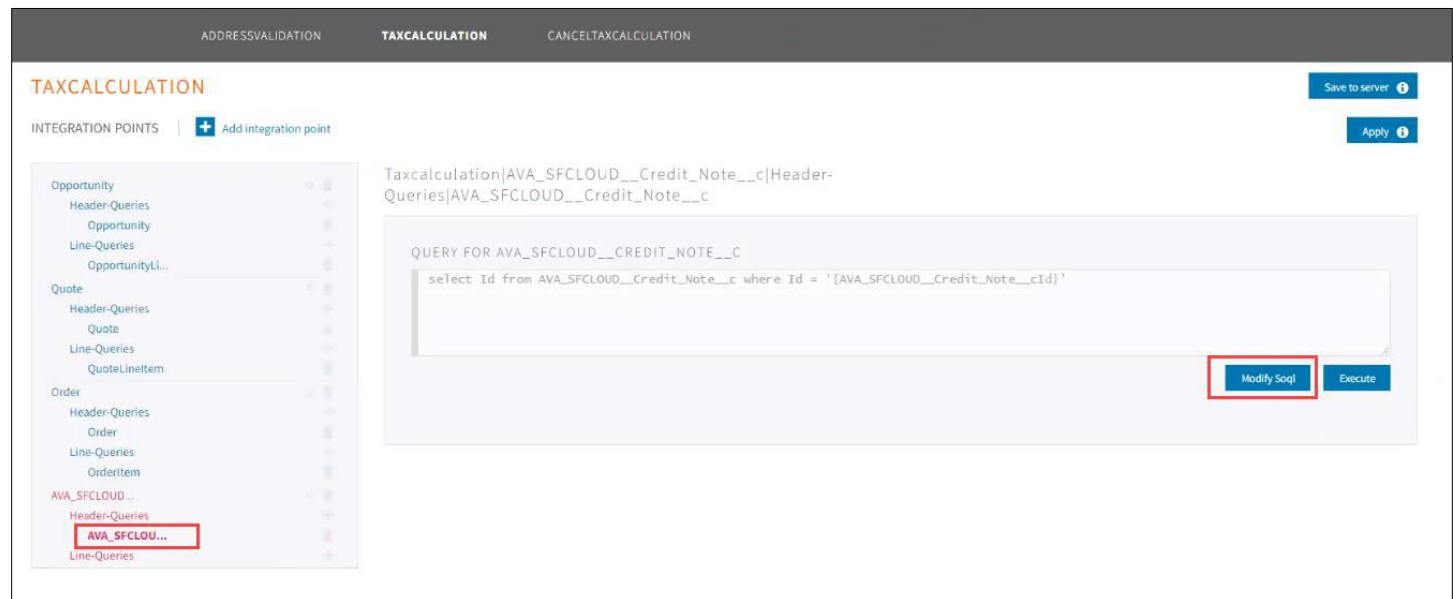
10. Browse to the recently added Custom Object and click on the Header Queries Plus icon to open.



11. Select **AVA_SFCLCLOUD__Credit_Note__c** from the ADD NEW INTEGRATION POINT drop down option.



12. Click on the recently created address type and then click on Modify Soql.



13. From the Add Fields list, select the fields and then click on Execute.

The screenshot shows the TAXCALCULATION interface. On the left, there is a list of integration points including Opportunity, Quote, and Order. The main area is titled 'TAXCALCULATION' and contains a 'QUERY FOR AVA_SFCLCLOUD__CREDIT_NOTE__C' section. A red box highlights the 'Add Fields' list, which includes 'Shipping Country', 'Shipping Postal Code', 'Account', 'Status', and 'Sales Tax'. Below this, the 'Add Where Clause' section shows a query: 'select Id,AVA_SFCLCLOUD__Street__c,AVA_SFCLCLOUD__State__c,AVA_SFCLCLOUD__Country__c,AVA_SFCLCLOUD__City__c,AVA_SFCLCLOUD__Zip__c,AVA_SFCLCLOUD__Tax__c from AVA_SFCLCLOUD__Credit_Note__c where Id = '{AVA_SFCLCLOUD__Credit_Note__c}''. A red arrow points to the 'Execute' button at the bottom right.

14. Browse down to the "AvaTax Request Mappings" and "AvaTax Response Mappings" to map the Salesforce custom fields with the AvaTax fields.

The screenshot shows the 'AvaTax Request Mappings' interface. It displays two columns of fields. The left column, labeled 'AVA_SFCLCLOUD__Credit_Note__c', contains fields: Id, AVA_SFCLCLOUD__Street__c, AVA_SFCLCLOUD__State__c, AVA_SFCLCLOUD__Country__c, AVA_SFCLCLOUD__City__c, AVA_SFCLCLOUD__Zip__c, and AVA_SFCLCLOUD__Tax__c. The right column, labeled 'AvaTax', contains fields: code, type, totalTax, companyCode, date, salespersonCode, customerCode, customerUsageType, and customerUsageCode. A red box highlights the 'AvaTax Request Mappings' title. A blue arrow points from the 'Id' field in the left column to the 'code' field in the right column.



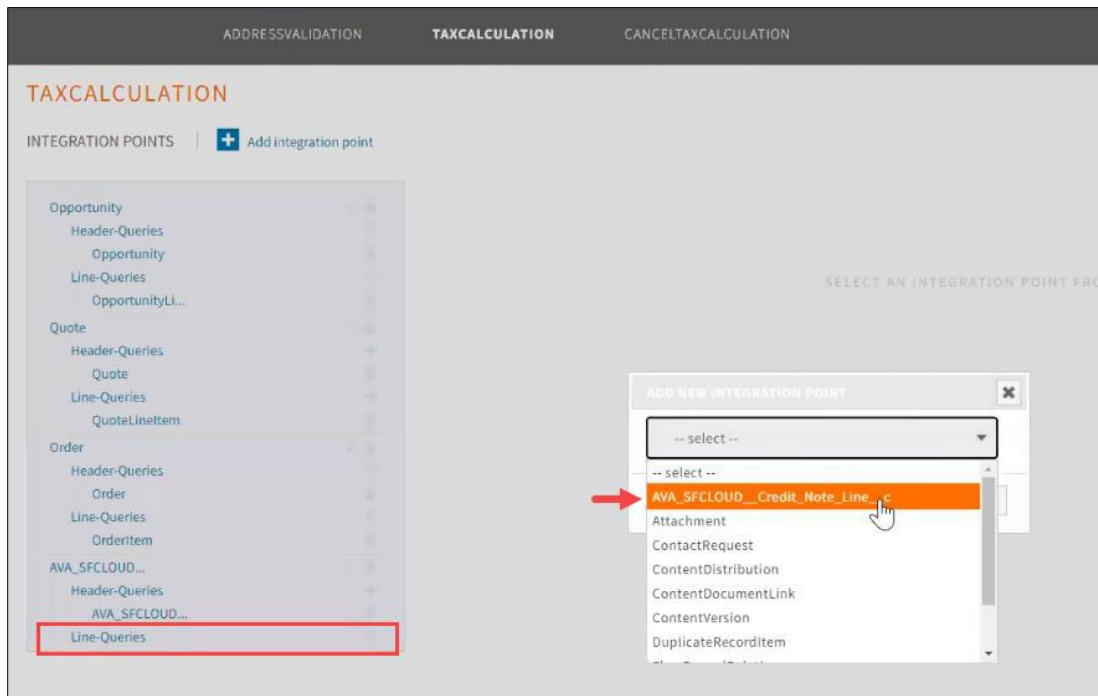
AvaTax Request Mappings

Sr. No.	Salesforce Custom Field	AvaTax Service Field
1.	Id	Code
2.	AVA_SFCLCLOUD__Street__c ShipTo	Line1
3.	AVA_SFCLCLOUD__City__c ShipTo	City
4.	AVA_SFCLCLOUD__State__c ShipTo	Region
5.	AVA_SFCLCLOUD__Country__c ShipTo	Country
6.	AVA_SFCLCLOUD__Zip__c ShipTo	PostalCode
7.	AVA_SFCLCLOUD__Account__r.Name	CustomerCode

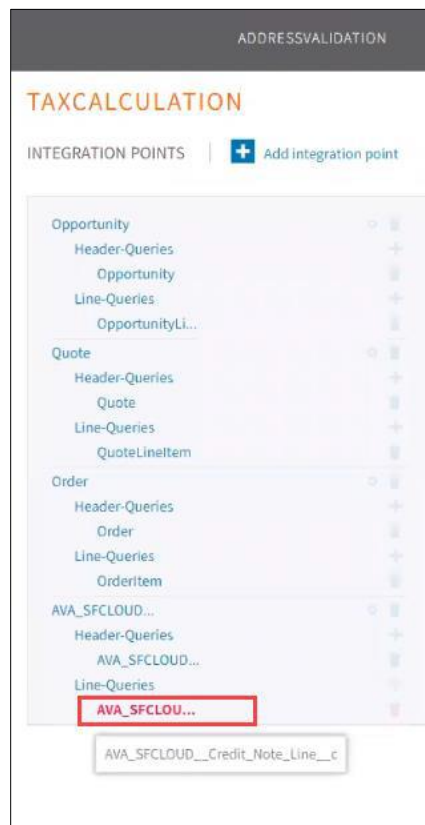
AvaTax Response Mappings

Sr. No.	Salesforce Custom Field	AvaTax Service Field
1.	totalTax	AVA_SFCLCLOUD__Tax__c

15. Click on the Line-Queries Plus icon to select the Credit Note Line object from the ADD NEW INTEGRATION POINT dialog box and then click on the Add button.



16. Click on the recently created integration point to open the QUERY FOR AVA_SFCLCLOUD__CREDIT_NOTE_LINE__C dialog box.



17. Click Modify Soql button.

The screenshot shows the TAXCALCULATION interface. On the left, there is a tree view of integration points. The 'Line-Queries' under 'AVA_SFCLCLOUD...' is highlighted. The main area displays the current SQL query: `select Id from AVA_SFCLCLOUD__Credit_Note_Line__c where AVA_SFCLCLOUD__Credit_Note__c = '{AVA_SFCLCLOUD__Credit_Note__cId}'`. The 'Modify Soql' button is highlighted with a red box.

18. From the Add Fields list, select the fields and then click on Execute.

The screenshot shows the TAXCALCULATION interface with the 'Add Fields' dialog box open. The dialog box has two columns of fields. The left column contains 'Credit Note >', 'Product', and 'Product >'. The right column contains 'Product ID', 'Product Name', 'Product Code', 'Product Description', 'Active', and 'Created Date'. The fields 'Quantity', 'Amount', 'Tax', 'Product Name', and 'Product Code' are selected with checkmarks. The 'Execute' button is highlighted with a red box.

19. Browse down to the “AvaTax Request Mappings” and “AvaTax Response Mappings” to map the Salesforce custom fields (AVA_SFCLCLOUD__Credit_Note_Line__c) with the AvaTax fields.

AvaTax Request Mappings

AVA_SFCLCLOUD__Credit_Note_Line__c

Id

AVA_SFCLCLOUD__Tax__c

AVA_SFCLCLOUD__Amount__c

AVA_SFCLCLOUD__Quantity__c

AVA_SFCLCLOUD__Product__r.Name

AVA_SFCLCLOUD__Product__r.ProductCode

AvaTax

number

quantity

amount

tax

Single Location LocationCode

Single Location Line1

Single Location Line2

Single Location Line3

Product Location #1

AvaTax Response Mappings

AvaTax

number

quantity

amount

tax

Single Location LocationCode

Single Location Line1

Single Location Line2

Single Location Line3

Product Location #1

AVA_SFCLCLOUD__Credit_Note_Line__c

+ ADD NEW

Id

AVA_SFCLCLOUD__Tax__c

AVA_SFCLCLOUD__Amount__c

AVA_SFCLCLOUD__Quantity__c

AVA_SFCLCLOUD__Product__r.Name

AVA_SFCLCLOUD__Product__r.ProductCode

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20. Click Save to server button to save the changes in the Salesforce local environment that is SFCConfig file and new node will be added for the Credit Note under <avatax><taxcalculation> node.

TAXCALCULATION

ADDRESSVALIDATION TAXCALCULATION CANCEL TAXCALCULATION

INTEGRATION POINTS | + Add integration point

Opportunity
Header-Queries
Opportunity
Line-Queries
OpportunityLi...

Quote
Header-Queries
Quote
Line-Queries
QuoteLineItem

Order
Header-Queries
Order
Line-Queries
OrderItem

AVA_SFCLCLOUD...
Header-Queries
AVA_SFCLCLOUD...
Line-Queries
AVA_SFCLCLOUD...

Taxcalculation|AVA_SFCLCLOUD__Credit_Note__c|Line-Queries|AVA_SFCLCLOUD__Credit_Note_Line__c

QUERY FOR AVA_SFCLCLOUD__CREDIT_NOTE__C

Modify your SOQL

Add Fields

Credit Note >
Product >
Product >
Quantity
Amount
Tax

Product ID
Product Name
Product Code
Product Description
Active
Created Date

Add Query Params

Add Where Clause

AVA_SFCLCLOUD__CREDIT_NOTE__C = '{ava_sfcloud_credit_note__c}'

select Id,AVA_SFCLCLOUD__Tax__c,AVA_SFCLCLOUD__Amount__c,AVA_SFCLCLOUD__Quantity__c,AVA_SFCLCLOUD__Product__r.Name,AVA_SFCLCLOUD__Product__r.ProductCode from AVA_SFCLCLOUD__Credit_Note_Line__c where AVA_SFCLCLOUD__Credit_Note__c = '{AVA_SFCLCLOUD__Credit_Note__cId}'

Modify Sql Execute

Save to server Apply

AvaTax Request Mappings

Sr. No.	Salesforce Custom Field	AvaTax Service Field
1.	AVA_SFCLCLOUD__Product__r.Name Note: By default, the ProductCode (Salesforce custom field) is mapped as the itemCode (AvaTax service field). If the ProductCode value is unavailable, then the Product Name value is picked as the itemCode.	itemCode
2.	AVA_SFCLCLOUD__Product__r.ProductCode	itemCode
3.	AVA_SFCLCLOUD__Amount__c	Amount
4.	AVA_SFCLCLOUD__Quantity__c	Quantity
5.	AVA_SFCLCLOUD__Tax__c	Tax

AvaTax Response Mappings

Sr. No.	Salesforce Custom Field	AvaTax Service Field
1.	Tax	AVA_SFCLCLOUD__Tax__c

Sample XML File

```

<avatax>
  <taxcalculation>
    <AVA_SFCLCLOUD__Credit_Note__c>
      <hooks>
        <overriderequestline />
        <beforetaxcalculation />
        <beforeupdate />
        <afterupdate />
      </hooks>
      <Header>
        <Queries>
          <AVA_SFCLCLOUD__Credit_Note__c>
            <query>select
Id,AVA_SFCLCLOUD__Street__c,AVA_SFCLCLOUD__City__c,AVA_SFCLCLOUD__State__c,AVA_SFCLCLOUD__Cou
ntry__c,AVA_SFCLCLOUD__Zip__c,AVA_SFCLCLOUD__Account__r.Name,AVA_SFCLCLOUD__Tax__c from
  AVA_SFCLCLOUD__Credit_Note__c where Id =
'{AVA_SFCLCLOUD__Credit_Note__cId}'</query>
            <parameters>
              <AVA_SFCLCLOUD__Credit_Note__cId>Id</AVA_SFCLCLOUD__Credit_Note__cId>
            </parameters>
            <Mapping>
              <code>Id</code>
              <addresses.shipTo.line1>AVA_SFCLCLOUD__Street__c</addresses.shipTo.line1>
              <addresses.shipTo.city>AVA_SFCLCLOUD__City__c</addresses.shipTo.city>
              <addresses.shipTo.region>AVA_SFCLCLOUD__State__c</addresses.shipTo.region>
              <addresses.shipTo.country>AVA_SFCLCLOUD__Country__c</addresses.shipTo.country>
              <addresses.shipTo.postalCode>AVA_SFCLCLOUD__Zip__c</addresses.shipTo.postalCode>
              <customerCode>AVA_SFCLCLOUD__Account__r.Name</customerCode>
            </Mapping>
            <UpdateMapping>
              <AVA_SFCLCLOUD__Tax__c>totalTax</AVA_SFCLCLOUD__Tax__c>
            </UpdateMapping>
          </AVA_SFCLCLOUD__Credit_Note__c>
        </Queries>
      </Header>
      <Line>
        <Queries>
          <AVA_SFCLCLOUD__Credit_Note_Line__c>
            <query>select
Id,AVA_SFCLCLOUD__Product__r.Name,AVA_SFCLCLOUD__Product__r.ProductCode,AVA_SFCLCLOUD__Amou
nt__c,AVA_SFCLCLOUD__Quantity__c,AVA_SFCLCLOUD__Tax__c from
  AVA_SFCLCLOUD__Credit_Note_Line__c where AVA_SFCLCLOUD__Credit_Note__c =
'{AVA_SFCLCLOUD__Credit_Note__cId}'</query>
            <parameters>
              <AVA_SFCLCLOUD__Credit_Note__cId>AVA_SFCLCLOUD__Credit_Note__c.Id</AVA_SFCLCLOUD__Credit_Note__cId>
            </parameters>
            <Mapping>
              <quantity>AVA_SFCLCLOUD__Quantity__c</quantity>
              <amount>AVA_SFCLCLOUD__Amount__c</amount>
              <tax>AVA_SFCLCLOUD__Tax__c</tax>

```

```

        <itemCode>AVA_SFCLLOUD__Product__r.Name</itemCode>
        <taxCode>AVA_SFCLLOUD__Product__r.ProductCode</taxCode>
    </Mapping>
    <UpdateMapping>
        <AVA_SFCLLOUD__Tax__c>tax</AVA_SFCLLOUD__Tax__c>
    </UpdateMapping>
</AVA_SFCLLOUD__Credit_Note_Line__c>
</Queries>
</Line>
</AVA_SFCLLOUD__Credit_Note__c>
</taxcalculation>
</avatax>

```

21. To verify the mapping, navigate to the Credit Note object and click on the Calculate Tax button, the tax amount must be displayed in the Sales Tax field at the header level and Tax field at the line level.

5.3. Post-Tax Calculation

The Post-Tax Calculation follows an easy and effective approach using the AvaTax Mapper.

1. Create a visualforce page and add apex class as controller extension.

Below is an example to implement the Cancel Tax functionality on the Order object:

Visualforce Page

```

<apex:page standardController="Order" extensions="PostTaxCalculatorClassName"
action="{!postTaxCalculatorMethodName}">
</apex:page>

```

2. The user needs to create an Apex Class to fetch the AvaTax configuration values. The usage is shown in the following Apex Class:

Apex Class

```

Global class PostTaxCalculatorClassName {
    public static Id ordId = null;
    public PostTaxCalculatorClassName(){ }
    public PostTaxCalculatorClassName(ApexPages.StandardController controller) {
        ordId = controller.getRecord().id;
    }
    public PageReference postTaxCalculatorMethodName() {
        postTax();
        return redirectPage();
    }
    Public void postTax() {
        AVA_SFCLLOUD.ConfigurationProvider config = new AVA_SFCLLOUD.ConfigurationProvider();
        Map<String,String> ordParam = new Map<String,String>();
    }
}

```

```

    AVA_MAPPER.PostTaxCalculator postTaxObj = new
        AVA_MAPPER.PostTaxCalculator(config.hookExtension(),'AVA_
SFCLCLOUD__AvaTax__Mapping__c','AVA_SFCLCLOUD__Field_Mapping__c');
    AVA_MAPPER.QueryInput queryInput = new AVA_MAPPER.QueryInput();
    queryInput.recordId = ordId;
    queryInput.controller = ordId.getSObjectType().getDescribe().getName().toLowerCase();
    queryInput.optionalParams = ordParam;
    postTaxObj.executeQuery(queryInput);
    AVA_MAPPER.PostTaxCalculationInput postTaxInput = new
        AVA_MAPPER.PostTaxCalculationInput();
    postTaxInput.companyCode =
AVA_SFCLCLOUD__AvaTax__c.getInstance('AvaTaxConfig').AVA_SFCLCLOUD__Company_Code__c;
    postTaxInput.transactionCode = String.valueOf(ordId);
    postTaxInput.commitFlag = False;
    postTaxInput.docType = AVA_MAPPER.DocumentType.SalesInvoice;
    AVA_MAPPER.TransactionModel postTaxResult = postTaxObj.postTax(postTaxInput);
    if (postTaxResult.statusCode == 200 || postTaxResult.statusCode == 201) {
        System.debug('success');
    } else {
        System.debug('error');
        System.debug('transResult.statusCode: ' + postTaxResult.statusCode);
        System.debug('AVA_MAPPER.ErrorInfo error: ' + postTaxResult.error);
    }
}
}
public PageReference redirectPage() {
    PageReference pageRef = new PageReference('/' + ordId);
    pageRef.setRedirect(true);
    return pageRef;
}
}

```

3. Add the Visualforce Page (by following above Visualforce Page example).

Visualforce Page

Page Edit Save Quick Save Cancel Where is this used? Component Reference Preview

Page Information

Label

Name

Description

Available for Lightning Experience, Experience Builder sites, and the mobile app ☒

Require CSRF protection on GET requests ☐

Visualforce Markup **Version Settings**

```
1 <apex:page standardController="Order" extensions="PostTaxCalculatorClassName" action="{!postTaxCalculatorMethodName}">
2 </apex:page>
```

4. Create a button on an Object, and then select the Visualforce Page on the "Content Source" field.

5. On the Content, click on the drop-down arrow and select the Visualforce Page.

The screenshot shows the 'Custom Button or Link Edit' dialog box. The 'Content Source' dropdown is highlighted with a red box, and a red arrow points to its dropdown arrow. The 'Content Source' is currently set to 'Visualforce Page'. Other fields include 'Label' (Post-Tax Calculation), 'Name' (Post-Tax Calculation), 'Namespace Prefix' (SalesCloud_Sand), 'Description' (empty), 'Display Type' (radio buttons for 'Detail Page Link View example', 'Detail Page Button View example' (selected), and 'List Button View example'), 'Behavior' (Display in existing window without sidebar), and 'View Behavior Options'.

AvaTax XML

```
<posttaxcalculation>
  <Order>
    <hooks>
    </hooks>
    <Queries>
      <Order>
        <query>select Id, AccountId, Description from Order where id = '{OrderId}'</query>
        <parameters>
          <OrderId>Id</OrderId>
        </parameters>
        <Mapping>
          <transactionCode>Id </transactionCode>
        </Mapping>
      </Order>
    </Queries>
  </Order>
</posttaxcalculation>
```

5.4. Cancel Tax Calculation

The Cancel Tax Calculation follows an easy and effective approach using the AvaTax Mapper. Below is an example to implement the Cancel Tax functionality on the Order object.

1. Create a Visualforce page and add Apex Class as a controller extension.

Visualforce Page

```
<apex:page standardController="Order" extensions="CancelTaxCalculatorClassName"
action="{!cancelTaxCalculatorMethodName}">
</apex:page>
```

2. The user needs to create an Apex Class to fetch the AvaTax configuration values. The usage is shown in the following Apex Class:

Apex Class

```
Global class CancelTaxCalculatorClassName{
    public static Id ordId = null;
    public CancelTaxCalculatorClassName(ApexPages.StandardController controller) {
        ordId = controller.getRecord().id;
    }
    public PageReference cancelTaxCalculatorMethodName() {
        cancelTax();
        return redirectPage();
    }
    Public void cancelTax() {
        AVA_SFCLLOUD.ConfigurationProvider config = new AVA_SFCLLOUD.ConfigurationProvider();
        Map<String,String> ordParam = new Map<String,String>();
        AVA_MAPPER.CancelTaxCalculator ordObj = new
            AVA_MAPPER.CancelTaxCalculator(config.hookExtension(),'AVA_
SFCLLOUD__AvaTax__Mapping__c','AVA_SFCLLOUD__Field_Mapping__c');
        AVA_MAPPER.CancelTaxCalculationInput cancelTaxInput = new
            AVA_MAPPER.CancelTaxCalculationInput();
        cancelTaxInput.recordId = ordId;
        cancelTaxInput.controller = ordId.getSObjectType().getDescribe().getName().toLowerCase();
        cancelTaxInput.optionalParams = ordParam;
        cancelTaxInput.companyCode =
AVA_SFCLLOUD__AvaTax__c.getInstance('AvaTaxConfig').AVA_SFCLLOUD__Company_Code__c;
        cancelTaxInput.transactionCode = String.valueOf(ordId);
        cancelTaxInput.code = AVA_MAPPER.VoidReasonCode.DocVoided;
        cancelTaxInput.docType = AVA_MAPPER.DocumentType.SalesInvoice;
```



```

AVA_MAPPER.TransactionModel cancelTaxResult = ordObj.cancelTax(cancelTaxInput);
if (cancelTaxResult.statusCode == 200 || cancelTaxResult.statusCode == 201) {
    System.debug('success');
} else {
    System.debug('error');
    System.debug('transResult.statusCode: ' + cancelTaxResult.statusCode);
    System.debug('AVA_MAPPER.ErrorInfo error: ' + cancelTaxResult.error);
}
}
public PageReference redirectPage() {
    PageReference pageRef = new PageReference('/') + ordId);
    pageRef.setRedirect(true);
    return pageRef;
}
}

```

3. Add the Visualforce Page (by following the above Visualforce Page example).

Page Information ! = Required Information

Label

Name

Description

Available for Lightning Experience, Experience Builder sites, and the mobile app ☒

Require CSRF protection on GET requests ☐

Visualforce Markup **Version Settings**

```

1 <apex:page standardController="Order" extensions="CancelTaxCalculatorClassName" action="{!cancelTaxCalculatorMethodName}">
2 </apex:page>

```

4. Create a button on an Object, and then select the Visualforce Page on the "Content Source" field.
5. On the Content, click on the drop-down arrow and select the Visualforce Page.

Custom Button or Link Edit

Save Quick Save Preview Cancel

Label Cancel Tax Calculation

Name Cancel Tax Calculation ⓘ

Namespace Prefix SalesCloud_Sand

Description

Display Type

☐ Detail Page Link View example

☒ Detail Page Button View example

☐ List Button View example

Behavior Display in existing window without sidebar View Behavior Options

Content Source Visualforce Page

Content

Save Quick Save Preview Cancel

AvaTax XML

```
<canceltaxcalculation>
  <Order>
    <Queries>
      <Order>
        <query>select Id From Order where id = '{OrderId}' and isDeleted = true all rows</query>
        <parameters>
          <OrderId>Id</OrderId>
        </parameters>
        <Mapping>
          <transactionCode>Id</transactionCode>
        </Mapping>
      </Order>
    </Queries>
  </Order>
</canceltaxcalculation>
```

5.5. Hooks Manager for Tax Calculation

The Hooks Manager considers the extensions provided in the AvaTax XML configuration file. The following AvaTax XML informs about the hooks that are available for the Tax Calculation:

Four of hooks have been provided:

- **OverrideRequestLine:** Use this hook to override data at the line-item level during request.
- **BeforeTaxCalculation:** Use this hook to override data at header level during request
- **BeforeUpdate:** Use this hook to modify response before updating on records.
- **AfterUpdate:** Use this hook to modify response after updating on records.

AvaTax XML

```
<taxcalculation>
  <Order>
```

```

    <hooks>
      <overriderequestline></overriderequestline>
      <beforetaxcalculation></beforetaxcalculation>
      <beforeupdate></beforeupdate>
      <afterupdate></afterupdate>
    </hooks>
  </Order>
</taxcalculation>

```

Example for Hooks Definition

BeforeTaxCalculationExtender:

```

global class BeforeTaxCalculationExtender implements AVA_MAPPER.IBeforeTaxCalculation {
    // model consists of the request mappings
    // headerResults consists of the queried data for each object based on Config XML file mapped by
    // object api name
    // Object API name should be all small letters (ex: opportunity, quote, order, blng__invoice__c)
    // The source field should be added to the query on the xml file
    global void hookExtention(AVA_MAPPER.CreateTransactionModel model, Map<String,List<SObject>>
    headerResults) {
        // 1. Grab data from headerResult
        // 2. Update mapping in the model
        //model.customerCode = yourCustomFieldData;

    }
}

```

To provide the extension in XML file you need to update the XML file as follows:

```

<taxcalculation>
  <Order>
    <hooks>
      <overriderequestline></overriderequestline>
      <beforetaxcalculation>BeforeTaxCalculationExtender</beforetaxcalculation>
      <beforeupdate></beforeupdate>
      <afterupdate></afterupdate>
    </hooks>
  </Order>
</taxcalculation>

```

OverrideRequestLineExtender:

```

global class OverrideRequestLineExtender implements AVA_MAPPER.IOverrideRequestLine {

```

```
// lineItemModel consists of the request mappings at line level
// lineQueryResult consists of the queried data for each object's line item based on Config XML file
mapped by object api name
// Id consists of the record id of the line item
// Object API name should be all small letters (ex: orderitem, blng__invoiceline__c)
// The source field should be added to the query on the xml file
global void hookExtention(AVA_MAPPER.LineItemModel lineItemModel, Map<String,List<SObject>>
lineQueryResult, Id id) {
    // 1. Grab data from lineQueryResult
    // 2. Update mapping in the model
    //lineItemModel.amount = yourCustomFieldData;
}
}
```

To provide the extension in XML file you need to update the XML file as follows:

```
<taxcalculation>
  <Order>
    <hooks>
      <overriderequestline>OverrideRequestLineExtender</overriderequestline>
      <beforetaxcalculation></beforetaxcalculation>
      <beforeupdate></beforeupdate>
      <afterupdate></afterupdate>
    </hooks>
  </Order>
</taxcalculation>
```

6.0 AvaTaxClient

Description	Initializes the Rest REST based client with the supplied parameters.
Returns	A new object of AvaTaxClient class.
Parameters	
AppName	It is a name of Salesforce package which calls the AvaTax Mapper package. Name of calling application. Mostly it is ERP/Salesforce Package Name which is willing to call AvaTax Services.
AppVersion	It is a version of Salesforce package which calls the AvaTax Mapper package A version of the calling application.
MachineName	It is a name of the machine Name of the machine from which this call is being made. In case of Salesforce, it can be the hostname.
Environment	Select whether to make calls to the AvaTax Sandbox or Production environment. Select whether calls to be made to AvaTax Sandbox or Production environment.

6.1. Methods

WithSecurity

Description	- Initializes the AvaTaxClient class with the security information. - If you want to send username and password for authentication, then use this method.
Returns	An instance of the AvaTaxClient class.
Parameters	
Username	Username provided by Avalara.
Password	Password provided by Avalara.

WithSecurity

Description	- Initializes the AvaTaxClient class with the security information. - If you want to send Account ID and License Key for authentication, then use this method.
Returns	An instance of the AvaTaxClient class.
Parameters	
AccountId	10-digit Account ID provided by Avalara.
LicenseKey	16-character License Key provided by Avalara.

resolveAddressPost

Description	It retrieves geolocation information for a specified address.
Returns	- If address is validated, then the resolveAddressPost API provides the latitude and longitude of the validated address location. - The value 'resolutionQuality' can be used to identify how closely this address can be located. - If the address cannot be located, then use the 'messages' structure to learn more about the problems with the address.
Parameters	
Model	Instance of AddressValidationInfo.

CreateTransaction

Description	- It records a new transaction in AvaTax. - A transaction represents a unique potentially taxable action that the company has recorded, the transactions include actions like sales, purchases, inventory transfer, and returns (also called refunds).
Returns	TransactionModel as transaction object.
Parameters	
Model	Instance of CreateTransactionModel.
Include	Specifies objects to include in the response after the transaction is created. - Lines - Details (implies lines) - Summary (implies details) - Addresses - SummaryOnly (omit lines and details - reduces API response size) - LinesOnly (omit details - reduces API response size) - ForceTimeout - Simulates a timeout. This adds a 30 second delay and error to your API call. This can be used to test your code to ensure that it can respond correctly in case of a dropped connection.

CommitTransaction

Description	- Marks a transaction by changing its status to Committed. - The transactions which are committed are available for the Avalara Managed Returns to report to a tax authority. - A transaction represents a taxable action that your company has recorded, and the transactions include the actions like sales, purchases, inventory transfer, and returns (also called refunds). - If you have more than one document with the same code, specify the documentType parameter to choose between them. - Any changes made to a committed transaction will generate a transaction history.
Returns	TransactionModel as transaction object.
Parameters	
CompanyCode	The company code of the company that records the transaction.
TransactionCode	The transaction code to commit.
Model	Instance of CommitTransactionModel.
DocumentType	The document type of the transaction to void. If not provided, then the default is SalesInvoice.

VoidTransaction

Description	- A transaction represents a taxable action that the company has recorded, and the transactions include the actions like sales, purchases, inventory transfer, and returns (also called refunds). - When you void a transaction, the transaction's status is recorded as DocVoided. - If you have more than one document with the same code, specify the documentType parameter to choose between them. - The transactions that have been previously reported to a tax authority by Avalara Managed Returns are no longer available to be voided.
Returns	TransactionModel as transaction object.
Parameters	
CompanyCode	The company code of the company that recorded this transaction.
TransactionCode	The transaction code to commit.
Model	Instance of VoidTransactionModel.
DocumentType	The document type of the transaction to void. If not provided, then the default is SalesInvoice.

getCertificateSetup

Description	Checks whether this company is configured to use exemption certificates in AvaTax.
Returns	ProvisionStatusModel as object.
Parameters	
CompanyId	The company ID to check.

CreateCertExpressInvitation

Description	Creates an invitation for a customer to self-report certificates using the CertExpress website.
Returns	FetchResultCertInvitationStatusModel as object.
Parameters	
CompanyId	The unique ID number of the company that will record certificates
customerCode	The number of the customer where the request is sent to
model	List of CreateCertExpressInvitationModel.

getCertExpressInvitationById

Description	Retrieve an existing CertExpress invitation sent to a customer.
Returns	CertExpressInvitationModel as object.
Parameters	
CompanyId	The unique ID number of the company that issued this invitation
customerCode	The number of the customer where the request is sent to
Id	The unique ID number of this CertExpress invitation

listCertificatesForCustomer

Description	List all customers linked to this certificate.
Returns	FetchResultCertificateModel as object.
Parameters	
CompanyId	The unique ID number of the company that recorded this customer.
customerCode	The unique code representing this customer.
Include	- OPTIONAL: A comma separated list of special fetch options. You can specify one or more of the following: - customers - Retrieves the list of customers linked to the certificate. - po_numbers - Retrieves all PO numbers tied to the certificate.

	attributes - Retrieves all attributes applied to the certificate.
filter	A filter statement to identify specific records to retrieve
top	If nonzero, return no more than this number of results. Used with skip to provide pagination for large datasets. Unless otherwise specified, the maximum number of records that can be returned from an API call is 1,000 records.
skip	If nonzero, skip this number of results before returning data. Used with \$top to provide pagination for large datasets.
orderBy	A comma separated list of sort statements in the format (fieldname) [ASC DESC], for example id ASC.

6.2. Models

AddressResolutionModel

Description	- This API provides address resolution and does not calculate the tax. - It returns success if it has been able to identify and provide the address information. - If insufficient information has been provided, then this API will respond with an error code.
--------------------	--

Properties

The following are the address resolution model properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Address	AddressInfo	It is an original address.
2.	Coordinates	CoordinateInfo	- It is a geospatial coordinate of the address. - For Example, latitude: 33.684884, longitude: -117.851321
3.	Messages	AvaTaxMessage	It is a list of address information and warning messages.
4.	resolutionQuality	resolutionQuality	It is a list of address information and warning messages.
5.	taxAuthorities	TaxAuthorityInfo	The resolution quality of the geospatial coordinates = ['NotCoded', 'External', 'CountryCentroid', 'RegionCentroid', 'PartialCentroid', 'PostalCentroidGood', 'PostalCentroidBetter', 'PostalCentroidBest', 'Intersection', 'Interpolated', 'Rooftop', 'Constant'].
6.	validatedAddresses	ValidatedAddressInfo	It is a list of tax authorities associated with the address.

ValidatedAddressInfo**Description**

- It represents the validated address.

Properties

The following are the validate address info model properties and their information:

Sr. No.	Properties	Data Type	Description
1.	addressType	String	• Firm or company address. • General Delivery address. • High-rise or business complex. • PO Box address. • Rural route address.
2.	city	String	• It is a city component of the address. For Example, Irvine.
3.	country	String	• It is a name or ISO 3166 code identifying the country. For Example, US.
4.	latitude	Double	• It is a geospatial latitude measurement, in Decimal Degrees, floating point format.
5.	line1	String	• It is a first line of the street address. For Example, 200 Main Street.
6.	line2	String	• It is a second line of the street address.
7.	line3	String	• It is a third line of the street address.
8.	longitude	Double	• It is a geospatial longitude measurement, in Decimal Degrees, floating point format.
9.	postalCode	String	• It is a postal code / zip code component of the address. For Example, 92614.
10.	region	String	• It is name or ISO 3166 code identifying the region within the country. For Example, CA.

AddressInfo**Description**

- It represents address for validation or tax calculation.

Properties

The following are the address info properties and their information:

Sr. No.	Properties	Data Type	Description
1.	city	String	• It is a city component of the address. For Example, Irvine.
2.	country	String	• It is a name or ISO 3166 code identifying the country. For Example, US.
3.	latitude	Double	• It is a geospatial latitude measurement, in Decimal Degrees, floating point format.
4.	line1	String	• It is a first line of the street address. For Example, 200 Main Street.
5.	line2	String	• It is a second line of the street address.

6.	line3	String	It is a third line of the street address.
7.	longitude	Double	It is a geospatial longitude measurement, in Decimal Degrees, floating point format.
8.	postalCode	String	It is a postal code / zip code component of the address. For Example, 92614.
9.	region	String	It is name or ISO 3166 code identifying the region within the country. For Example, CA.

CoordinateInfo

Description

- It represents the geospatial latitude and longitude information.

Properties

The following are the CoordinateInfo properties and their information:

Sr. No.	Properties	Data Type	Description
1.	latitude	Double	It is a geospatial latitude measurement, in Decimal Degrees floating point format. For Example, Latitude: 33.684884.
2.	longitude	Double	It is a geospatial longitude measurement, in Decimal Degrees floating point format. For Example, Longitude: -117.851321.Longitude.

AvaTaxMessage

Description

- An Informational or warning messages returned by AvaTax with a transaction.

Properties

The following are the AvaTaxMessage properties and their information:

Sr. No.	Properties	Data Type	Description
1.	details	String	It is a detailed information that explains what was provided in the summary.
2.	refersTo	String	It is an information about the object to which the message refers in your request. For Example, Addresses [0] - TaxDate: 2017-09-21 00:00:00Z.
3.	severity	String	It is a category that indicates how severely the message affects the results. For Example, Success.
4.	source	String	It is a name of the code or service that generated the message. For Example, Avalara.AvaTax.Services.Tax.Steps.
5.	summary	String	It is an overview of the message.

Description

- Information about the tax authority that is relevant to the address.

Properties

The following are the TaxAuthorityInfo properties and their information:

Sr. No.	Properties	Data Type	Description
1.	avalaraId	String	It is a unique ID number assigned by Avalara to this tax authority. For Example, AGAM.
2.	jurisdictionName	String	The friendly jurisdiction name for the tax authority. For Example, California.
3.	jurisdictionType	jurisdictionType	It is a type of jurisdiction referenced by the tax authority. For Example, State.
4.	signatureCode	String	It is an Avalara-assigned signature code for the tax authority. For Example, AGAM.

TransactionModel**Description**

- This object represents a single transaction; For Example, a sales invoice or purchase order.

Properties

The following are the TransactionModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Addresses	TransactionAddressModel	It is a list of line items addresses in the transaction.
2.	adjustmentDescription	String	If the transaction has been reported to a tax authority, then the transaction is considered as locked and may not be adjusted after reporting.
3.	adjustmentReason	AdjustmentReason	If the transaction was adjusted, then it indicates the unique ID number with the reason of transaction adjustment. For Example, NotAdjusted.
4.	batchCode	String	If the transaction was created as part of a batch, then the code indicates about that batch.
5.	businessIdentificationNo	String	It is a VAT business identification number used for the transaction.
6.	code	String	It is a unique customer-provided code is identifying the transaction.
7.	companyId	Long	It is a unique ID number of the company that records the transaction.
8.	country	String	It is a two-character ISO 3166 code of the country for the transaction. For Example, US.

Sr. No.	Properties	Data Type	Description
9.	currencyCode	String	It is a three-character ISO 4217 currency code that was used for payment for this transaction. For Example, USD.
10.	customerUsageType (DEPRECATED)	String	- It is a client application as customer or usage type. - Use the EntityUseCode, instead of customerUsageType.
11.	customerVendorCode	String	It is a Customer/Vendor Code.
12.	description	String	Field permits the unicode values and allows to add the description of the transaction.
13.	destinationAddressId	Long	The unique number of the destination address for this transaction.
14.	email	String	The email address associated with this transaction.
15.	exchangeRate	Decimal	It is used when the transaction includes foreign currency
16.	exchangeRateEffectiveDate	DateTime	It is a date to which the exchange rate was calculated when the transaction includes foreign currency exchange.
17.	exemptNo	String	This field will be updated as "Exempt" if the transactions are exempted.
18.	history	TransactionModel	It stores the list of all the previous version of the documents when the transaction was adjusted.
19.	id	Long	It is a unique ID number of the transaction.
20.	isSellerImporterOfRecord	Boolean	- The "importer of record" is liable to pay customs and import duties for products shipped internationally. - If you specify that the seller is the importer of record, then estimates of customs and import duties will be added as tax details to the transaction. - If you specify that the buyer is the importer of record, then customs and import duties will not be added to the tax details to the transaction. - If the transaction is not an international transaction, then this field is left blank.
21.	lines	TransactionLineModel	It is a list of line items in this transaction.
22.	locationCode (DEPRECATED)	String	- This field has been replaced by the reportingLocationCode field. - Use reportingLocationCode instead of locationCode.
23.	locationTypes	TransactionLocationTypeModel	It is a list of location types in the transaction.
24.	locked	Boolean	It is a two or three-character ISO region code for the transaction.

Sr. No.	Properties	Data Type	Description
25.	messages	AvaTaxMessage	- It is a list of information and warning messages for the API call. - These messages are only relevant to the current API call.
26.	modifiedDate	DateTime	It is a date/time that the record was last modified.
27.	modifiedUserId	Long	It is a user ID of the user that last modified the record.
28.	originAddressId	Long	It is a unique ID number of the origin address for the transaction.
29.	parameters	String	It contains a list of extra parameters that were set when the transaction was created. This is deprecated. Use parameterList instead.
30.	parameterList	ParameterModel	It contains a list of extra parameters that were set when the transaction was created.
31.	paymentDate	DateTime	- The date when payment was made on the transaction. - The default date is the date of the transaction.
32.	purchaseOrderNo	String	It is a purchase order number supplied by the customer for the transaction.
33.	reconciled	Boolean	The value is set to true when the transaction has been reconciled against the company's ledger. For Example, True.
34.	referenceCode	String	It is a user-defined reference code for the transaction.
35.	region	String	It is a two-character ISO 3166 code of the country for the transaction. For Example, CA.
36.	reportingLocationCode	String	It is a code string of the location when the transaction was made from a specific reporting location.
37.	salespersonCode	String	The salesperson who provides the transaction.
38.	softwareVersion	String	It is a software version used to calculate the transaction.
39.	status	DocumentStatus	It displays the status of the transaction. For Example, Committed.
40.	summary	TransactionSummary	It displays the tax summary of the transaction.
41.	taxDate	DateTime	It displays the tax date for the transaction.
42.	taxOverrideAmount	Decimal	It displays the tax override amount.
43.	taxOverrideReason	String	It displays the tax override reason.
44.	taxOverrideType	TaxOverrideTypeId	It displays the tax override type.

Sr. No.	Properties	Data Type	Description
45.	totalAmount	Decimal	It displays the total amount of the transaction.
46.	totalExempt	Decimal	It displays the total exempt amount of the transaction.
47.	totalTax	Decimal	It displays the total tax calculated for all the lines in the transaction.
48.	totalTaxable	Decimal	It is a portion of the total amount of the transaction that was taxable.
49.	totalTaxCalculated	Decimal	It is the amount of tax that AvaTax calculated for the transaction.
50.	transactionDate	DateTime	It is a date on which the transaction occurred.
51.	type	DocumentType	- It is a type of the transaction. - For Example, for returns customers, a transaction type of "Invoice" will be reported to the tax authorities.
52.	version	Integer	- If the transaction was adjusted, it indicates the version number of the transaction. - Incremented each time the transaction is adjusted.
53.	pendingLineUpdates	SObject	This is a collection (List<SObject>) that stores line-level record updates that couldn't be immediately applied to the database.
54.	pendingHeaderUpdates	SObject	This is a collection (List<SObject>) that stores header-level record updates that couldn't be immediately applied to the database.

TransactionLocationTypeModel

Description	Information about the location type.
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Properties

The following are the TransactionLocationTypeModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	documentAddressId	Long	It is an address ID for the transaction.
2.	documentId	Long	It is a transaction ID of the document. Example: 456
3.	documentLocationTypeId	Long	It is a location type ID for the location type in transaction.
4.	locationTypeCode	String	It is a location type code. Example: SingleLocation

TransactionSummary

Description	None
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Properties

The following are the TransactionSummary properties and their information:

Sr. No.	Properties	Data Type	Description
1.	country	String	It is a two-character ISO-3166 country code. For Example, US.
2.	exemption	Decimal	It is an amount of the transaction that was exempt.
3.	jurisCode	String	It is a jurisdiction code for the tax jurisdiction.
4.	jurisName	String	It is a name of the jurisdiction that collects the tax.
5.	jurisType	JurisdictionType	It is a type of jurisdiction that collects the tax.
6.	nonTaxable	Decimal	The amount of the transaction that was non-taxable.
7.	rate	Decimal	Tax Rate - The rate of taxation, as a fraction of the amount.
8.	rateType (DEPRECATED)	RateType	- It is used to indicates the tax ratetype. - Use the rateTypeCode instead of rateType.
9.	rateTypeCode	String	- It indicates the code of the ratetype. - Use `/api/v2/definitions/ratetypes` for a full list of ratetypecodes.
10.	region	String	Two or three-character ISO region, state, or province code, if applicable.
11.	stateAssignedNo	String	The state assigned the number of the jurisdiction that collects the tax.
12.	tax	Decimal	Tax amount - The calculated tax (Base x Rate).
13.	taxable	Decimal	Tax Base - The adjusted taxable amount.
14.	taxAuthorityType	Integer	The unique ID of the tax authority type that collects the tax.
15.	taxCalculated	Decimal	It is a tax calculated by AvaTax.
16.	taxGroup	String	It is a group code when special grouping is enabled.
17.	taxName	String	It is a name of the tax.
18.	taxSubType	String	The tax sub type of this tax.

TransactionAddressModel**Description**

- An address used withing this transaction.

Properties

The following are the TransactionAddressModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	boundaryLevel	BoundaryLevel	• The boundary level at which the address was validated.
2.	city	String	• It is a city component of the address. For Example, Irvine.
3.	country	String	• It is a name or ISO 3166 code identifying the country. For Example, US.
4.	id	Long	• It is a unique ID number of the address.
5.	latitude	String	• Latitude for the address.
6.	line1	String	• It is a first line of the street address. For Example, 200 Main Street.
7.	line2	String	• It is a second line of the street address.
8.	line3	String	• It is a third line of the street address.
9.	longitude	String	• It is a geospatial longitude measurement, in Decimal Degrees, floating point format.
10.	postalcode	String	• It is a postal code / zip code component of the address. For Example, 92614.
11.	region	String	• It is a name or ISO 3166 code identifying the region within the country. For Example, CA.
12.	taxRegionId	Long	• It is a unique ID number of the tax region for the address.
13.	transactionId	Long	• It is a unique ID number of the document to which the address belongs.

TransactionLineModel

Description

- It represents the line items on the transaction.

Properties

The following are the TransactionLineModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	boundaryOverrideId	Long	• It is a unique ID number of the boundary override applied to the line item.
2.	businessIdentificationNo	String	• VAT business identification number used for the transaction.
3.	certificateId	String	• It is a CertCapture Certificate ID.
4.	customerUsageType (DEPRECATED)	String	• The customer usage type for the line item. • The usage type often affects taxability rules.
5.	description	String	• A description of the item or service represented by the line.
6.	destinationAddressId	Long	• It is a unique ID number of the destination address where the line item was delivered or sold.
7.	details	TransactionLineDetailModel	• It is a list of location types for the line item.
8.	discountAmount	Decimal	• The amount of discount that was applied to the line item.
9.	discountTypeId	Long	• It is a type of discount that was applied to the line item.
10.	exemptAmount	Decimal	• The amount of the line item that was exempt.
11.	exemptCertId	Long	• It is a unique ID number of the exemption certificate that applied to the line item.
12.	exemptNo	String	• The customer Tax ID Number (tax_number) associated with a certificate. • The sales tax calculation requests first determine if there is an applicable ECMS entry available and will utilize it for exemption processing. • If no applicable ECMS entry is available, then the AvaTax service will determine whether an Exemption Number field is populated, or an Entity/Use Code is included in the sales tax calculation request and will perform the exemption processing using either of these two options.
13.	id	Long	• It is a unique ID number of the transaction line item.
14.	isItemTaxable	Boolean	• It True, then the item is taxable.

Sr. No.	Properties	Data Type	Description
15.	isSSTP	Boolean	If True, then the item is a Streamlined Sales Tax line item.
16.	itemCode	String	It is a code string of the item represented by the line item.
17.	lineAmount	Decimal	It is a total amount of the transaction, including both taxable and exempt.
18.	lineLocationTypes	TransactionLineDetailModel	It is a list of location types for the line item.
19.	lineNumber	String	The line number or code indicating the line on the invoice, receipt or document.
20.	originAddressId	Long	It is a unique ID number of the origin address where the line item was delivered or sold.
21.	parameters	String	This is deprecated. Use parameterList instead. It contains a list of extra parameters that were set when the transaction was created.
22.	parameterList	ParameterModel	It contains a list of extra parameters that were set when the transaction was created.
23.	quantity	Decimal	It is a quantity of products added on the line item.
24.	ref1	String	It is a user-defined reference identifier for the transaction line item.
25.	ref2	String	It is a user-defined reference identifier for the transaction line item.
26.	reportingDate	DateTime	It is a date when the transaction is reported.
27.	revAccount	String	The revenue account number for the line item.
28.	sourcing	Sourcing	Indicates whether the line item was taxed according to the origin or destination.
29.	tax	Decimal	The amount of tax generated for the line item.
30.	taxableAmount	Decimal	The taxable amount of the line item.
31.	taxCalculated	Decimal	The tax calculated for the line by Avalara AvaTax.
32.	taxCode	String	The code string for the tax code that was used to calculate this line item.
33.	taxCodeId	Long	It is a unique ID number for the tax code that was used to calculate the line item.
34.	taxDate	DateTime	It is a date that was used for calculating tax amounts for the line item.
35.	taxEngine	String	It is a tax engine identifier that was used to calculate the line item.

Sr. No.	Properties	Data Type	Description
36.	taxIncluded	Boolean	If True, then the tax was included in the purchase price of the item.
37.	taxOverrideAmount	Decimal	If a tax override was specified, then it indicates the amount of tax that was requested.
38.	taxOverrideReason	String	If a tax override was specified, then it represents the reason for the tax override.
39.	taxOverrideType	TaxOverrideTypeId	If a tax override was specified, then it indicates the type of tax override.
40.	transactionId	Long	It is a unique ID number of the transaction to which the line item belongs.
41.	vatCode	String	The code string for the VAT code that was used to calculate this line item.
42.	hscode	String	The harmonized system code applicable for the product used by the customs authorities to calculate the tariff rates for internationally traded goods.
43.	category	String	This custom field is from the AvaTax for Salesforce CPQ managed package for Product category.

TransactionLineDetailModel

Description	- It is an individual tax detail element. - It represents the tax amount calculated for a particular jurisdiction, and for a particular line in an invoice.
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Properties

The following are the TransactionLineDetailModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	addressId	Long	It is a unique ID number of the address used for the tax detail.
2.	country	String	It is a two-character ISO 3166 country code of the country where the tax detail is assigned. For Example, US.
3.	countyFIPS	String	It is a Federal Information Processing Standard (FIPS) code for the county where the tax detail is assigned.
4.	exemptAmount	Decimal	The amount of the line that was considered exempt in the tax detail.
5.	exemptReasonId	Long	It is a unique ID number of the exemption reason for the tax detail.
6.	exemptUnits	Decimal	The number of units in the line item that were calculated to be exempt according to the rating detail.
7.	id	Long	It is a unique ID number of the tax detail.
8.	inState	Boolean	If True, then the detail element represented an in-state transaction.

Sr. No.	Properties	Data Type	Description
9.	jurisCode	String	It is a code of the jurisdiction to which the tax detail applies.
10.	jurisdictionId	Long	It is a unique ID number of the jurisdiction to which the tax detail applies.
11.	jurisName	String	It is a name of the jurisdiction to which the tax detail applies.
12.	jurisType	JurisTypeId	It is a type of the jurisdiction to which the tax detail applies.
13.	nonTaxableAmount	Decimal	The amount of the line item that was considered nontaxable in the tax detail.
14.	nonTaxableRuleId	Long	The rule according to which portion of the detail was considered nontaxable.
15.	nonTaxableType	TaxRuleTypeId	It is a type of nontaxability that was applied to the tax detail.
16.	nonTaxableUnits	Decimal	It is the number of units in the line item that were calculated to be nontaxable according to the rating detail.
17.	rate	Decimal	It is a rate at which the tax detail was calculated.
18.	rateRuleId	Long	It is a unique ID number of the rule according to which the tax detail was calculated.
19.	rateSourceId	Long	It is a unique ID number of the source of the rate according to which the tax detail was calculated.
20.	rateType (DEPRECATED)	RateType	- It is a ratetype for the tax detail. - Use the rateTypeCode instead of rateType.
21.	rateTypeCode	String	It indicates the code of the ratetype that was used to calculate the tax detail.
22.	region	String	It is a two or three-character ISO region code for the region where the tax detail is assigned. For Example, CA.
23.	serCode	String	For Streamlined Sales Tax customers, the SST Electronic Return code under which the tax detail should be applied.
24.	signatureCode	String	It is an Avalara-specified signature code of the jurisdiction to which the tax detail applies.
25.	sourcing	Sourcing	It indicates whether the tax detail applies to the origin or destination address of the transaction.
26.	stateAssignedNo	String	It is a state assigned number of the jurisdiction to which the tax detail applies.
27.	stateFIPS	String	It is a Federal Information Processing Standard (FIPS) is a code for the state where the tax detail is assigned.
28.	tax	Decimal	It is a tax on the amount of the tax detail.
29.	taxableAmount	Decimal	The taxable amount of the tax detail.

Sr. No.	Properties	Data Type	Description
30.	taxableUnits	Decimal	It is the number of units in the line item that were calculated to be taxable according to the rating detail.
31.	taxAuthorityTypeId	Long	It is a type of the tax authority to which the tax will be remitted.
32.	taxCalculated	Decimal	The amount of tax that was calculated.
33.	taxName	String	It is a name of the tax against which the tax amount was calculated.
34.	taxOverride	Decimal	The amount of tax override that was specified for the tax line.
35.	taxRegionId	Long	It is a unique ID number of the tax region.
36.	transactionId	Long	It is a unique ID number of the transaction.
37.	transactionLineId	Long	It is a unique ID number of the line within the transaction.

AddressValidationInfo

Description	REST based client for Avalara address validation and tax calculation services. @author Avalara Salesforce Connector Team.
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Properties

The following are the AddressValidationInfo properties and their information:

Sr. No.	Properties	Data Type	Description
1.	city	String	It is a city component of the address. For Example, Irvine.
2.	country	String	It is a name or ISO 3166 code identifying the country. For Example, US.
3.	latitude	Double	It is a geospatial latitude measurement, in Decimal Degrees, floating point format.
4.	line1	String	It is a first line of the street address.
5.	line2	String	It is a second line of the street address.
6.	line3	String	It is a third line of the street address.
7.	longitude	Double	It is a geospatial longitude measurement, in Decimal Degrees, floating point format.
8.	postalcode	String	It is postal code / zip code component of the address. For Example, 92614.
9.	region	String	It is name or ISO 3166 code identifying the region within the country. For Example, CA.
10.	textcase	TextCase	- It specifies the text case for the validated address result. - If not specified, then it returns address in uppercase.

CreateTransactionModel

Description

- Create a transaction.
- This is used to calculate the tax.

Properties

The following are the CreateTransactionModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Addresses	AddressesModel	• It is a default addresses for all lines in the document.
2.	batchCode	String	• It is a BatchCode for the batch operations.
3.	businessIdentificationNo	String	• VAT business identification number for the customer for the transaction.
4.	Code	String	• It is an internal reference code used by the client application.
5.	commitDocument	Boolean	• If True, then the document is committed. • This option is only applicable for invoice document types, not orders.
6.	companyCode	String	• It specifies the code of the company creating the transaction.
7.	currencyCode	String	• It is a three-character ISO 4217 currency code for the transaction. For Example, USD.
8.	customerCode	String	• The client application customer reference code.
9.	customerUsageType (DEPRECATED)	String	• It is a client application as customer or usage type. • Use the EntityUseCode, instead of customerUsageType.
10.	debugLevel	TaxDebugLevel	• It specifies a level higher than 'normal' if the user wishes to request additional debug information from the transaction.
11.	Description	String	• It is a user-supplied description for the transaction.
12.	Discount	Decimal	• The discount amount to apply to the document.
13.	documentDate	DateTime	• The date on the invoice and purchase order.
14.	Email	String	• It is a user-supplied email address relevant for the transaction.
15.	exchangeRate	Decimal	• It is currency exchange rate from the transaction to the company base currency.
16.	exchangeRateEffectiveDate	DateTime	• It is an effective date of the exchange rate.
17.	exemptionNo	String	• It is an exemption number for the document.

Sr. No.	Properties	Data Type	Description
18.	isSellerImporterOfRecord	Boolean	- The "importer of record" is liable to pay customs and import duties for products shipped internationally. - If you specify that the seller is the importer of record, then estimates of customs and import duties will be added as tax details to the transaction. - If you specify that the buyer is the importer of record, then customs and import duties will not be added to the tax details to the transaction. - If the transaction is not an international transaction, then this field is left blank.
19.	Lines	LineItemModel	It is a list of line items that appears on the transaction.
20.	Parameters	String	This property is deprecated. User parameter instead.
21.	parameterList	ParameterModel	- It is a special parameter for the transaction. - Use the ListParameters endpoint to get a full list of available parameters.
22.	posLaneCode	String	It sets the point of sale lane code sent by the user for the document.
23.	purchaseOrderNo	String	It is a purchase order number for the document.
24.	referenceCode	String	It is a customer-provided reference code with information about the transaction.
25.	reportingLocationCode	String	It sets the sale location code (Outlet ID) for reporting the document to the tax authority.
26.	salespersonCode	String	It is the client application salesperson reference code.
27.	serviceMode	ServiceMode	It specifies whether the tax calculation is handled Local, Remote, or Automatic (default).
28.	taxOverride	TaxOverrideModel	It specifies a tax override for the entire document.
29.	type	DocumentType	It specifies the type of document to create.

ParameterModel

Description	It represents the special parameters for the transactions/transaction line.
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Properties

The following are the ParameterModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	name	String	The name of the parameter. For Example, ScreenSize
2.	value	String	The value of the parameter. For Example, 22.0.

Sr. No.	Properties	Data Type	Description
3.	unit	Decimal	The unit of measure of the parameter value. For Example, Inch

AddressesModel

Description	- Information about all the addresses involved in the transaction. - For a physical in-person transaction at a retail point-of-sale location, specify one address using the singleLocation field. - Specify the shipFrom and shipTo addresses for a transaction that was shipped, delivered, or provided from an origin location such as a warehouse to a destination location. - In the United States, some jurisdictions recognize the address types pointOfOrderOrigin and pointOfOrderAcceptance. These address types affect the sourcing models of some transactions. - If latitude and longitude information is provided for any addresses along with the line, city, region, country, and postal code information, then only the latitude and longitude will be considered and will discard the line, city, region, country, and postal code information for the transaction. - You need to ensure that you have the correct latitude/longitude information for the addresses prior to using the API. If you provide either latitude or longitude information but not both, then the line, city, region, country and postal code information for the addresses will be used.
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Properties

The following are the AddressesModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	pointOfOrderAcceptance	AddressLocationInfo	It is a place of business where you accept/approve the customer's order, thereby becoming contractually obligated to make the sale.
2.	pointOfOrderOrigin	AddressLocationInfo	It is a place of business where you receive the customer's order.
3.	shipFrom	AddressLocationInfo	It is an origin address where the products were shipped from, or from where the services originated.
4.	shipTo	AddressLocationInfo	It is a destination address where the products were shipped to, or where the services were delivered.
5.	singleLocation	AddressLocationInfo	If the transaction occurred at a retail point-of-sale location, and then it provides the single address in AddressLocationInfo and leave all other address types null.

AddressLocationInfo

Description	It represents the address to resolve.
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Properties

The following are the AddressLocationInfo properties and their information:

Sr. No.	Properties	Data Type	Description
1.	city	String	It is a city component of the address. For Example, Irvine.
2.	country	String	It is a name or ISO 3166 code identifying the country. For Example, US.
3.	latitude	Decimal	It is a geospatial latitude measurement in Decimal Degrees floating point format.
4.	line1	String	The first line of the street address.
5.	line2	String	The second line of the street address.
6.	line3	String	The third line of the street address.
7.	locationCode	String	It allows you to specify the address of an existing location of the company.
8.	longitude	Decimal	Geospatial longitude measurement in Decimal Degrees floating point format.
9.	postalCode	String	It is postal code/zip code component of the address. For Example, 92614.
10.	region	String	It is name or ISO 3166 code identifying the region within the country. For Example, CA.

LineItemModel

Description	It represents the address to resolve.
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Properties

The following are the AddressLocationInfo properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Addresses	AddressesModel	The addresses to use for the transaction line.
2.	Amount	Decimal	The total amount for the line.
3.	businessIdentificationNo	String	VAT business identification number for the customer for the line item.
4.	customerUsageType (DEPRECATED)	String	Use the entityUseCode.
5.	entityUseCode	String	It allows you to specify a type of usage that makes the transaction to be considered as exempt with a reason of exempt.
6.	Description	String	- It is an item description. - This field is required if an unmapped itemcode is used for the Streamlined Sales Tax (SST) customers.

Sr. No.	Properties	Data Type	Description
7.	Discounted	Boolean	If True, then the document discount should be applied to the line.
8.	exemptionCode	String	- It is a customer Tax ID Number (tax_number) associated with the certificate. - The sales tax calculation requests first determine if there is an applicable ECMS entry available, and then utilize it for the exemption processing. - If no applicable ECMS entry is available, then the AvaTax service determines whether the Exemption Number field is populated, or an Entity/Use Code is included in the sales tax calculation request and then performs the exemption processing using either of these two options.
9.	itemCode	String	- It is an Item Code (SKU). - If you provide an itemCode field, then the AvaTax API will look up the item you created with the CreateItems API call and use all the information available about that item for the transaction.
10.	lineNumber	String	It is a line number within the document.
11.	Quantity	Decimal	It is a quantity of items in the line.
12.	ref1	String	- It is a customer defined field. - This field is used to provide the customer preferred information. It does not affect tax calculation.
13.	ref2	String	- It is a customer defined field. - This field is used to provide the customer preferred information. It does not affect tax calculation.
14.	revenueAccount	String	- It is a customer defined field. - This field is used to provide the customer preferred information. It does not affect tax calculation.
15.	taxCode	String	It is a system or custom tax code.
16.	taxIncluded	Boolean	- It indicates if the line item has tax included. - The default value is set to false.
17.	parameterList	ParameterModel	It is a special parameter for the transaction.
18.	hscode	String	The harmonized system code applicable for the product used by the customs authorities to calculate the tariff rates for internationally traded goods.
19.	category	String	This custom field is from the AvaTax for Salesforce CPQ managed package for Product category.

TaxOverrideModel

Description	It represents the address to resolve.
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Properties

The following are the TaxOverrideModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Reason	String	It provides the reason for a tax override for audit purposes.
2.	taxAmount	Decimal	It indicates a total override of the calculated tax on the document.
3.	taxDate	Decimal	- It is a tax override date. - If the date is not overridden, then it is set as the DocDate.
4.	Type	TaxOverrideType	It identifies the type of tax override.

CreateOrAdjustTransactionModel

Description	It creates or adjust the transaction model.
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Properties

The following are the CreateOrAdjustTransactionModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	createTransactionModel	createTransactionModel	- It creates a transaction, if the transaction does not exist, and if the transaction exists, then it adjusts the existing transaction. - If True, then it commits the transactionInstance of createTransactionModel.

CommitTransactionModel

Description	It commits the transaction as a permanent, so that it can be reported on a tax filing.
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Properties

The following are the CommitTransactionModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	commitTransaction	Boolean	If True, then it commits the transaction.

CustomerModel

Description	Represents a customer to whom you sell products and/or services.
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Properties

The following are the CustomerModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Id	Long	It is a unique ID number of the customer.
2.	companyId	Long	It is a unique ID number of the AvaTax company.
3.	customerCode	String	- It is a unique code to identify the customer. - This code is used in the customerCode field of any call that creates or adjusts a transaction to ensure that all the exemptions that applies to the customer are correctly considered.
4.	AlternateId	String	- It is a customer-configurable alternate ID number for a customer. - The customer can set this value to match any other system that would like to reference to the customer record.
5.	Name	String	It is a name identifying the customer.
6.	attnName	String	It indicates the "Attn:" component of the address for the customer, if the customer requires mailings to be shipped to the attention of a specific person or department name.
7.	line1	String	It is a first line of the street address.
8.	line2	String	It is a second line of the street address.
9.	City	String	It is a city component of the address. For Example, Irvine.
10.	postalCode	String	It is postal code / zip code component of the address. For Example, 92614.
11.	PhoneNumber	String	It is a main phone number for the customer.
12.	FaxNumber	String	It is a fax phone number for the customer.
13.	emailAddress	String	It is a main email address for the customer.
14.	contactName	String	It is a name of the main contact person for the customer.
15.	lastTransaction	DateTime	A date when the customer has last executed a transaction.
16.	createdDate	DateTime	The date when the record was created.
17.	modifiedDate	DateTime	The date/time that the record was last modified.
18.	country	String	It is a name or ISO 3166 code identifying the country. For Example, US.

Sr. No.	Properties	Data Type	Description
19.	region	String	It is name or ISO 3166 code identifying the region within the country. For Example, CA.
20.	isShip	Boolean	If True, then the customer record is specifically used for ship-to purposes.
21.	taxpayerIdNumber	String	- It is an ID number for the taxpayer/ customer. - For the United States customers, this field is referred as the Federal Taxpayer ID Number. - For businesses, this field is referred as the Federal Employer Identification Number. - For individuals, this field is referred as the Social Security Number.
22.	certificates	List<CertificateModel> (Link to the CertificateModel)	- It is a list of exemption certificates that applies to the customer. - The customer can fetch the data by specifying '\$include=certificates' when calling a customer fetch API.

CertificateModel

Description	- A certificate is a document stored in either AvaTax Exemptions or CertCapture. - A certificate is a document containing information on the customer's eligibility for exemption from sales or use taxes based on the criteria specified by the customer when the customer stores the certificate. - Login to the administrative website for the product you have purchased and view or manage the customers certificates directly.
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Properties

The following are the CertificateModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Id	Long	It is a unique ID number of the certificate.
2.	companyId	Long	It is a unique ID number of the AvaTax company that recorded the certificate.
3.	signedDate	DateTime	The date when the certificate was signed.
4.	expirationDate	DateTime	It is an expiration date when the certificate will no longer be valid.
5.	filename	String	- It is a file name of the exemption certificate. - If you do not upload a PDF or JPG image, you must specify the filename of the certificate as it is tracked in your repository.

Sr. No.	Properties	Data Type	Description
			- To create a certificate, you must provide either a filename, a pdf file, or an array of JPG pages. - The API returns an error if you omit these fields or if you attempt to put more values.
6.	Valid	Boolean	- If True, then the certificate is marked as valid. - A valid certificate is considered for the exemption. - When a certificate is marked invalid, it will no longer be considered to calculate exemption for the customer.
7.	verified	Boolean	- If True, then it means that the certificate has gone through the certificate validation process. - For more information on the certificate validation process, see the Avalara Help Center.
8.	exemptPercentage	Decimal	- It describes the tax exempt in percent. - For a fully exempt certificate, the percentage should be 100.
9.	isSingleCertificate	Boolean	- If True, then the certificate is a single (or standalone) certificate. - The value is set during the audit stage of the certificate validation process.
10.	exemptionReason	ExemptionReasonModel (Link to ExemptionReasonModel)	- It is an exemption reason associated with the certificate. - For a list of exemption reasons, call ListCertificateExemptReasons.
11.	createdDate	DateTime	The date/time when the record was created.
12.	ModifiedDate	DateTime	The date/time that the record was last modified.
13.	pageCount	Long	It is number of pages contained within the exemption certificate.
14.	Customers	List<CustomerModel> (Link to CustomerModel)	- It is a list of customers to which the certificate applies. - You can fetch the data by specifying \$include=customers when calling a certificate fetch API.
15.	poNumbers	List<PoNumberModel> (Lnk to PoNumberModel)	- It is a list of purchase order numbers that are valid to use with the certificate. - If the certificate is applicable for all purchase order numbers, then the field will be empty. - You can fetch the data by specifying \$include=po_numbers when calling a certificate fetch API.

Sr. No.	Properties	Data Type	Description
16.	exposureZone	ExposureZoneModel (Link to ExposureZoneModel)	- It is an information about a physical area or zone in which a certificate can apply. - An exposure zone for an exemption certificate is a tax authority such as a state, country, or local government entity.
17.	attributes	List<CertificateAttributeModel> (Link to CertificateAttributeModel)	- It is a list of certificate attributes that applies to the certificate. - You can fetch the data by specifying \$include=attributes when calling a certificate fetch API.
18.	pdf	String	- This field is available for input only. - To retrieve the image after creation, use the DownloadCertificateImage API. - When creating a certificate, you can optionally provide a PDF image in Base64 URLEncoded format. - PDFs are automatically parsed into individual page JPG images and can be retrieved back later as either the original PDF or the individual pages. - To create a certificate, you must provide either a filename, a pdf file, or an array of JPG pages. - The API returns an error if you omit these fields or if you attempt to put values in more than one of them.
19.	pages	List<String>	- This is an input field. - To retrieve the image after creation, use the DownloadCertificateImage API.
20.	documentExists	Boolean	This value is true if there exists scanned PDF copy of this certificate or the PDF version of the form that the customer filled via the CertCapture wizard on S3 bucket.
21.	exemptionNumber	String	Indicates the tax number passed in for the certificate.
22.	validatedExemptionReason	ExemptionReasonModel	An exemption reason defines why a certificate allows a customer to be exempt for purposes of tax calculation. For a full list of defined exemption reasons, please call the ListCertificateExemptionReasons API.
23.	Status	String	The status of the certificate.
24.	taxNumberType	String	The tax number type for the certificate. For example, FEIN, Social Security Number, or Employer Identification Number.

Sr. No.	Properties	Data Type	Description
25.	businessNumberType	String	Description of business for the certificate. For example, Retail trade, Professional services, or Construction.
26.	ecmsId	Integer	The unique ID number of current AvaTax Exemption Certificate that refers this certificate.
27.	documentExists	Boolean	This value is true if there exists scanned PDF copy of this certificate or the PDF version of the form that the customer filled via the CertCapture wizard on S3 bucket.

TransactionLineLocationTypeModel

Description	Represents information about the location types stored in a line.
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Properties

The following are the TransactionLineLocationTypeModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	documentLineLocationTypeId	Long	It is a unique ID number of the line location address model.
2.	documentLineId	Long	It is a unique ID number of the document line associated with the line location address model.
3.	documentAddressId	Long	It is an address ID corresponding to the model.
4.	locationTypeCode	String	It is a locationTypeCode corresponding to the model.

CertificateAttributeModel

Description	- A certificate attribute can be thought of as a feature or flag that is applied to a certificate. - A single certificate can be linked to zero, one, or many certificate attributes. - The full list of attributes can be obtained by calling the ListCertificateAttributes API.
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Properties

The following are the CertificateAttributeModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	id	Long	It is a unique ID number representing the certificate attribute.
2.	name	String	It is a name of the certificate.
3.	description	String	It is a full help text description of the certificate attribute.

Sr. No.	Properties	Data Type	Description
4.	isSystemCode	Boolean	- If True, then it is a system-defined certificate attribute. - The system-defined attributes cannot be modified or deleted on the CertCapture website.

CompanyModel

Description	It represents about the company or a business entity.
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Properties

The following are the CompanyModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	id	Integer	It is a unique ID number of the company.
2.	accountId	Integer	It is a unique ID number of the account to which the company belongs.
3.	parentCompanyId	Integer	It is a unique identity of the parent company.
4.	sstPid	String	If the company files a Streamlined Sales Tax, then the PID of a company is defined by the Streamlined Sales Tax governing board.
5.	companyCode	String	It is a unique code that references the company within your account.
6.	name	String	It is a name of the company.
7.	isDefault	Boolean	- If True, then the company is considered as a default company for the account. - One company can be set as default.
8.	DefaultLocationId	Integer	It is the unique ID number of the default location for the company.
9.	IsActive	Boolean	- It indicates whether tax activity can occur for the company. - Set the flag to true to permit the company to process transactions.
10.	taxpayerIdNumber	String	- For United States companies, the field contains the customers Taxpayer Identification Number. - It is a nine-digit number that is usually called an EIN for an Employer Identification Number if the company is a corporation, or SSN for a Social Security Number if the company is a person. - The value is required if you subscribe to Avalara Managed Returns or the SST Certified Service Provider

Sr. No.	Properties	Data Type	Description
			services, but it is optional if you do not subscribe to either of those services.
11.	hasProfile	Boolean	- Set the flag to true to give the company its own unique tax profile. - If the flag is true, then the company will have its own Nexus, TaxRule, TaxCode, and the Item definitions. - If the flag is false, then the company will inherit all profile values from its parent.
12.	isReportingEntity	Boolean	- Set the flag to true if the company files its own tax returns. - For the users who have Returns enabled, the flag turns on monthly worksheet generation for the company.
13.	sstEffectiveDate	DateTime	It is a date when the company has joined the Streamlined Sales Tax program.
14.	defaultCountry	String	It is a two-character ISO-3166 country code of the default country for the company. For Example, US.
15.	baseCurrencyCode	String	It is a three-character ISO-4217 currency code of the default currency used by the company. For Example, USD
16.	roundingLevelId	RoundingLevelId	It indicates whether the company prefers to round amounts at the document level or line level.
17.	warningsEnabled	Boolean	Set the value to true to receive warnings in API calls via SOAP.
18.	isTest	Boolean	- If the flag is set to true, then it indicates that the company is a Test company. - If you have Returns enabled, then the Test companies will not file tax returns and can be used for validation purposes.
19.	taxDependencyLevelId	TaxDependencyLevelId	It is used to apply the tax detail dependency at a jurisdiction level.
20.	InProgress	Boolean	- Set the value to true to indicate that you are still working to finish the company configuration. - While the value is true, no tax reporting will occur, and the company will not be usable for transactions.

Sr. No.	Properties	Data Type	Description
21.	businessIdentificationNo	String	It is a company Business Identification Number.
22.	createdDate	DateTime	It is a date when the record was created.
23.	createdUserId	Integer	It is a user ID of the user who have created the record.
24.	modifiedDate	DateTime	It is a date/time that the record was last modified.
25.	modifiedUserId	Integer	It is a user ID of the user that last modified the record.
26.	Contacts	List<ContactModel> (Link to Contact Model)	It is a list of contacts defined for a company, to retrieve the contact list use the ContactModel attributes.
27.	items	List<ItemModel> (Link to Item Model)	It is a list of items defined in the company's product catalog.
28.	locations	List<LocationModel> (Link to LocationModel)	It is a list of locations defined for the company's business location.
29.	Nexus	List<NexusModel> (Link to NexusModel)	It is a list of nexuses defined for the company and declares a nexus within a particular taxing jurisdiction.
30.	settings	List<SettingModel> (Link to SettingModel)	It is a list of settings defined for the company to keep a track of custom information about a company.
31.	taxCodes	List<TaxCodeModel> (Link to TaxCodeModel)	- It is a list of TaxCodes defined for the company that can be applied to items on a transaction. - A tax code can have specific rules for specific jurisdictions that change the tax calculation behavior.
32.	taxRules	List<TaxRuleModel> (Link to TaxRuleModel)	It is a list of tax rules defined for the company that changes the behavior of Avalara's tax engine for the products.
33.	Upcs	List<UPCModel> (Link to UPCModel)	It is a list of Universal Product Code object defined for the products of the company.
34.	exemptCerts (DEPRECATED)	List<EcmsModel> (Link to EcmsModel)	Use the `ListCertificates` API, instead of exemptCerts.

TaxRuleModel**Description**

- It represents a tax rule that changes the behavior of Avalara's tax engine for certain products and/or Entity/Use Codes in certain jurisdictions.
- The Avalara supports different types of tax rules. For information about tax rule types, see TaxRuleTypeId.
- Because the different types of tax rules have different behavior, some fields may change their behavior based on the type of tax rule selected.
- For each below mentioned properties ensure that the value you send is appropriate for the tax rule type.

Properties

The following are the TaxRuleModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	id	Integer	• It is a unique ID number of the tax rule.
2.	companyId	Integer	• It is a unique ID number of the company that owns the tax rule.
3.	taxCodeId	Integer	• It is a unique ID number of the taxcode for the rule. • When creating or updating a tax rule, you need to specify either the taxCodeId value or the taxCode value.
4.	taxCode	String	• It is a code string of the taxcode for the rule. • When creating or updating a tax rule, you need to specify either taxCodeId value or the taxCode value.
5.	stateFIPS	String	• For U.S. tax rules, it is a state's Federal Information Processing Standard (FIPS) code.
6.	jurisName	String	• It is a name of the jurisdiction to which the tax rule applies.
7.	jurisCode	String	• It is a code of the jurisdiction to which the tax rule applies.
8.	jurisTypeId	JurisTypeId	• It is a type of the jurisdiction to which the tax rule applies.
9.	customerUsageType	String	• It is a type of customer usage to which the rule applies.
10.	taxTypeId	MatchingTaxType	• It indicates which tax rule is applied to which tax types.
11.	rateTypeId (DEPRECATED)rateTypeId	RateType	• Use the rateTypeCode instead of rateTypeId.
12.	rateTypeCode	String	• It indicates the code of the ratetype that applies to the rule. • If you specify a value in the rateTypeCode field, this rule will cause tax lines that are affected by the rule to change to a different ratetypecode.

Sr. No.	Properties	Data Type	Description
			Use ListRateTypesByCountry API for a full list of ratetypecodes.
13.	taxRuleTypeId	TaxRuleTypeId	- It is a type of value which determines the behavior of the tax rule. - You can specify that the tax rule controls the product's taxability or exempt / nontaxable status and the product's rate.
14.	isAllJuris	Boolean	Set the value to true if the tax rule applies in all jurisdictions.
15.	value	Decimal	- This field has different behavior based on the type of the tax rule. - For a product taxability rule, this value is either 1 or 0, indicating taxable or non-taxable. - For a rate override rule, this value is the verified rate stored as a decimal, For Example, a rate of 5% would be stored as 0.05 decimal. If you use the special value of 1.0, then the cap and threshold values will be applied, and the rate will be left alone.
16.	Cap	Decimal	It is a maximum cap for the price of the item according to the rule.
17.	Threshold	Decimal	It is a per-unit threshold that must be met before the rule applies.
18.	Options	String	It supports custom options for the rule. For Example, All value indicates that the entire amount of the line becomes taxable when the line amount exceeds the threshold.
19.	effectiveDate	DateTime	- The first date (start date) at which the rule applies. - If null, then the rule will apply to all dates prior to the end date.
20.	endDate	DateTime	- The last date (end date) for which this rule applies. - If null, then the rule will apply to all dates after the effective date.
21.	Description	String	It is a description for the tax rule.
22.	countyFIPS	String	It is a Federal Information Processing Standard (FIPS) code for the county where the tax detail is assigned.
23.	isSTPro	Boolean	If True, then it indicates a rule for the Sales Tax Pro.
24.	Country	String	It is a name or ISO 3166 code for the locations where the tax rule is applied. For Example, US.

Sr. No.	Properties	Data Type	Description
25.	Region	String	It is a name or ISO 3166 code for the region, state, or province for the locations where the rule applies. For Example, AZ.
26.	sourcing	Sourcing	It is a sourcing types to which the rule applies.
27.	createdDate	DateTime	It is a date when the record was created.
28.	createdUserId	Integer	It is a user ID of the user who created the record.
29.	modifiedDate	DateTime	It is a date/time that the record was last modified.
30.	modifiedUserId	Integer	It is a user ID of the user that last modified the record.
31.	taxTypeGroup	String	It is a group ID of tax types supported by Avalara.
32.	taxSubType	String	It is an ID of sub tax types supported by Avalara.
33.	taxTypeMappingId	Integer	It is an ID for the TaxTypeMapping object.
34.	rateTypeTaxTypeMappingId	Integer	It is an ID for the RateTypeTaxTypeMapping object.

ECMSModel**Description**

- It represents about the exemption certificate used internally by AvaTax to track information about the exemptions.

Properties

The following are the ECMSModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	ExemptCertId	Integer	• It is a calc_id associated with a certificate in CertCapture.
2.	companyId	Integer	• It is a unique ID number of the company that records the transaction.
3.	customerCode	String	• It is a unique code which identifies the customer who requested the transaction.
4.	CustomerName	String	• It is a name of the customer which is saved in the Avalara AvaTax. For Example: ABC.
5.	address1	String	• It is a first line of the street address. For Example, 200 Main Street.
6.	address2	String	• It is a second line of the street address.
7.	address3	String	• It is a third line of the street address.
8.	city	String	• It is a city component of the address. For Example, Irvine.
9.	region	String	• It is name or ISO 3166 code identifying the region within the country. For Example, CA.
10.	postalCode	String	• It is a postal code / zip code component of the address. For Example, 92614.
11.	country	String	• It is a name or ISO 3166 code identifying the country. For Example, US.
12.	exemptCertTypeId	ExemptCertTypeId	• It is a type of exemption certificate. • The permitted values are Blanket and Single.
13.	documentRefNo	String	• It is an exemption certificate document reference number.
14.	BusinessTypeId	Byte	• It is a type of business to which the customer belongs.
15.	businessTypeOtherDescription	String	• It is a description for the business type.
16.	exemptReasonId	String	• It is an exemption reason which is associated with the certificate, coded by CustomerUsageType. For Example, A – Federal Government.

Sr. No.	Properties	Data Type	Description
17.	exemptReasonOtherDescription	String	The other description for exemption reason that is populated if the exemptReasonId is 'L' – Other or Custom.
18.	effectiveDate	DateTime	It is a start date for the exemption certificate.
19.	regionsApplicable	String	- It is a list of applicable regions for the exemption certificate. - To list more than one applicable region, separate the list of region codes with commas.
20.	exemptCertStatusId	ExemptCertStatusId	It is a status of the exemption certificate.
21.	createdDate	DateTime	It is date when the exemption certificate was created.
22.	lastTransactionDate	DateTime	It is a date when the last transaction with the exemption certificate has been created.
23.	expiryDate	DateTime	It indicates the expiry date of the exemption certificate.
24.	createdUserId	Integer	It is an ID of a user that creates the certificate.
25.	modifiedDate	DateTime	It is a date when the exemption certificate was modified.
26.	modifiedUserId	Integer	It is a user ID of the user that last modifies the record.
27.	countryIssued	String	It is a name or ISO 3166 code identifying the country that issued the ECMS certificate.
28.	AvaCertId	String	- If the certificate record was synced from an AvaTax Certs account, the unique AvaTax Certs identifies the certificate record. - It is an ID of a certificate.
29.	exemptCertReviewStatusId	ExemptCertReviewStatusId	It is a review status for the exemption certificate.
30.	Details	List<EcmsDetailModel> (Link to EcmsDetailModel)	It is an exemption certificate detail.

LocationModel**Description**

- A location where the company does business.
- Some jurisdictions require you to list all the locations where your company does the business.

Properties

The following are the LocationModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Id	Integer	• It is a unique ID number of this location.
2.	companyId	Integer	• It is a unique ID number of the company that operates at the location.
3.	locationCode	String	• It is a code that identifies the location and it must be unique within your company.
4.	Description	String	• It is a name for the location.
5.	addressTypeId	AddressTypeId	• It specifies whether the venue is a permanent place of business or a place of temporary salesperson.
6.	AddressCategoryId	AddressCategoryId	• It indicates the type of place of business that the location represents.
7.	line1	String	• It is a first line of the street address.
8.	line2	String	• It is a second line of the street address.
9.	line3	String	• It is a third line of the street address.
10.	city	String	• It is a city component of the address. For Example, Irvine.
11.	county	String	• It is a name or ISO 3166 code identifying the country. For Example, US.
12.	region	String	• It is name or ISO 3166 code identifying the region within the country. For Example, CA.
13.	postalCode	String	• It is postal code / zip code component of the address. For Example, 92614.
14.	country	String	• It is a two-character ISO-3166 country code of the physical address of the location.
15.	isDefault	Boolean	• If the flag is set to true, then it indicates the default location for the company.
16.	isRegistered	Boolean	• If the flag set to true, then it indicates that the location has been registered with a tax authority.

Sr. No.	Properties	Data Type	Description
17.	DbName	String	If the location has a different business name from its legal entity name, specify the "Doing Business As" name for the location.
18.	outletName	String	It is an outlet/ main office name.
19.	EffectiveDate	DateTime	It is a date when the location was opened for business, or null if not known.
20.	endDate	DateTime	It is a date when the business is closed.
21.	lastTransactionDate	DateTime	It is a recent date when a transaction was processed for the location.
22.	RegisteredDate	DateTime	It is a date when the location was registered with a tax authority.
23.	createdDate	DateTime	It is a date when the record was created.
24.	createdUserId	Integer	It is a user ID of the user who created the record.
25.	modifiedDate	DateTime	It is a date/time that the record was last modified.
26.	modifiedUserId	Integer	It is a user ID of the user that last modified the record.
27.	settings	List<LocationSettingModel> (Link to LocationSettingModel)	- It is an extra information required by certain jurisdictions for filing. - For a list of settings recognized by Avalara, query the endpoint <code>"/api/v2/definitions/locationquestions"</code>. - Query the endpoint <code>"/api/v2/companies/(id)/locations/(id)/validate"</code> to determine the list of settings required for the location.

NexusModel

Description

- It represents a declaration of nexus within a particular taxing jurisdiction.
- You must first call the Definitions API ListNexus to obtain a list of Avalara-defined nexus and create a nexus declaration for your company. Once you have determined which nexus you wish to declare, you should customize only the user-selectable fields in the object.
- It is a user selectable fields for the nexus object as companyId, effectiveDate, endDate, localNexusTypeId, taxId, nexusTypeId, hasPermanentEstablishment, and isSellerImporterOfRecord.
- When calling CreateNexus or UpdateNexus, all values in your nexus object except for the user-selectable fields must match an Avalara-defined system nexus object.
- You can retrieve a list of Avalara-defined system nexus objects by calling ListNexus. If any data does not match, then the AvaTax may not recognize your nexus declaration.

Properties

The following are the NexusModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	id	Integer	• It is a unique ID number of the declared nexus.
2.	companyId	Integer	• It is a unique ID number of the company that has declared its nexus. • This field is user-selectable and should be provided when creating or updating a nexus object.
3.	country	String	• It is a two-character ISO-3166 country code of the country in which this company has declared its nexus.
4.	region	String	• It is a two or three-character ISO region code of the region, state, or province in which this company has declared its nexus.
5.	jurisTypeId	JurisTypeId	• The type of jurisdiction in which the company has declared the nexus.
6.	jurisCode	String	• It is a code that defines jurisdiction under which the company has declared its nexus.
7.	jurisName	String	• The common name of the jurisdiction in which the company declared nexus.
8.	EffectiveDate	DateTime	• It is a date when the nexus is active. If not known, then set to null.
9.	EndDate	DateTime	• It is an end date of the nexus.
10.	shortName	String	• It is a short name of the jurisdiction.
11.	SignatureCode	String	• It is a signature code of the boundary region as defined by Avalara.
12.	stateAssignedNo	String	• It is a state assigned number of the jurisdiction.

Sr. No.	Properties	Data Type	Description
13.	nexusTypeId (DEPRECATED)	NexusTypeId	- It is a type of nexus that the company is declaring. - Use the NexusTaxTypeGroupId instead of nexusTypeId.
14.	Sourcing	Sourcing	It indicates whether the nexus is defined as origin or destination nexus.
15.	hasLocalNexus	Boolean	If True, then you are declaring local nexus within the jurisdiction.
16.	localNexusTypeId	LocalNexusTypeId	- If you are declaring local nexus within the jurisdiction, then it indicates whether you are declaring only a specified list of local jurisdictions, all state-administered local jurisdictions, or all local jurisdictions. - The user define field and can be provide while creating or updating the nexus object.
17.	hasPermanentEstablishment	Boolean	- Set the value to true if your company has a permanent establishment within this jurisdiction. - This field is user-selectable and should be provided when creating or updating a nexus object.
18.	taxId	String	It is a tax identification number under which you declare the nexus.
19.	streamlinedSalesTax	Boolean	- For the United States, this flag indicates whether the particular nexus falls within a U.S. State that participates in the Streamlined Sales Tax program. - For countries other than the US, the flag is null.
20.	CreatedDate	DateTime	It is a date when the record was created.
21.	createdUserId	Integer	It is a user ID of the user who created this record.
22.	modifiedDate	DateTime	It is a date/time that the record was last modified.
23.	modifiedUserId	Integer	It is a user ID of the user that last modified the record.
24.	nexusTaxTypeGroup	String	- The type of nexus that this company is declaring.Replaces NexusTypeId. - Use '/api/v2/definitions/taxtypegroups' for a list of tax type groups.
25.	taxAuthorityId	Integer	- A unique ID number of the tax authority that is associated with this nexus. - This field is defined by Avalara. All the Avalara defined fields must match an Avalara defined nexus object by calling ListNexus.

LocationSettingModel

Description

- It describes the response of a local jurisdiction's question about the location.

Properties

The following are the LocationSettingModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	questionId	Integer	It is a unique ID number of the location question answered.
2.	Value	String	It is a response provided by the user.

SubscriptionModel

Description

- It represents a service to which the account has subscribed.

Properties

The following are the SubscriptionModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	id	Integer	It is a unique ID number of the subscription.
2.	accountId	Integer	It is a unique ID number of the account the subscription belongs.
3.	subscriptionTypeId	Integer	- It is a unique ID number of the service that the account is subscribed. - If it is provided, then the subscription description is ignored.
4.	subscriptionDescription	String	- It is a description of the service to which the account is subscribed. - You can either provide the subscription type ID or the subscriptionDescription but not both. - If the subscription type ID is provided, then the information is ignored, and the field will be updated with the information from subscription type ID.
5.	effectiveDate	DateTime	It is a date when the subscription gets active.
6.	EndDate	DateTime	If the subscription has ended or will end, the date indicates when the subscription ends.
7.	createdDate	DateTime	It is a date when the record was created.
8.	createdUserId	Integer	It is a user ID of the user who created the record.
9.	modifiedDate	DateTime	It is a date/time that the record was last modified.
10.	ModifiedUserId	Integer	It is a user ID of the user that last modified the record.

SettingModel

Description

- This object is used to keep track of custom information about a company.
- The company settings system is a metadata system that you can use to store extra information about a company.
- Your integration or connector could use this data storage to keep track of preference information, reminders, or any other storage that would need to persist even if the customer uninstalls your application.
- A setting can refer to any type of data you need to remember about the company object.
- While creating this object, you may define your own set, name, and value parameters.
- Choose a set name which begins with X to indicate an extension and to define your own values.

Properties

The following are the SettingModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	id	Integer	• It is a unique ID number of the setting.
2.	companyId	Integer	• It refers to the unique ID number of the company.
3.	set	String	• It is a user-defined "set" containing this setting. • Avalara defines some sets that cannot be changed. To create your own set, choose a set name that begins with X, to indicate that it is an extension value. • It is recommended that you must choose a set name that clearly identifies your application, and then store data within the name/value pairs within that set. For Example, if you were creating an application called MyApp, you can choose to create a set name as X-MyCompany-MyApp.
4.	names	String	• A user-defined "name" for the name-value pair.
5.	Value	String	• It is a value of the name-value pair.

EntityUseCodeModel**Description**

- It represents a code describing the intended use for a product that may affect its taxability.

Properties

The following are the SettingModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	code	String	• It is an Avalara-recognized Entity/ Use Code for this definition Entity/Use Code value. For Example, A.
2.	name	String	• It is the name of the Entity/Use Code Entity/Use Code name. For Example, Federal Government.EDERAL GOV
3.	description	String	• It is an Entity/Use Code description. describing the meaning of this use code.
4.	validCountries	List<String>	• It is a list of countries where this Entity/Use Code is valid list of the countries.

ContactModel

Description

- A contact person for a company.

Properties

The following are the ContactModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Id	Integer	• It is a unique ID number of the contact.
2.	companyId	Integer	• It is a unique ID number of the company to which the contact belongs.
3.	ContactCode	String	• It is a unique code of the contact.
4.	firstName	String	• The first or given name of this contact.

ExemptionReasonModel

Description

- It is an exemption reason that defines why a certificate allows a customer to be exempt for the tax calculation.
- For a full list of defined exemption reasons, call the ListCertificateExemptionReasons API.

Properties

The following are the ExemptionReasonModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Id	Long	• It is a unique ID number representing the exemption reason.
2.	Name	String	• It describes the exemption reason.

UPCModel**Description**

- It is a Universal Product Code object defined for the company.

Properties

The following are the UPCModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Id	Integer	The unique ID number for the UPC code.
2.	CompanyId	Integer	The unique ID number of the company to which this UPC code belongs.
3.	upc	String	It is a 12-14-character Universal Product Code, European Article Number, or Global Trade Identification Number.
4.	legacyTaxCode	String	The legacyTaxCode applies to any product that is sold with the UPC code.
5.	description	String	It is a description of the product to which the UPC code applies.
6.	EffectiveDate	DateTime	It shows the date on which the UPC code was active.
7.	endDate	DateTime	It an endDate of the UPC code was active.
8.	Usage	Integer	It is a usage identifier for the UPC code.
9.	isSystem	Integer	It is a flag that indicates whether the UPC code is attached to the AvaTax system or to a company.
10.	createdDate	DateTime	It is a date when the record was created.
11.	createdUserId	Integer	It is a user ID of the user who created the record.
12.	modifiedDate	DateTime	It is a date/time that the record was last modified.
13.	modifiedUserId	Integer	It is a user ID of the user that last modified the record.

PoNumberModel**Description**

- It represents a purchase order number for a transaction.

Properties

The following are the PoNumberModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	id	Long	It is unique ID number for a transaction.
2.	PoNumber	String	It is a purchase order number for a transaction.

ExposureZoneModel**Description**

- It is an information about a physical area or zone in which a certificate can apply.
- An exposure zone for an exemption certificate is generally a tax authority such as a state, country, or local government entity.

Properties

The following are the ExposureZoneModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	id	Long	• It is a unique ID number of the exposure zone.
2.	CompanyId	Long	• It is a unique ID number of the AvaTax company.
3.	Name	String	• It is a short name of the exposure zone, suitable for use in a drop-down list.
4.	tag	String	• It is a tag indicating the exposure zone.
5.	description	String	• It is a description of the exposure zone suitable to use as a tooltip or help text.
6.	created	DateTime	• It is a date when the record was created.
7.	modified	DateTime	• It is a date/time when the record was last modified.
8.	Region	String	• It is a two or three-character ISO 3166 region, province, or state name of the exposure zone.
9.	Country	String	• It is a two-character ISO 3166 county code for the country component of the exposure zone.

ItemModel**Description**

- It represents an item in your company product catalog.

Properties

The following are the ItemModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Id	Integer	• It is a unique ID number of the item.
2.	companyId	Integer	• It is a unique ID number of the company that contains the item.
3.	itemCode	String	• It is a unique code that represents the item.
4.	taxCodeId	Integer	• It is a unique ID number of the tax code that is applied when selling the item.

Sr. No.	Properties	Data Type	Description
			When creating or updating an item, you can either specify the tax code ID number or the tax code string; you do not need to specify both values.
5.	taxCode	String	- It is a unique code string of the tax code that is applied when selling the item. - When creating or updating an item, you can either specify the tax code ID number or the tax code string; you do not need to specify both values.
6.	Description	String	It is a description of the item.
7.	CreatedDate	DateTime	It is a date when the record was created.
8.	createdUserId	Integer	It is a user ID of the user who created the record.
9.	ModifiedDate	DateTime	It is a date/time that the record was last modified.
10.	modifiedUserId	Integer	It is a user ID of the user that last modified the record.

IsoCountryModel

Description	It represents an ISO 3166 recognized country.
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Properties

The following are the IsoCountryModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Code	String	It is a two-character ISO 3166 country code.
2.	Name	String	It is a full name of the country.
3.	isEuropeanUnion	Boolean	If True, then the country is a member of the European Union.

Description

- It represents an ECMS record, and it is used internally by AvaTax to track information about exemptions.

Properties

The following are the ECMSDetailModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	exemptCertDetailId	Integer	• It is a unique, system -assigned identifier of an ExemptCertDetail record.detail id
2.	exemptCertId	Integer	• The calc_id associated with a certificate in CertCapture.exempt certificate id
3.	StateFips	String	• A State FIPS (Federal Information Processing Standard) value associated towith the certificate.State FIPS
4.	Region	String	• It is name or ISO 3166 code identifying the region within the country. For Example, CA.Name or ISO 3166 code identifying the region within the country.
5.	idNo	String	• It is a customer Tax IdID Number (tax_number) associated with a certificate. This is same as the exemptionNo applied in their Ttransactions.ID number
6.	Country	String	• It is a Nname or ISO 3166 code identifying the country. For Example, US.Country that this exempt certificate is for
7.	endDate	DateTime	• It is an end date of the exemption certificate.
8.	IdType	String	• It is an ID type of the exemption certificate.
9.	isTaxCodeListExclusionList	Byte	• It specifies whether the tax code list is an exclusion list.
10.	taxCodes	List<EcmsDetailTaxCodeModel> (Link to EcmsDetailTaxCodeModel)	• It is a list of tax codes that is associated with the exemption certificate detail.

TaxCodeModel**Description**

- It represents a tax code that can be applied to items on a transaction.
- A tax code can have specific rules for specific jurisdictions that change the tax calculation behavior.

Properties

The following are the TaxCodeModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Id	Integer	• It is a unique ID number of the tax code.
2.	companyId	Integer	• It is a unique ID number of the company that contains the tax code.
3.	taxCode	String	• It is a code string that identifies the tax code.
4.	taxCodeTypeId	String	• It is a type of the tax code.
5.	description	String	• It is a description for a tax code.
6.	parentTaxCode	String	• If the tax code is a subset of a different tax code, then it identifies the parent code.
7.	isPhysical	Boolean	• If True, then the tax code type refers to a physical object. • It is a Read only field.
8.	goodsServiceCode	Integer	• The Avalara goods and service codes are represented by this tax code.
9.	entityUseCode	String	• The Avalara Entity/Use Code is represented by this tax code.
10.	isActive	Boolean	• If True, then the tax code is active and can be used in transactions.
11.	isSSTCertified	Boolean	• If True, then the tax code has been certified by the Streamlined Sales Tax governing board.
12.	createdDate	DateTime	• It is a date when this record was created.
13.	createdUserId	Integer	• It is a user ID of the user who created the record.
14.	modifiedDate	DateTime	• It is a date/time that the record was last modified.
15.	modifiedUserId	Integer	• It is a user ID of the user that last modified the record.

ECMSDetailTaxCodeModel**Description**

- It represents the ECMS tax code ID.

Properties

The following are the ECMSDetailTaxCodeModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	exemptCertDetailTaxCodeId	Integer	• It is an dID of the exempt certificate detail tax coded of the exempt certificate detail tax code.
2.	ExemptCertDetailId	Integer	• It is an exempt certificate detail ID.
3.	taxCodeId	Integer	• It is a tax code ID.

LinkCustomerModel**Description**

- It represents a customer to whom you sell products and/or services.

Properties

The following are the LinkCustomerModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Customers	List<String>	• It is an array of customerCodes that are exempted by the certificate.

InvoiceMessage**Description**

- It represents the message to be displayed on an invoice.

Properties

The following are the InvoiceMessage properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Content	String	• The content of the invoice message.
2.	lineNumbers	List<String>	• The applicable tax line numbers and codes.

VoidTransactionModel

Description

- It is a request to void a previously created transaction.

Properties

The following are the VoidTransactionModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Code	VoidReasonCode	The user needs to specify the reason for voiding or canceling the transaction.

CertExpressInvitationModel

Description

- It is a request to create certexpress invitation.

Properties

The following are the CertExpressInvitationModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	Id	Integer	A unique ID number representing this CertExpress invitation.
2.	companyId	Integer	The unique ID number of the AvaTax company that sent this invitation.
3.	recipient	String	The email address to which this invitation was sent. If this invitation was presented as a weblink, this value will be null.
4.	customerCode	String	The unique code of the customer that received this invitation. Note: This field is case sensitive. To have exemption certificates apply, this value should be the same as the one passed to create a customer.
5.	customer	CustomerModel	The customer who received this invitation.
6.	coverLetter	CoverLetterModel	The attached cover letter object to this request.
7.	emailStatus	String	The status of the emails associated with this invitation. If this invitation was sent via email, this value will change to `Sent` when the email message has been sent.
8.	coverLettersOnly	Boolean	True if this invitation contained a cover letter only.
9.	exposureZones	List<Integer>	When an invitation is sent, it contains a list of exposure zones for which the customer is invited to upload their exemption certificates. This list contains the ID numbers of the exposure zones identified. For a list of exposure zones, please call `ListCertificateExposureZones`.
10.	exemptReasons	List<Integer>	The list of exemption reasons identified by this CertExpress invitation. For a list of reason codes, please call `ListCertificateExemptReasons`.

Sr. No.	Properties	Data Type	Description
11.	deliveryMethod	CertificateRequest DeliveryMethod	Indicates the method that was used to deliver this CertExpress invitation.
12.	message	String	The custom message delivered with this invitation.
13.	invitationDate	DateTime	The date of the invitation.
14.	requestLink	String	The web link (URL) that a customer can click on or visit to begin using this CertExpress invitation. This value is only usable if the status of this invitation is `Ready` and the request was created with type `Download`. NOTE: This link usually takes a few minutes to be available.

CertExpressInvitationStatusModel

Description	The status of the CertExpress invitation for this customer.
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Properties

The following are the CertExpressInvitationStatusModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	status	CertExpressInvitationStatus	The status of the CertExpress invitation for this customer.
2.	invitation	CertExpressInvitationModel	The CertExpress invitation for the customer. If you specified an email address in the invitation request, this invitation will be sent via email. Otherwise, you are expected to direct the customer using a hyperlink directly in your application.

CreateCertExpressInvitationModel

Description	None
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Properties

The following are the CreateCertExpressInvitationModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	recipient	String	If the value of `deliveryMethod` is set to `Email`, please specify the email address of the recipient for the request.
2.	coverLetterTitle	String	If this invitation is sent via email or download, please specify the cover letter to use when building this invitation. For a list of cover letters, please call `ListCoverLetters`.
3.	exposureZones	List<Integer>	You may optionally specify a list of exposure zones to request in this CertExpress invitation. If you list more than one exposure zone, the customer will be prompted to provide an exemption certificate for each one. If you do not provide a list of exposure zones, the customer will be prompted to select an exposure zone. For a list of available exposure zones, please call `ListCertificateExposureZones`.
4.	exemptReasons	List<Integer>	You may optionally specify a list of exemption reasons to pre-populate in this CertExpress invitation. If you list exemption reasons, the customer will have part of their form already filled in when they visit the CertExpress website. For a list of available exemption reasons, please call `ListCertificateExemptReasons`.
5.	deliveryMethod	CertificateRequestDeliveryMethod	- Specify the type of invitation. CertExpress invitations can be delivered via email, web link, or facsimile. - If you specify `Email`, the invitation will be delivered via email. Please ask the customer to ensure that - If you specify `Fax`, the invitation will be sent via fax to the customer's fax number on file. - If you specify `Download`, the invitation will be prepared as a web link that you can display to the customer.

CoverLetterModel**Description**

- It represents a cover letter for cert express invitation.

Properties

The following are the CoverLetterModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	id	CertExpressInvitationStatus	• A unique ID number representing a cover letter sent with a CertExpress invitation.
2.	companyId	CertExpressInvitationModel	• The unique ID number of the AvaTax company that received this certificate.
3.	title	String	• The title used when sending the cover letter.
4.	subject	String	• The subject message used when sending the cover letter via email.
5.	description	String	• A full description of the cover letter's contents and message.
6.	createdDate	DateTime	• The date when this record was created.
7.	modifiedDate	DateTime	• The date/time when this record was last modified.
8.	active	Boolean	• Is this cover letter active.
9.	pageCount	Integer	• How many pages this cover letter encompasses.
10.	templateFilename	String	• The file name of the cover letter template.
11.	version	Integer	• The version number of the template.

FetchResultCertInvitationStatusModel**Description**

- It represents cert express invitation response.

Properties

The following are the FetchResultCertInvitationStatusModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	value	List<CertExpressInvitationStatusModel>	• It represents cert express invitation response.

ProvisionStatusModel

Description

- It represents a company's exemption certificate status.

Properties

The following are the ProvisionStatusModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	status	String	- The status of exemption certificate setup for this company. - Provisioned, InProgress, NotProvisioned.
2.	accountId	Integer	The accountId of the company represented by this status.
3.	companyId	Integer	The AvaTax company represented by this status.

6.3. Enums

BoundaryLevel

Description

- The boundary level at which this address was validated.

Properties

The following are the address model properties and their information:

Sr. No.	Properties	Description
4.	Address	It is a Street address precision.
5.	Zip9	It is a 9-digit zip code precision.
6.	Zip5	It is a 5-digit zip code precision.

DocumentType

Description

- It is a document type of the transaction to commit. If not provided, then the default document type is SalesInvoice.Document Type for Tax Calculation.

Return

- NA

Properties

The following are the DocumentType properties and their information:

Sr. No.	Properties	Description
1.	SalesOrder	- It represents an estimate, or a sales order, or a quote. - The document type is used before a sale has occurred to estimate the final amount of tax to be paid when the sale is completed.

Sr. No.	Properties	Description
		- For a sales order, the <code>companyCode</code> of the transaction refers to the seller and the <code>customerVendorCode</code> refers to the buyer. - It is a temporary document type and is not saved in AvaTax history.
2.	SalesInvoice	- It Represents a sale that has been finalized. - The document type is used to record a final transaction and calculate the final tax amount with all necessary information. - For a sales invoice, the <code>companyCode</code> of the transaction refers to the seller and the <code>customerVendorCode</code> refers to the buyer. - It is a permanent document and is recorded in AvaTax.
3.	PurchaseOrder	- It Represents an estimate, or a purchase order, or a quote. - The document type is used before a sale has occurred to estimate the final amount of tax to be paid when the sale is completed. - For a purchase order, the <code>companyCode</code> of the transaction refers to the buyer and the <code>customerVendorCode</code> refers to the seller. - It is a temporary document type and is not saved in AvaTax history.
4.	PurchaseInvoice	- It represents a sale that has been finalized. - The document type is used to record a final transaction and calculate the final tax amount with all necessary information. - For a purchase invoice, the <code>companyCode</code> of the transaction refers to the buyer and the <code>customerVendorCode</code> refers to the seller. - It is a permanent document and is recorded in AvaTax.
5.	ReturnOrder	- It represents an estimate of tax to be refunded if a refund or return is processed. - The document type is used before a customer chooses to request a refund for a previous sale, and it estimates the final amount of tax to be refunded when the refund is completed. - For a return order, the <code>companyCode</code> of the transaction refers to the seller who is giving the refund and the <code>customerVendorCode</code> refers to the buyer who is requesting the refund. - It is a temporary document type and is not saved in AvaTax history.
6.	ReturnInvoice	- It represents a refund that has been finalized. - The document type is used when a customer chooses to request a refund for a previous sale, and it records the final amount of tax that has been refunded when the refund is completed. - For a return invoice, the <code>companyCode</code> of the transaction refers to the seller who is giving the refund and the <code>customerVendorCode</code> refers to the buyer who is requesting the refund. - It is a permanent document, and it is recorded in AvaTax. Please note that many jurisdictions around the United States have special rules about how to report refunds on a sales tax filing, and some refund transactions may be unable to file immediately due to credit netting rules.
7.	InventoryTransferOrder	- It represents a proposed movement of inventory from one jurisdiction to another. - The document type is used when physical goods are shipped from one jurisdiction to another, and it may cause updates in the tax liability for various jurisdictions.

Sr. No.	Properties	Description
		- For an inventory transfer invoice, the companyCode of the transaction refers to the owner of the inventory that will be moved from one location to another. - It is a temporary document type and is not saved in AvaTax history.
8.	InventoryTransferInvoice	- It represents a movement of inventory from one jurisdiction to another that has been finalized. - The document type is used when physical goods are shipped from one jurisdiction to another, and it may cause updates in the tax liability for various jurisdictions. - For an inventory transfer invoice, the companyCode of the transaction refers to the owner of the inventory that will be moved from one location to another. - It is a permanent document and is recorded in AvaTax.
9.	ReverseChargeOrder	- It represents a VAT reverse charge mechanism transaction as defined in the European Union. - The reverse charge transaction moves the responsibility for VAT reporting from the seller to the buyer and is intended to allow simplification of commerce and reduce the reporting requirements for sellers that deliver goods and services into the single market. - It is a temporary document type and is not saved in AvaTax history.
10.	ReverseChargeInvoice	- It represents a VAT reverse charge mechanism transaction as defined in the European Union. - The reverse charge transaction moves the responsibility for VAT reporting from the seller to the buyer and is intended to allow simplification of commerce and reduce the reporting requirements for sellers that deliver goods and services into the single market. - It is a permanent document and is recorded in AvaTax.
11.	CustomsInvoice	- Represents a VAT "Customs Invoice". - The user will enter this DocType with a TaxOverride when the VAT and ImportDuty are calculated by Customs at the border, and then these documents are entered later into AvaTax to create the balancing entries for VAT and Import Duty. - This is a permanent document and is recorded in AvaTax.
12.	CustomsOrder	- Represents a VAT "Customs Order". - The user will enter this DocType with a TaxOverride when the VAT and ImportDuty are calculated by Customs at the border, and then these documents are entered later into AvaTax to create the balancing entries for VAT and Import Duty. - This is a temporary document type and is not saved in tax history.
13.	Any	- This value is used to query documents. - You can specify the type "Any" in some cases to permit the system to find any document matching other criteria.

TaxOverrideTypeId

Description	It is a TaxOverrideTypeId for the tax calculation.
Return	NA

Properties

The following are the TaxOverrideTypeId properties and their information:

Sr. No.	Properties	Description
1.	None	No override has been applied.
2.	TaxAmount	The tax was overridden by the client.
3.	Exemption	Entity/Use Code and exemption was ignored. (For example For Example, consumed item).
4.	TaxDate	Only the tax date was overridden.
5.	AccruedTaxAmount	It is used to support Consumer Use Tax.
6.	DeriveTaxable	It derives the taxable amount from the tax amount.
7.	OutOfHarbor	It is for the documents for which AvaTax service is not used for tax calculation but for reporting / return purposes the documents are passed into the AvaTax.
8.	VendorChargedTax	This describes the vendor-charged tax.

VoidReasonCode

Description	- It specifies the reason for voiding or cancelling the transaction. - Specify the reason DocVoided to void the transaction. If no void reason is provided, then the void command will fail.
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Properties

The following are the VoidReasonCode properties and their information:

Sr. No.	Properties	Description
1.	Unspecified	If selected, then the document status will be set to unspecified reason.
2.	PostFailed	The document status will be changed to unposted.
3.	DocDeleted	If committed, document status will be changed to Cancelled.
4.	DocVoided	The document has been voided, and DocStatus will be set to Cancelled.
5.	AdjustmentCancelled	The adjust tax operation has been canceled. The adjustment will be reversed.

Description	If a tax override was applied to the transaction, and then it indicates what type of tax override was applied.
Return	NA

Properties

The following are the TaxOverrideType properties and their information:

Sr. No.	Properties	Description
1.	None	AvaTax calculated the tax for the transaction and no override occurred.
2.	TaxAmount	- AvaTax calculated tax for the transaction, but the final tax amount on the transaction was determined outside of AvaTax to see the tax amounts determined by AvaTax, at the taxCalculated field. - The taxAmount field is used to see the tax amounts determined by the customer's outside tax engine. - The behavior can also occur when a customer requests a refund. - For refunds calculated using the RefundTransaction API, AvaTax will ensure that the exact tax charged to the customer is refunded to the customer using a tax amount override.
3.	Exemption	Entity exemption was ignored. For Example, consumed item.
4.	TaxDate	- AvaTax was instructed to calculate this transaction using the tax rules that were in effect on a different day than the transaction occurred. - It occurs during the refunds. - If the customer attempts to return a product without a receipt that shows the exact tax amount paid, then the AvaTax can calculate tax on the date when they believed that the product was purchased.
5.	AccruedTaxAmount	It is used to support Consumer Use Tax.
6.	DeriveTaxable	It is used to derive the taxable amount from the tax amount.
7.	OutOfHarbor	It is for the documents for which AvaTax service is not used for tax calculation but for reporting / return purposes the documents are passed into the AvaTax.
8.	TaxAmountByTaxType	To support Tax Override Amount with Tax Type feature.
9.	VendorChargedTax	This describes the vendor-charged tax.

TaxDebugLevel

Description	If the user wants to request additional debug information from the transaction, then specify a level higher than the Normal property.
Return	NA

Properties

The following are the TaxDebugLevel properties and their information:

Sr. No.	Properties	Description
1.	Normal	It is a normal level of debugging information when creating a tax transaction.
2.	Diagnostic	It is an additional diagnostic information when creating a tax transaction.

AdjustmentReason

Description	A reason code indicates the transaction adjustment reason.
Return	NA

Properties

The following are the AdjustmentReason properties and their information:

Sr. No.	Properties	Description
1.	NotAdjusted	It is a transaction that has not been adjusted.
2.	SourcingIssue	A sourcing issue existed which caused the transaction to be adjusted.
3.	ReconciledWithGeneralLedger	The transaction was adjusted to reconcile it with a general ledger.
4.	ExemptCertApplied	The transaction was adjusted after an exemption certificate was applied.
5.	PriceAdjusted	The transaction was adjusted when the price of an item changed.
6.	ProductReturned	The transaction was adjusted due to a product return.
7.	ProductExchanged	The transaction was adjusted due to a product exchange.
8.	BadDebt	The transaction was adjusted due to bad or uncollectable debt.
9.	Other	The transaction was adjusted for another reason not specified.

DocumentStatus

Description	It shows the status of the transaction.
Return	NA

Properties

The following are the DocumentStatus properties and their information:

Sr. No.	Properties	Description
1.	Temporary	The temporary document which is not saved (SalesOrder, PurchaseOrder). This document has not been recorded to AvaTax.
2.	Saved	- Saved document (SalesInvoice or PurchaseInvoice) ready to be posted. - The status indicates that the transaction has been saved to AvaTax but is not available to be reported on a tax filing and has not yet been verified by a business process that posts transactions. - Use the VerifyTransaction or SettleTransaction to mark the transaction as Posted. - Use the CommitTransaction or SettleTransaction to mark the transaction as Committed - Use the AdjustTransaction, CreateOrAdjustTransaction or VoidTransaction to adjust or void the transaction.
3.	Posted	It is a posted document that has not committed, indicates the status of the transaction has been saved to AvaTax and has been verified by a business process that posts transactions but is not ready to report on a tax filing.
4.	Committed	- A posted document that has been committed and indicates the status of the transaction has been saved to AvaTax and can be reported on a tax filing. - If you use Avalara's Managed Returns Service, then these transactions will be captured and reported on a tax return. - When it occurs, the transaction's locked flag will be set to true. Once the transaction is locked, no further changes may occur. - If the transaction has not been locked, you can adjust or void the transaction using AdjustTransaction, CreateOrAdjustTransaction, or VoidTransaction.
5.	Cancelled	- A document that has been cancelled and indicates the status of the transaction has been cancelled or voided. - When a transaction has been cancelled, it is considered to no longer exist. - User can create a new transaction with the same code.
6.	Adjusted	- An older version of a document that has been adjusted. - When you call AdjustTransaction, AvaTax preserves a record of the old document as well as a record of the new document. - The old document is changed to the status Adjusted, and the new document is created in the status you requested.
7.	Queued (DEPRECATED)	It is a document which is queued for later processing.
8.	PendingApproval	It indicates that the document requires approval from a government agency before it can be finalized.

Sr. No.	Properties	Description
		- Some governments require active verification of transactions before they can be finalized. - When AvaTax detects that a transaction must be verified with a government agency, then the transaction will be flagged as PendingApproval until the government certifies the transaction.
9.	AnyStatus (DEPRECATED)	- It represents "a document in any status" while searching. - Search using the [Filtering in REST] (/avatax/filtering-in-rest/) documentation.

JurisdictionType

Description	The type of the jurisdiction in which this tax detail applies.
Return	NA

Properties

The following are the JurisdictionType properties and their information:

Sr. No.	Properties	Description
1.	Country	It represents a country code that is recognized by ISO 3166.
2.	Composite	It is an administrative jurisdiction representingValue for the Composite Jurisdiction the composite value.
3.	State	It represents an administrative jurisdiction within the country.
4.	County	A smaller administrative jurisdiction than a state/region/province, but typically larger than a city. Not all countries have the county jurisdictions.
5.	City	It is an administrative jurisdiction representing a single city.
6.	Special	It is an administrative jurisdiction created for the purpose of taxing transactions within a boundary.

JurisTypeId (Deprecated)

Description	- The type of jurisdiction in which the company has declared its nexus. - Use the jurisdictionTypeId instead of JurisTypeId.
Return	NA

Properties

The following are the JurisTypeId properties and their information:

Sr. No.	Properties	Description
1.	STA	It is a state value considered for the tax calculation.
2.	CTY	It is a county value considered for the Tax Calculation.
3.	CIT	It is a city value considered for the tax calculation.
4.	STJ	It is a special type of value considered for the tax calculation.
5.	CNT	It is country value considered for the tax calculation.

RateType

Description	The RateType for tax calculation is the class of possible RateType values.
Return	NA

Properties

The following are the RateType properties and their information:

Sr. No.	Properties	Description
1.	ReducedA	The ReducedA is a ratetype response from the AvaTax.
2.	ReducedB	The ReducedB is a ratetype response from the AvaTax.
3.	Food	The Food is a ratetype response from the AvaTax.
4.	General	The General is a ratetype response from the AvaTax.
5.	IncreasedStandard	The IncreasedStandard is a ratetype response from the AvaTax.
6.	LinenRental	The LinenRental is a ratetype response from the AvaTax.
7.	Medical	The Medical is a ratetype response from the AvaTax.
8.	Parking	The Parking is a ratetype response from the AvaTax, when it matches with Rate Type Tax Code -- P.

Sr. No.	Properties	Description
9.	SuperReduced	The SuperReduced is a ratetype response from the AvaTax, when it matches with Rate Type Tax Code -- Q.
10.	ReducedR	The ReducedR is a ratetype response from the AvaTax, when it matches with Rate Type Tax Code -- R.
11.	Standard	The Standard is a ratetype response from the AvaTax, when it matches with Rate Type Tax Code -- S.
12.	Zero	The Zero is a ratetype response from the AvaTax, when it matches with the Rate Type Tax Code -- Z.
13.	Services	The Services is a ratetype response from the AvaTax, when it matches with the RateType Tax Code – X.
14.	Leasing	The Leasing is a ratetype response from the AvaTax, when it matches with the RateType Tax Code – T.
15.	LeasingLow	The Leasing is a ratetype response from the AvaTax, when it matches with the RateType Tax Code – U.
16.	Construction	The Construction is a ratetype response from the AvaTax, when it matches with the RateType Tax Code – N.

ResolutionQuality

Description	The resolution quality of the geospatial coordinates.
Return	NA

Properties

The following are the ResolutionQuality properties and their information:

Sr. No.	Properties	Description
1.	NotCoded	The location was not geocoded.
2.	External	The location was already geocoded on the request.
3.	CountryCentroid	It is an Avalara-defined country centroid.
4.	RegionCentroid	It is Avalara-defined state / province centroid.
5.	PartialCentroid	The geocoded at a level coarser than a PostalCentroid1.
6.	PostalCentroidGood	It is a standard postal code (zip5 in US, left three in CA), etc.
7.	PostalCentroidBetter	The better postal code (zip7 in US).
8.	PostalCentroidBest	The best postal code (zip9 in US, complete postal code elsewhere).

Sr. No.	Properties	Description
9.	Intersection	It indicates theN nearest intersection points of the location or street.
10.	Interpolated	- It indicates that the returned result reflects an approximation (usually on a road) interpolated between two precise points (such as intersections). - The interpolated results are generally returned when the rooftop geocodes are unavailable for a street address.
11.	Rooftop	Assumed to be rooftop level, non-interpolated.
12.	Constant	Pulled from a static list of geocodes for specific jurisdictions.

TaxRuleTypeId

Description	- This type of value determines the behavior of the tax rule. - You can specify that the rule controls the product's taxability or exempt / nontaxable status, the product's rate. - For Example, if you have been granted an official ruling for the product's rate that differs from the official rate or other types of behavior.
--------------------	--

Properties

The following are the TaxRuleTypeId properties and their information:

Sr. No.	Properties	Description
1.	RateRule	It is reserved for Avalara internal usage.
2.	RateOverrideRule	- It specifies a rule that changes the rate of tax for a specified set of criteria. - The rule can apply to one specific type of product (identified by TaxCode) or all products (identified by a null TaxCode).
3.	BaseRule	- It specifies a rule that changes the tax base of a specific product. - This rule can apply to one specific type of product (identified by TaxCode) or all products (identified by a null TaxCode).
4.	ExemptEntityRule	- It indicates that a particular product is exempt from a tax due to an exempt usage reason. - The amounts exempted by this rule are stored in the exemptAmount field on the transaction.
5.	ProductTaxabilityRule	- It specifies a rule that changes whether a product is taxable or not. - This rule must be applied to a specific type of product. If you attempt to create a product taxability rule while leaving the TaxCode and TaxCodeId fields empty, then you will get an error. - A value field of 1 means that this tax code is taxable; 0 means it is nontaxable. The amounts that are considered not taxable according to this rule are stored in the nonTaxableAmount column in a transaction. - This type of rule can also determine the rate type for a product or to apply a cap or threshold.

Sr. No.	Properties	Description
6.	NexusRule	It is reserved for Avalara internal usage.
7.	RateCapRule	RateCapRule tax rule type
8.	TaxOverrideRule	TaxOverrideRule tax rule type id
9.	FeeRule	FeeRule tax rule type id
10.	OtherRule	OtherRule tax rule type id

Sourcing

Description	The types of sourcing to which the tax calculation rule apply.
--------------------	--

Properties

The following are the Sourcing properties and their information:

Sr. No.	Properties	Description
1.	Mixed	Mixed sourcing, for states that do both origin and destination calculation.
2.	Destination	Apply the sales tax to the products based on the point of destination (Ship To address).
3.	Origin	Apply the sales tax to the products based on the point of origin (Ship From address).

ServiceMode**Description**

- It specifies whether the tax calculation is handled Local, Remote, or Automatic (default). This only applies when using an AvaLocal server.

Properties

The following are the ServiceMode properties and their information:

Sr. No.	Properties	Description
1.	Automatic	• It Automatically uses local or remote (default) automatically.
2.	Local	• It specifies a local server only.
3.	Remote	• It specifies a remote server only.

CertExpressInvitationStatus**Description**

- It specifies the status of cert express invitation status.

Properties

The following are the CertExpressInvitationStatus properties and their information:

Sr. No.	Properties	Description
1.	InProgress	• Cert invitation is in in progress state.
2.	Ready	• Cert invitation is in ready state.

CertificateRequestDeliveryMethod**Description**

- It specifies the delivery method for certificate request.

Properties

The following are the CertificateRequestDeliveryMethod properties and their information:

Sr. No.	Properties	Description
1.	Email	• The invitation will be delivered via email
2.	Fax	• The invitation will be sent via fax to the customer's fax number on file
3.	Download	• The invitation will be prepared as a web link that you can display to the customer.

7.0 AddressCategoryId

Description	It indicates the type of place of business represented by the location.
--------------------	---

Properties

The following are the AddressCategoryId properties and their information:

Sr. No.	Properties	Description
1.	Storefront	It is an address that refers to a storefront location.
2.	MainOffice	It is an address refers to a main office of the company.
3.	Warehouse	It is an address refers to a warehouse or other non-public location.
4.	Salesperson	It is an address refers to a location for a single salesperson.
5.	Other	It is an address is a type which does not reflect in the other lists.
6.	SellerRemitsTax	- The marketplace vendor does not collect and remit tax for the transactions linked to this location. - Use this option if you use a marketplace vendor to manage your transactions, then it is your company's responsibility to collect and remit the taxes on transactions linked to this site.
7.	MarketplaceRemitsTax	- The marketplace vendor collects and remits the collected tax for the transactions linked to this location. - Use this option if the marketplace already pays the collected sales and uses taxes. - If this option is selected, then the transactions connected to this location will be considered as already filed and will be reported as amounts already paid on each sales tax return.
8.	NonPhysical	The address refers to the company's mailing address, which is not a physical location.

AddressTypeId

Description	It indicates whether the location is a physical place of business or a temporary salesperson location.
--------------------	--

Properties

The following are the AddressTypeId properties and their information:

Sr. No.	Properties	Description
1.	Location	It is a business location type., for example, a store, warehouse, or office.
2.	Salesperson	A person's address who performs sales tasks for the company remotely from an office.
3.	Marketplace	This is a marketplace vendor location who manages the transactions of the company.

Sr. No.	Properties	Description
		If the location address type is selected as marketplace, then you must choose either 'SellerRemitsTax' or 'MarketplaceRemitsTax' to indicate which business entity is responsible for collecting and remitting tax for linked location.

CertificatePreviewType

Description	It is a type of certificate preview to download.
-------------	--

Properties

The following are the CertificatePreviewType properties and their information:

Sr. No.	Properties	Description
1.	Pdf	It is a File type property of the format Pdf.
2.	Jpeg	It is a File type property of the format Jpeg.

ExemptCertReviewStatusId

Description	Review status for the exempt certificate.
-------------	---

Properties

The following are the ExemptCertReviewStatusId properties and their information:

Sr. No.	Properties	Description
1.	Pending	It is a status of the certificate as certificate review pending.
2.	Accepted	It is a status of the certificate as accepted.
3.	Rejected	It is a status of the certificate as rejected.

ExemptCertStatusId

Description	Status for the exempt certificate.
-------------	--

Properties

The following are the ExemptCertStatusId properties and their information:

Sr. No.	Properties	Description
1.	Inactive	It is an exemption certificate status as inactive.
2.	Active	The exemption is active and is applied to transactions that matches the certificate details.
3.	Expired	It is exemption certificate status as expired.

Sr. No.	Properties	Description
4.	Revoked	When the exemption certificate withdraws or takes back, the status of the certificate is considered as revoked.

ExemptCertTypeId

Description	It is a type of exemption certificate and the permitted values are Blanket and Single.
-------------	--

Properties

The following are the ExemptCertTypeId properties and their information:

Sr. No.	Properties	Description
1.	Blanket	It is an exemption certificate type as Blanked certificate.
2.	SingleUse	It is an exemption certificate type as Single use.

LocalNexusTypeId

Description	It describes the local nexus type ID.
-------------	---

Properties

The following are the LocalNexusTypeId properties and their information:

Sr. No.	Properties	Description
1.	Selected	- It is a user-selectable value and should be provided when creating or updating a nexus object. - It is the specific nexus object declared for the company.
2.	StateAdministered	- The customer declares the nexus in all state administered taxing authorities. - The value only takes effect if you set `hasLocalNexus` = true.
3.	All	- The customer declares the nexus in all local taxing authorities. - The value only takes effect if you set `hasLocalNexus` = true.

NexusTypeId

Description	- The type of nexus that the company is declaring. - If you are voluntarily declaring nexus in a jurisdiction, then you should select SalesOrSellersUseTax for your nexus type option. - It allows you to calculate tax correctly whether you are selling in-state or shipping from an out-of-state location. - If you are legally obligated to declare nexus due to physical presence or another sufficient nexus, then you should select SalesTax. - It indicates that as a legal requirement, your company must collect and remit full sales tax in this jurisdiction. - If you are participating in the Streamlined Sales Tax program, your SST administrator will select nexus settings for you in all SST jurisdictions. Do not select any SST options by yourself. - This field is user-selectable and should be provided when creating or updating a nexus object.
--------------------	--

Properties

The following are the NexusTypeId properties and their information:

Sr. No.	Properties	Description
1.	None	It indicates no nexus.
2.	SalesOrSellersUseTax	It indicates that the entity is StateAdministered voluntarily collecting tax (default).
3.	SalesTax	It indicates that the entity is required to collect tax in the state.
4.	SSTVolunteer	- It indicates that the entity is registered as a Volunteer in an SST state. - The SST administrator may set this option.
5.	SSTNonVolunteer	- It indicates that the entity is registered as a Non-Volunteer in an SST state. - The SST administrator may set this option.

RoundingLevelId

Description	It indicates whether the company prefers to round amounts at the document level or line level.
--------------------	--

Properties

The following are the RoundingLevelId properties and their information:

Sr. No.	Properties	Description
1.	Line	It is a round amount at the line level.
2.	Document	It is a round amount at the document level

Description	Used to apply tax detail dependency at a jurisdiction level.
-------------	--

Properties

The following are the TaxDependencyLevelId properties and their information:

Sr. No.	Properties	Description
1.	Document	It defines the tax dependency type.
2.	State	It defines tax dependency type when the value matches with the State value.
3.	TaxRegion	It defines tax dependency type when the value matches with the TaxRegion value.
4.	Address	It defines tax dependency type, when the value matches with the Address value.

8.0 PostTaxCalculator

Description	- It contains the methods for posting tax and update it to the Salesforce object. - It reads the AvaTax XML configuration file value and generates the post-tax calculation request from the mapping provided in the AvaTax XML configuration file. - It supports several hooks that can be subscribed to modify the tax calculation request as per the business requirement.
--------------------	---

Properties

The following are the PostTaxCalculator properties and their information:

Sr. No.	Properties	Data Type	Description
1.	headerQueryResult	String	It stores query result of SOQL that is defined under the TaxCalculator, SalesforceObject, Header, and Query node.
2.	lineQueryResult	String	It stores the query result of SOQL that is defined under the TaxCalculator, SalesforceObject, Header, and Query node.

8.1. Methods

executeQuery

Description	- Retrieves Data from Salesforce. - It reads all the Queries in XML config provided for the calling package. Also, keeps it ready for further usage in tax calculation.
Returns	NA
Parameters	
Model	An instance of QueryInput.

postTax

Description	- The post-tax method is responsible for document posting. - It formulates the posting document for the calling package. - It provides the Post Tax Calculation result to the Connector.
Returns	An instance of TransactionModel.
Parameters	
Model	An instance of PostTaxCalculationInput.

PostTaxCalculator

Description	- It initializes an instance of PostTaxCalculator class for the provided Salesforce object. - It reads the AvaTax XML configuration file provided for the calling package.
Returns	An instance of TaxCalculator class.
Parameters	
CallingPackage	It is a name of the package which wants to calculate the tax.
Model	AvaTax Configuration details.

setConfigProviderListener

Description	Hook Method for getting the config before Post Tax Calculation.
Returns	None
Parameters	
Model	Instance of IConfigManagerPost.

8.2. Models**8.2.1. QueryInput****Properties**

The following are the QueryInput properties and their information:

Sr. No.	Properties	Data Type	Description
1.	controller	String	It is a controller for the cancel tax calculation.
2.	optionalParams	Map<String, String>	The data required for creating parameters in the query.
3.	recordId	Id	It is an ID of the Salesforce record.

PostTaxCalculationInput

Description	Post-tax details essential for Committing the document.
-------------	---

Properties

The following are the PostTaxCalculationInput properties and their information:

Sr. No.	Properties	Data Type	Description
1.	commitFlag	VoidReasonCode	The flag to commit document, the value can either true or false
2.	companyCode	String	It specifies the code of the company creating the transaction.
3.	docType	DocumentType	It is an AvaTax document type.
4.	transactionCode	String	It is a transaction code for cancel tax calculation.

9.0 Tax Calculator

Description	- It contains the methods for tax calculation and update it to the Salesforce object. - It reads the AvaTax XML configuration file, and generates the tax calculation request from the mapping provided in the AvaTax XML configuration file. - It supports several hooks that can be subscribed to modify the tax calculation request as per the business requirement.
--------------------	---

Properties

The following are the TaxCalculator properties and their information:

Sr. No.	Properties	Data Type	Description
1.	headerQueryResult	String	It stores the query result of SOQL that is defined under the TaxCalculator, SalesforceObject, Header, and Query node.
2.	lineQueryResult	String	It stores the query result of SOQL that is defined under the TaxCalculator, SalesforceObject, Header, and Query node.

9.1. Methods

TaxCalculator

Description	- It initializes an instance of the TaxCalculator class for the Salesforce object provided. - It reads the AvaTax XML configuration file provided for the calling package.
Returns	An instance of TaxCalculator class.
Parameters	
CallingPackage	It is a name of the package which wants to calculate the tax.
Config	AvaTax Configuration details.

calculateTax

Description	- This method is responsible for tax calculation. - It formulates the tax for the calling package and provides the tax calculation result to the connector.
Returns	An instance of TransactionModel.
Parameters	
Model	An instance of TaxCalculationInput.

setBeforeTaxCalculationListener

Description	The hook method is used for editing the header data of tax request before the tax calculation.
Returns	None
Parameters	
Model	Instance of IBeforeTaxCalculation.

setBeforeTaxValidationListener

Description	Hook method for validations needs to be done before actual Tax Calculation.
Returns	None
Parameters	
Model	Instance of IBeforeTaxValidation.

setConfigProviderListener

Description	Hook method for getting the config before the Tax Calculation.
Returns	None
Parameters	
Model	Instance of IBeforeTaxValidation.

setOverrideRequestLineListener

Description	Hook method for editing the line data of Tax Request before Tax Calculation.
Returns	None
Parameters	
Model	Instance of IOverrideRequestLine.

9.2. Models

9.2.1. TaxCalculationInput

Properties

The following are the TaxCalculationInput properties and their information:

Sr. No.	Properties	Data Type	Description
1.	commitFlag	Boolean	The flag for committing document on AvaTax.
2.	controller	String	It is a name of Salesforce object.
3.	docType	DocumentType	It is an AvaTax Document Type.
4.	optionalParams	String	The data required for creating parameters in the query.
5.	recordId	Id	It is an ID of the Salesforce record.
6.	skipDml	Boolean	It is a Flag to Control DML Operations.

10.0 Tax Calculation Batch Preprocessor

Description	- It contains the methods for tax calculation in batches and update it to the Salesforce object. - It reads the AvaTax XML configuration file, and generates the tax calculation request from the mapping provided in the AvaTax XML configuration file and executes the Tax calculation for line items in batches. - It supports several hooks that can be subscribed to modify the tax calculation request as per the business requirement.
--------------------	---

Properties

The following are the TaxCalculator properties and their information:

Sr. No.	Properties	Data Type	Description
1.	headerQueryResult	String	It stores the query result of SOQL that is defined under the TaxCalculator, SalesforceObject, Header, and Query node.
2.	lineQueryResult	String	It stores the query result of SOQL that is defined under the TaxCalculator, SalesforceObject, Header, and Query node.

10.1. Methods

TaxCalculationBatchPreProcessor

Description	- It initializes an instance of the TaxCalculator class for the Salesforce object provided. - It reads the AvaTax XML configuration file provided for the calling package.
Returns	An instance of TaxCalculator class.
Parameters	
CallingPackage	It is a name of the package which wants to calculate the tax.
Config	AvaTax Configuration details.

calculateTax

Description	- This method is responsible for tax calculation. - It formulates the tax for the calling package and provides the tax calculation result to the connector.
Returns	An instance of TransactionModel.
Parameters	
Model	An instance of TaxCalculationInput.

setCustomBusinessCaseListener

Description	The hook method is used for editing the header data of tax request before the tax calculation.
Returns	None
Parameters	
Model	Instance of ICustomBusinessCase.

10.2. Models**10.2.1. TaxCalculationInput****Properties**

The following are the TaxCalculationInput properties and their information:

Sr. No.	Properties	Data Type	Description
1.	commitFlag	Boolean	The flag for committing document on AvaTax.
2.	controller	String	It is a name of Salesforce object.
3.	docType	DocumentType	It is an AvaTax Document Type.
4.	optionalParams	String	The data required for creating parameters in the query.
5.	recordId	Id	It is an ID of the Salesforce record
6.	itemsPerBatch	Integer	Defaulted to 50, meaning 50 items per batch for Tax calculation

11.0 CancelTaxCalculator

Description	It contains the methods for canceling tax and updating back to the Salesforce object.
--------------------	---

Properties

The following are the CancelTaxCalculator properties and their information:

Sr. No.	Properties	Data Type	Description
1.	headerQueryResult	Map<String,List<SObject>>	It stores the query result of SOQL that is defined under the TaxCalculator, SalesforceObject, Header, and Query node.

11.1. Methods

cancelTax

Description	This The cancel tax method is responsible for Tax Calculation.
Returns	An instance of TransactionModel.
Parameters	An instance of CancelTaxCalculationInput.

CancelTaxCalculator

Description	Initializes an instance of CancelTaxCalculator class for the Salesforce object provided.
Returns	An instance of TaxCalculator class.
Parameters	
CallingPackage	Name of the package which wants to calculate the tax.
Config	AvaTax Configuration details.

11.2. Models

CancelTaxCalculationInput

Description	Cancel Tax details are essential for voiding the document. The tax calculation is affected by the Cancel Tax details.
--------------------	---

Properties

The following are the CancelTaxCalculationInput properties and their information:

Sr. No.	Properties	Data Type	Description
1.	code	VoidReasonCode	It is a type of code used to cancel/void the document.
2.	companyCode	String	It specifies the code of the company creating the transaction.
3.	controller	String	It is a controller for the cancel tax calculation.
4.	docType	DocumentType	It is an AvaTax Document Type.
5.	optionalParams	Map<String, String>	- The data required for creating parameters in the query. - You can specify if any extra parameters need to be sent to the AvaTax Service.
6.	recordId	Id	It is an ID of the Salesforce record.
7.	transactionCode	String	TransactionCode for cancel tax calculation.

ConfigurationBase

Description	NA
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Properties

The following are the ConfigurationBase properties and their information:

Sr. No.	Properties	Data Type	Description
1.	accountId	Integer	It specifies the accountId validation in AvaTax.
2.	appName	String	It specifies the Application Name.
3.	appVersion	String	It specifies the Application Version.
4.	companyCode	String	It specifies the code of the company.
5.	defaultOriginAddress	AddressValidationInfo	It specifies the instance of AddressValidationInfo for the origin address.
6.	environment	AvaTaxEnvironment	AvaTaxEnvironment for tax calculation.

Sr. No.	Properties	Data Type	Description
7.	isAddressValidationEnabled	Boolean	It specifies whether the address validation is enabled.
8.	isDateOverrideAllowed	Boolean	It specifies whether the transaction date can be overridden.
9.	isTaxCalculationEnabled	Boolean	It specifies whether the tax calculation is enabled.
10.	isUPCEnabled	Boolean	It specifies whether the UPC tax code is enabled.
11.	licenseKey	String	It specifies the licenseKey validation in the AvaTax.
12.	password	String	It specifies the password validation in the AvaTax.
13.	restClientName	String	It specifies the Client Name.
14.	returnAddressInUpperCase	Boolean	It returns the validated address result in upper case.
15.	saveTransactionToAvaTax	Boolean	It records the document to the AvaTax.
16.	username	String	It specifies the username validation in the AvaTax.
17.	connectorId	String	It specifies the Connector Id of connectors.
18.	configId	String	It specifies the unique Id (Integration company Id) for the Avalara Portal.
19.	configSectionId	String	It specifies the unique Id for the configurations section on the Avalara Portal.

11.3. Enums

AvaTaxEnvironment

Description	AvaTaxEnvironment for Tax Calculation.
Return	NA

Properties

The following are the AvaTaxEnvironment properties and their information:

Sr. No.	Properties	Description
1.	Sandbox	Development Environment.
2.	Production	Production Environment.

12.0 CCS APIs

CCS APIs are used to create, post, delete, and get Config Id and Config Section Id in the AvaTax Portal from connectors.

12.1. AvaTaxClient

Description	Initializes the REST-based client with the supplied parameters.
Returns	A new object of AvaTaxClient class.
Parameters	
AppName	It is a name of Salesforce package which calls the AvaTax Mapper package. Name of calling application. Mostly it is ERP/Salesforce Package Name to call AvaTax Services.
AppVersion	It is a version of Salesforce package which calls the AvaTax Mapper package. A version of the calling application.
MachineName	It is a name of the machine from which this call is being made. In case of Salesforce, it can be the hostname.
ConfigurationBase	Holds AvaTax configuration details. Tax calculation is affected by these details.

12.2. Methods

getApplicationConfigId

Description	The getApplicationConfigId method is responsible for retrieving the configId for the CCS API.
Returns	An instance of ApplicationModel.

createConfigId

Description	The createConfigId method is responsible for posting the configId for the CCS API.
Returns	An instance of StatusCodeWrapper class.
Parameters	
Model	An instance of AppConfigModel.

updateConfigId

Description	The updateConfigId method is responsible for updating the configId for the CCS API.
Returns	An instance of StatusCodeWrapper class.
Parameters	
Model	An instance of AppConfigModel.

deleteAppConfigId

Description	The deleteAppConfigId method is responsible for deleting the configId for the CCS API.
Returns	An instance of StatusCodeWrapper class.

getConfigSectionId

Description	The getConfigSectionId method is responsible to get the config sectionId for the CCS API.
Returns	An instance of AppConfigGetSectionIdModel class.

postConfigSecId

Description	The postConfigSecId method is responsible for posting the config sectionId for the CCS API.
Returns	An instance of StatusCodeWrapper class.
Parameters	
String	A serialized JSON section in the form of a String.

deleteAppConfigSectionId

Description	The deleteAppConfigSectionId method is responsible for deleting the config sectionId for the CCS API.
Returns	An instance of StatusCodeWrapper class.

12.3. Models

ApplicationModel

Description	- It contains the method for retrieving the ConfigId for the CCS API. - It contains the erpCompany, AvaTaxCompany, Version, ApplicationId, and all the application attributes.
--------------------	---

Properties

The following are the ApplicationModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	erpCompany	String	It is a unique Id assigned for the respective company.
2.	avaTaxCompany	String	It specifies the code of the company creating the transaction.
3.	semanticVersion	String	It is the connector version of the Connector.
4.	applicationId	String	It is the application Id of the connector.
5.	attributes	Map<String, String>	You can specify if any extra parameters need to be sent to the AvaTax Service.
6.	applicationAttributes	Map<String, String>	You can specify if any extra parameters need to be sent to the AvaTax Service.
7.	tags	List<String>	You can specify if any extra parameters need to be sent to the AvaTax Service.
8.	metadata	String	Created and Updated time.

StatusCodeWrapper

Description	It contains the method for handling the error or success response from the CCS APIs.
--------------------	--

Properties

The following are the StatusCodeWrapper properties and their information:

Sr. No.	Properties	Data Type	Description
1.	sCode	Integer	It is the Status code of the response received from the API.
2.	status	String	It specifies the status of the response received from the API.
3.	ServiceCode	String	It is the code received when there is an error received from the API.
4.	Message	String	It is the error message received from the response of API.

Sr. No.	Properties	Data Type	Description
5.	error	String	It is the error received from the response of API.

AppConfigGetSectionIdModel

Description	- It contains the method for retrieving the Config sectionId for the CCS API. - It contains all the configurations for the config section and address setting.
--------------------	---

Properties

The following are the AppConfigGetSectionIdModel properties and their information:

Sr. No.	Properties	Data Type	Description
1.	enable_save_transactions	Boolean	It is the configuration for saving the transaction in AvaTax.
2.	enable_tax_calculation	Boolean	It is the configuration for enabling the tax calculation in AvaTax.
3.	enable_logging	Boolean	It is the configuration for enabling the logging in AvaTax.
4.	enable_ecm	Boolean	It is the configuration for enabling the exemption in AvaTax.
5.	enable_address_validation	Boolean	It is the configuration for enabling the address validation in AvaTax.
6.	disable_triggers	Boolean	It is the configuration for disabling the triggers in AvaTax.
7.	company_code	String	It specifies the code of the company.
8.	address_zipcode	String	It specifies the address zip code in AvaTax.
9.	address_state	String	It specifies the address state in AvaTax.
10.	address_line_two	String	It specifies the address line two in AvaTax.
11.	address_line_three	String	It specifies the address line three in AvaTax.
12.	address_line_one	String	It specifies the address line one in AvaTax.
13.	address_country_code	String	It specifies the address country code in AvaTax.
14.	address_city	String	It specifies the address city in AvaTax.
15.	customer_code	String	It specifies the customer code in AvaTax.
16.	shipping_code	String	It specifies the shipping code in AvaTax.

Sr. No.	Properties	Data Type	Description
17.	enable_upc_as_item_code	Boolean	It is the configuration for enabling the UPC as item code in AvaTax.
18.	enable_tax_override	Boolean	It is the configuration for enabling the tax override in AvaTax.
19.	Metadata	String	Created and updated time.
20.	Items_threshold	Integer	The threshold value for number of line items after which batch processing for tax calculation is triggered. If the count of line items exceeds this threshold value, tax calculations will be handled in batches.